

Pricing and hedging financial derivatives

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Preface

The goals of this course are:

- Understand the role and the basic principles of using mathematical models in finance
- Understand the method of valuing derivative products by arbitrage and the importance of hedging
- Review the most important methods for pricing and hedging derivatives in financial markets, from theoretical, practical and numerical point of view.
- The focus of the course will be on continuous time models for a single asset.

How to use these lecture notes The lecture notes consist of 8 chapters, which correspond, approximately, to the 8 lectures of the course. At the beginning of each chapter you will find a “bird’s eye view and key concepts” box which lists the essential notions of each chapter. Many chapters have a jupyter notebook with illustrations associated to them - the link to a read-only version can be found at the end of the “bird’s eye view” box. To run the codes, save the notebook in your own google drive.

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Chapter 1

Introduction

Key concepts of this chapter

- *Buyers and sellers of derivatives; structure of an investment bank.*
- *Role of models in finance.*
- *The MBS and the CDO.*
- *Role of rating agencies.*
- *The course and main causes of the subprime crisis.*
- *Basic principles of financial derivatives pricing: hedging, self-financing portfolios and no arbitrage; difference with the insurance approach.*

1.1 Derivative markets and products

Types of derivative products

- *Futures/forwards* allow to fix today the price of a transaction which will take place at a future date.
- The *swaps* allow to exchange a series of financial flows
- The *options* are nonlinear products which give more freedom to the buyer, for example the call option on a financial asset gives the right but not the obligation to buy this asset at a future date at a fixed price

Category	Marker size (billion\$)
<u>OTC markets</u>	
Foreign exchange	118,004
Fixed income	529,813
Stocks / indices	7,783
Commodities	2,203
Credit derivatives	8,709
<u>Organized markets</u>	
Fixed income	86,725
Foreign exchange	455

Table 1.1: Market sizes for different classes of derivative products in December 2023.
Source: Bank of International Settlements.

- The *structured products* are investment strategies based on options and other derivatives which present specific guarantees and risks to the buyers (example: CDO)

Derivative products are sold in organized markets or over the counter (OTC) by the banks. Table 1.1 shows approximate market sizes for different classes of derivative products in 2018 by category of underlying.

Uses of derivative products

- *Hedging of risks:*
 - an company protects itself against fluctuations in raw material prices with a futures contract;
 - an investment fund uses options as part of the a portfolio insurance strategy
- *Speculation:*
 - an investment fund uses options as part of an aggressive portfolio management strategy
- *Arbitrage:*
 - a trader takes advantage of an imbalance in the market making an immediate profit
- More generally, *risk transfer:*
 - a company pays its employees in stock options

Cash flow →	Front office	Middle office	Back office
	Consulting / in- termédiation	Recording of trans- actions	<i>Model validation</i>
	Trading / sales	Reporting	<i>Risk management</i>
	<i>Quantitative research</i>	Cash management	Accounting
	Equity research		Support functions

Table 1.2: Simplified structure of an investment bank

Key players of the derivative markets

- The *sellers of options* are investment banks, via their activity of *market making*.
- The *buyers* are financial institutions: hedge funds (for speculation or arbitrage), companies and even local authorities, (to optimize cash flow management).
- The *organized markets* improve liquidity by standardizing the contracts and eliminating the counterparty risk, since the *clearing house* serves as a single counterparty for all transactions.
- An important role is played by the *regulators*, who monitor the risk management methodologies and ensure that the capital held by banks is sufficient to ensure their solvency.

Investment banks An *investment bank* is a financial institution whose fields of activity are consulting and intermediation in initial public offerings, mergers/acquisitions/debt issues and trading for its own account or on behalf of its clients. In France, investment banks are generally subsidiaries of large banks (SGCIB, CACIB). The 2013 Banking Separation law imposes strong restrictions on proprietary trading by banks. In the United States, the Glass-Steagall Act of 1933 mandated the separation between investment banks and retail banks. It was repealed in 1999 but in 2010, the Volcker Rule prohibited proprietary trading for US banks (leaving an exception for market making). Table 1.2 presents a simplified structure of an investment bank and Table 1.3 presents a simplified balance sheet structure. There are teams of quants in the front office as well as in the risk management/model validation teams.

Role of Regulators While customers are interested in the solvency and soundness of the bank, the management and the traders have an incentive to take more risks to

Assets (actif)		Liabilities (passif)	
Cash and central bank balance	10M	Customer deposits	80M
Securities:		Bond issues:	
- bonds	20M	- senior bond issues	25M
- stocks	20M	- subordinated bond issues	15M
- derivatives	10M	Short-term borrowing	30M
Loans and mortgages:		Reserves (for losses on loans)	20M
- corporates	30M		
- retail	40M		
- government	30M	Debt (sum of the above)	170M
Other assets			
- property	10M		
- investment in companies	10M	Equity (fonds propres)	30M
Short-term lending	20M		
Total	200M	Total	200M

Table 1.3: Balance sheet structure of an investment bank

maximize their remuneration in case of gain. Gains are rewarded by bonuses, but the losses are borne by the shareholders, or even sometimes by the taxpayers. The presence of these *conflicts of interest* as well as the importance of the banks for the economy makes it necessary to introduce *banking supervision*, whose role is to *ensure the solvency* of the banks. In Europe, the tasks of banking supervision are divided among BCE and the national regulators (Autorité de Contrôle Prudentiel et de Résolution).

The principles of banking supervision The principles of banking supervision are developed in the Basel Accords (I, II and III). Regulators monitor and enforce limits on regulatory ratios, the most important of which is the *capital ratio*:

$$\text{Capital ratio} := \frac{\text{Capital}}{\text{Risk-weighted assets}} \geq 8\%$$

Mathematics is used to calculate the capital (the value of certain illiquid assets) as well as for the calculation of the value of risk-weighted assets (risk measures).

- The first Basel accord (1988) focuses on the *banking book*. It defines the capital as follows (see Table 1.3 for an illustration):

$$\text{Capital} \approx \text{Equity (Tier 1)} + \text{Reserves (Tier 2)}$$

and introduces simple weightings for asset classes shown in Table 1.4.

1.2. ROLE OF QUANTITATIVE MODELS IN INVESTMENT BANKS AND FINANCE IN GENERAL 11

Cash, ECB deposits and sovereign bonds	Mortgages	Other assets: corporate bonds etc.
0%	50%	100%

Table 1.4: Risk weightings in Basel I.

- The *Market risk amendment* of 1996 introduces a specific method for computing the regulatory capital associated to the market risk (trading book). It also introduces an important distinction between the *standardized approach* and the *internal model approach*. For banks using an internal model, capital charge is determined, roughly speaking, by the *Value at Risk* at the level of 99% calculated for a 10-day horizon.
- The second Basel Accord (2005) clarifies the rules for computing risk indicators and extends the possibility of using the internal models to the banking book.
- The Third Basel Accord (2011) adopted in the wake of the crisis of 2007-2008 reinforces the constraints on capital and liquidity.
- The Fundamental review of trading book (2016) modifies the rules for the calculation of risk measures and in particular introduces the *Expected Shortfall* risk measure.

1.2 Role of quantitative models in investment banks and finance in general

Mathematical models are used by traders and risk managers for various purposes:

- *Hedging of derivative products*: For example, a trader who makes a volatility bet using a delta-hedged option needs a model to calculate the hedge ratio.
- *Valuation of illiquid assets*: exotic options in the bank's portfolio, for which there is no market price, are valued using a model to determine the capital of the bank.
- *Risk management*: the models are used to value the bank's portfolio under stress scenarios in order to calculate risk measures and assess regulatory capital.

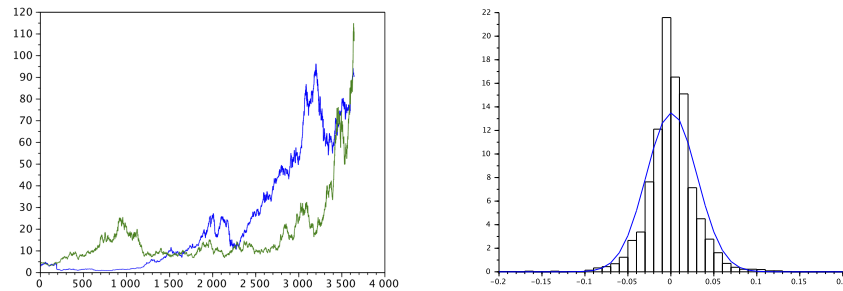


Figure 1.1: Left: APPL stock price over a 10-year period, compared with a geometric Brownian motion trajectory. Which is which? Right: Histogram of the daily returns of APPL superposed with the Gaussian distribution with the same mean and variance

The legitimacy of using mathematical models in the banking environment has been questioned after the crisis of 2007–2008. While that improper use of mathematical models may have been one of the reasons of this disaster, when properly used, the models remain an essential tool for financial risk management. Figure 1.1 shows a trajectory of Apple stock price (ticker APPL) together with a similar-looking trajectory of a the simplest mathematical model for stock price, that is, a geometric brownian motion. The readers will immediately convince themselves that it is very hard to tell “with a naked eye” which of the two trajectories corresponds to an actual stock price. And if this is the case, then clearly even such a simple model can already bring much useful information about the behavior of the stock price. Of course, if we take a “magnifying glass” and plot the histogram of the daily returns of the stock (Figure 1.1, right graph), we see that the law of stock returns does exhibit some differences from the Gaussian law: it has somewhat fatter tails and more mass in the middle. Thus, the geometric Brownian motion is not a perfect model, it should be accompanied with more precise tools to manage tail risk. Nevertheless, a model which is able to explain perhaps 95% of stock moves can be very useful in a variety of contexts.

Dangers of mathematical models Financial modeling is *fundamentally different* from modeling in natural sciences. The laws of physics do not change, but *asset prices are determined by the behaviour of market agents*, which changes according to the economic situation or even according to the models used. For example, before 2008, interbank market rates (Libor) were considered risk-free but a significant spread ap-

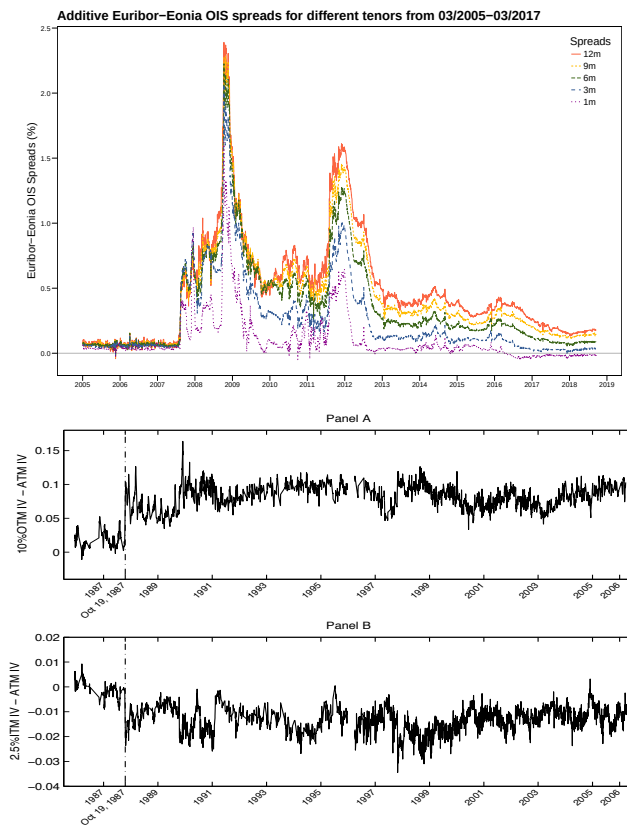


Figure 1.2: Top: spread between interbank market rates and the risk-free rate. Bottom: evolution of the implied volatility skew of S&P 500 options with one month to maturity. Source: [4]

peared during the crisis and persists to this day (see Figure 1.2, top graph). In a similar vein, a 'volatility smile' appeared after the 'Black Monday' of 1987 (as shown in the bottom graph of the same figure). This means that one needs to be extremely careful with financial models, especially those who attempt to make long-term forecasts or analyses. *All financial models are fundamentally wrong* in the sense that in addition to 'quantifiable' risks, there are uncertainties whose probability cannot be measured or estimated. The probabilistic approaches such as Value at Risk should therefore always be supplemented by an analysis aiming to *identify and take into account* possible loss scenarios without assigning them a specific probability.

One should also keep in mind that models developed by quants are used by the

people other than those who created them. Users of the models may not be aware of all of the assumptions / limitations, or deliberately decide not to take them into account. As a result, users of a financial model must always be informed, as accurately as possible, about the underlying assumptions of the model and its area of validity. These points are summarized in the "Financial Modeler's Manifesto", published by P. Wilmott and E. Derman, two famous quants, in the wake of the subprime financial crisis [8] (see also [7], [19] and the excellent film based on this book).

The Modelers' Hippocratic Oath

- I will remember that I didn't make the world, and it doesn't satisfy my equations.
- Though I will use models boldly to estimate value, I will not be overly impressed by mathematics.
- I will never sacrifice reality for elegance without explaining why I have done so.
- Nor will I give the people who use my model false comfort about its accuracy. Instead, I will make explicit its assumptions and oversights.
- I understand that my work may have enormous effects on society and the economy, many of them beyond my comprehension.

(source: Financial Modeler's Manifesto by Wilmott and Derman)

1.3 Lessons from the 2007-2008 subprime crisis

The global financial crisis of 2007–2008 which sparked the so called Great Recession (see Figure 1.3) has been largely caused by excessive growth of a particular type of credit derivatives: the *CDOs* (Collateralized Debt Obligations), and by major failures in understanding and management of the associated risks.

A real estate bubble The Fed's key rate cut following the internet (dot com) bubble of 2000 as well as U.S. government policies encouraging households to become homeowners, created a bubble in the real estate market (Figure 1.4, left graph). To meet the growing demand for housing loans, the banks used a method known as securitization,

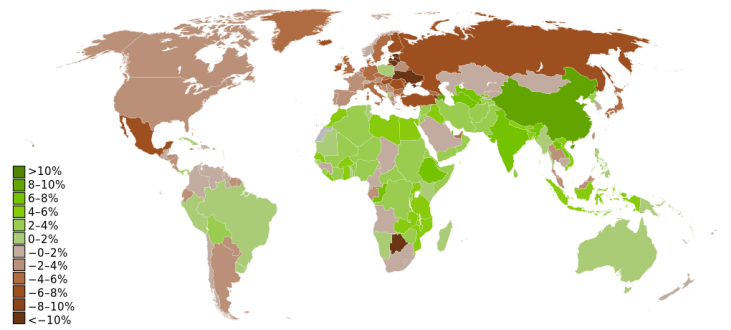


Figure 1.3: GDP growth in 2009. Source: Wikipedia

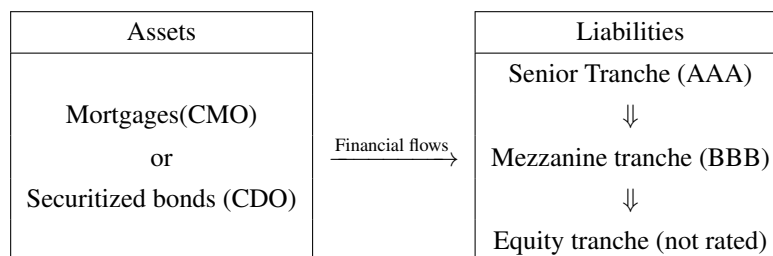


Table 1.5: Functioning of a CDO or CMO

or conversion of loans into securities (Table 1.5). In this method, individual loans are sold to an ad hoc company (known as special purpose vehicle), which issues different classes (tranches) of liabilities, which can then be sold to investors. The seniority of the tranche determines the order in which the holders of the corresponding liabilities are reimbursed and therefore its riskiness: the senior tranche is reimbursed first, is considered the least risky and therefore commands the smallest spread with respect to the risk-free rate, while the equity tranche is reimbursed last and is therefore the riskiest. These mortgage-backed bonds are called MBSs (mortgage-backed securities) or CMOs (collateralized mortgage obligations).

The senior tranches of MBSs were easily sold to investors, but the mezzanine and equity tranches were harder to sell. In response, the banks 'created the investor': the mezzanine and equity tranches were again securitized in CDOs (Figure 1.5). The senior tranches of CDOs were again highly rated and offered an attractive remuneration: the growing demand for CDOs fuelled the demand for real estate loans and led to the proliferation of subprime loans.

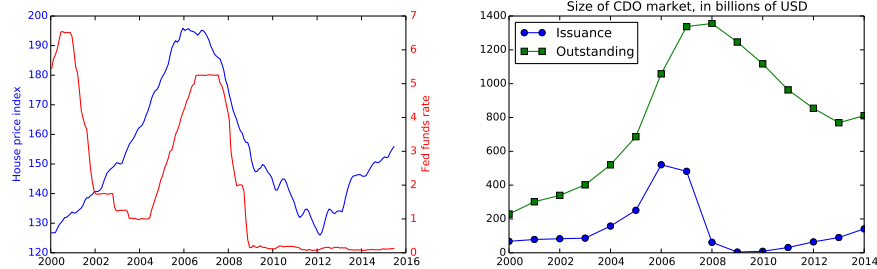


Figure 1.4: Left: House prices in the United States (data source: web page of Robert Shiller, the index is normalized to 100 in 1890) and the Fed funds rate (data source: Federal Reserve). Right: CDO market before and after the subprime crisis. Data source: Bank of International Settlements.

THE THEORY OF HOW THE FINANCIAL SYSTEM CREATED AAA-RATED ASSETS OUT OF SUBPRIME MORTGAGES

In the financial system, AAA-rated assets are the most valuable because they are the safest for investors and the easiest to sell. Financial institutions packaged and re-packaged securities built on high-risk subprime mortgages to create AAA-rated assets. The system

worked as long as mortgages all over the country and of all different characteristics didn't default all at once. When homeowners all over the country defaulted, there was not enough money to pay off all the mortgage-related securities.

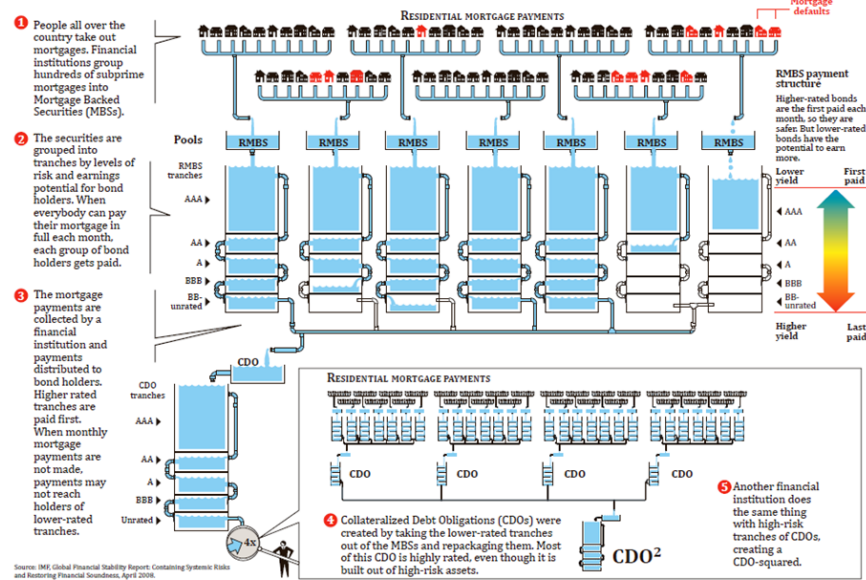


Figure 1.5: Creation of AAA assets from subprime loans

Rating agencies The credit quality of an issuer is measured by the credit rating: an assessment of the issuer's creditworthiness calculated internally by the bank or by an external rating agency. Sovereign and corporate debt ratings as well as the ratings of CDO issues are calculated by the rating agencies, the best known of which are Standard & Poor's, Moody's, and Fitch ratings. For example, the rating scale of Standard and Poor's is AAA, AA, A, BBB, BB, B, CCC, CC, C, D, where AAA is the best possible rating and D corresponds to default. Bonds rated at least BBB- are considered adequate for long-term investments ('investment grade'), while the others are considered as speculative investments. Rating agencies present their ratings as simple 'opinions' when in fact they play a very important role in the banking regulation. Some money market funds can only invest in AAA-rated bonds, and in the ratings determine asset weightings for the calculation of the capital ratio. For example under the Basel II framework, an AAA rating corresponds to a weighting of 0.6%, and a rating of BBB corresponds to a weighting of 4.6%. This means that a bank that has AAA-rated assets in its portfolio will have to hold almost 8 times less regulatory capital than a bank whose assets are rated BBB.

The unfolding of the crisis: phase 1

- With *the increase of the Fed funds rate* in 2006 the borrowers with floating-rate loans could no longer repay their debt and their homes were actioned off.
- With the saturation of the market due to the large number of houses under construction, this has first caused a slowdown and then a *sudden fall* in the housing market, which further increased the rate of defaults, because buyers could no longer refinance their loans.
- Rating agencies took notice of an increase in the rate of defaults and *lowered the ratings of MBS and CDO*.
- This had immediate repercussions on the *valuation of CDOs in the balance sheet of banks* and investments funds (accounting at market value).
- In July 2007 Bear Stearns announced unprecedented losses in two of its funds massively invested in CDO.
- In August, BNP Paribas temporarily suspended withdrawals of capital of three of its funds invested in CDOs.

- In October 2007, Merrill Lynch announced losses of 7.9 billion dollars related to its CDO business.
- The panic movement triggered by these losses then reduced the liquidity of the CDO market to almost zero.

The unfolding of the crisis: phase 2

- The liquidity crisis in the CDO market triggered a *crisis of confidence* in the interbank market:
- It became increasingly difficult for the investment banks to finance themselves on a day-to-day basis.
- To avoid the worst outcome, in March 2008, the Fed urgently negotiated the takeover of Bear Stearns by JP Morgan, providing guarantees of up to 30 billion dollars.
- The Bear Stearns shareholder bailout with taxpayer money put the Fed under strong pressure.
- In England, the Northern Rock Bank was nationalized in the spring 2008 after a rush on its deposits.
- Crisis peaked in September 2008 with the *default of Lehman Brothers* and the emergency buyout of Merrill Lynch by Bank of America.
- Goldman Sachs and Morgan Stanley converted themselves to retail banks to have access to emergency funding from the Fed.
- Fanny Mae and Freddie Mac Real Estate Lending Companies were taken over by the federal government, and AIG received more than 100 billion from the federal government.
- Fortis Bank was nationalized in Belgium on September 29.
- During the week of October 6 to 10, 2008, several major indices such as S&P 500 or CAC40 lost more than 20% of their value.
- In October 2008, a federal law put in place an unprecedented \$700 billion bank rescue program.

Failures in CDO risk management The *poor assessment and management of CDO risks*, by the investors, the banks and the rating agencies played a key role in the outbreak of the crisis. Most banks were using models too simple to account for the complex reality of simultaneous defaults that affect the pay-off of a CDO. For example, the Gaussian copula model of D. Li [20, 21] reduced all the complexity of the problem to a single correlation value (see box below). Other CDO buyers trusted the ratings issued by the rating agencies at the request of the vendors. Even the rating agencies could not really understand the complexity of CDOs and were obliged to make important simplifications in their modeling.

The Gaussian copula model.

Let $F_1, \dots, F_n$ be one-dimensional probability distribution functions. One can define an n -dimensional distribution function by

$$F(x_1, \dots, x_n) := C_\rho(F_1(x_1), \dots, F_n(x_n)),$$

where

$$C_\rho(u_1, \dots, u_n) = \Phi_\rho(N^{-1}(u_1), \dots, N^{-1}(u_n)),$$

with N the distribution function of the standard 1-dimensional Gaussian law and Φ_ρ the distribution function of an n -dimensional Gaussian vector whose components are standard normal random variables and the correlation between any two components is equal to ρ . C_ρ is called the Gaussian copula with parameter ρ .

D. Li's model hypothesizes that the dependence between the dates of default of n bonds of a CDO is given by the Gaussian copula.

The role of rating agencies The risk of a CDO depends on the probabilities of non-repayment of individual loans and correlations between the defaults. In their models, the rating agencies *never considered individual loans* but used the ratings of the MBS tranches, calculated over a too short historical period. The *systemic risk*, linked to a global real estate market downturn, was largely underestimated. The correlation between default events of the different tranches was not modeled correctly in the absence of reliable data: at Moody's, only 20 years of historical data was used [10]. In addition, the teams were overcharged and there existed an *important conflict of interest* since agencies were paid by the banks that structured the CDOs.

Subprime crisis: summary

- The risk of CDOs was not properly reflected in the ratings assigned to them by the rating agencies because of *gross underestimation of systemic risk* associated with the general downturn in the real estate market.
- Where this risk materialized in 2006-2007, the rating agencies have massively downgraded the CDO ratings, which *reduced almost to zero the liquidity* of these products and initiated a panic in the interbank market.
- It is not the mathematical models that are responsible for the crisis but on the one hand *their use outside their domain of validity* and without proper calibration by the rating agencies, and on the other hand a *blind trust contrary to all common sense* in these models by the banks and the investors.

1.4 Consequences of the crisis and the post-crisis evolution of the financial system

The 2007–2008 financial crisis triggered significant evolutions both in terms of improving our understanding of the nature and importance of various classes of financial risks and in terms of development of new regulations and risk management tools designed to prevent future crisis, which improved the overall resilience of banks and other financial institutions. This is one of the reasons why the COVID pandemic, which was a major shock for the world economy, did not lead to a global financial meltdown.

Consequences of the crisis: new risks A direct consequence of the financial crisis was to draw attention to the importance of certain types of financial risks, which were of course known before the crisis, but were not considered as prominent. For example, the counterparty risk, which was assumed to be negligible for transactions involving major banks, is now considered to be a part of every transactions.

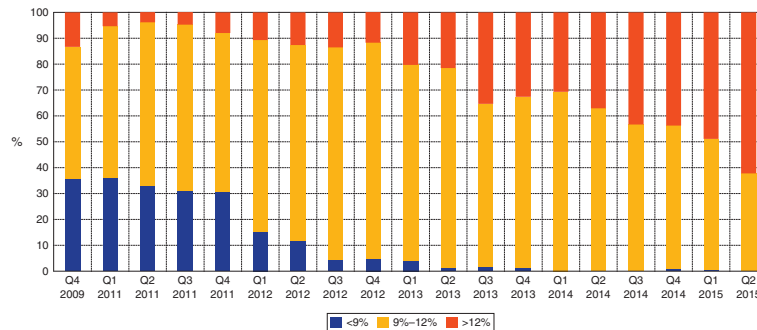
- The *liquidity risk*, which manifests itself in the form of funding liquidity risk (impossibility to obtain short-term funding to finance a transaction) and market liquidity risk (impossibility to find a buyer for an illiquid asset), is now the target of an explicit adjustment to the price of every derivative product: the funding valuation adjustment (FVA) and appears explicitly in the new risk management procedures in Basel III accords.

1.4. CONSEQUENCES OF THE CRISIS AND THE POST-CRISIS EVOLUTION OF THE FINANCIAL SYSTEM²¹

- The *default risk* and the *counterparty risk* (which are basically the same risks from different perspectives) were of course recognized before the crisis, but default spreads of large financial institutions and developed countries were almost non-existent. Following the financial crisis of 2007–2008 (and in particular the default of Lehman Brothers) and the Eurozone crisis of 2010–2012 it became common knowledge that default and counterparty risks are part of every transaction regardless of the size of the counterparties. The spread between the risk-free rate (EONIA) and the interbank market rate (EURIBOR) which appeared during the crisis (see Figure 1.2, top graph), still exists and represents the aggregate default risk of major banks. Option pricing methodologies were updated to reflect these risks, giving rise to debt valuation adjustment (DVA) and credit valuation adjustment (CVA).
- The *systemic risk*, which corresponds to the risk of a global collapse of the financial system caused by default contagion, was recognized by the regulator as a major source of financial instability. To limit this risk, new markets were created to improve liquidity and reduce counterparty exposure (such as organized markets for credit default swaps); and new products (contingent convertible bonds or CoCos) allowing to recapitalize the banks quickly were introduced.
- The *model risk* corresponds to using a model which does not provide an adequate representation of the risks essential for a particular product that is being priced: this may happen when the model is simply not precise enough, when the parameters are not well calibrated, or when the model is being used outside its domain of validity.

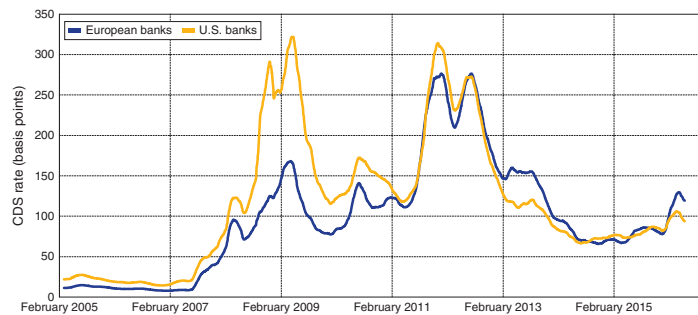
Consequences of the crisis: regulatory response

- Since 2007–2008, the banks have been regularly subject to *stress tests* conducted by national regulators and aiming to assess their financial stability.
- *Dodd-Frank act* (2010): Volker Rule: prohibition of proprietary trading by banks, creation of Financial Stability Supervisory Board...
- *European Market Infrastructure Regulation* (2012): new rules for the operation of over-the-counter derivatives markets.
- *Thirs Basel accord* (2013–2019): Reinforcement of capital and liquidity constraints, provisions for the model risk of illiquid products...



Data source: European Banking Authority.

Note: Fractions of European banks with CET1 ratios in the indicated ranges: less than 9%, between 9% and 12%, and more than 12%.



Data source: Bloomberg.

Figure 1.6: Improving the resilience of financial institutions. Top: CET1 capital ratios of European banks. Bottom: 5 year CDS rates on a sub-group of major European and American banks. Source: [9].

- *Fundamental Review of Trading Book* (FRTB, 2016, entry into force in 2019) - new standards of regulatory capital calculation ;
- *MIFID-II* (2014, entry into force in 2018) - transparency of markets, investor protection.

Consult [9] for a detailed review of financial regulatory reforms after the crisis and the annual reports of *Financial Stability Board* [11] to find out more about the state of their implementation in the G20 countries. Thanks to the implementation of more robust rules for calculating the capital ratio in Basel III, G20 banks are now better capitalized than before the crisis (see Figure 1.6).

Getting over with the “too big to fail” In the event of the bankruptcy of a financial institution of systemic importance, the government has a choice between

1.4. CONSEQUENCES OF THE CRISIS AND THE POST-CRISIS EVOLUTION OF THE FINANCIAL SYSTEM²³

- Letting the establishment go bankrupt with the consequences detrimental to the real economy
- Using taxpayers' money to recapitalize the business

The company is encouraged to take the risk through the assumption that it will be supported by the government no matter what happens. To avoid this wrong incentive, the supervisor must be able to safely resolve the failure of a systemically important company without using public funds. Following the crisis, controlled bankruptcy resolution mechanisms have been introduced in the US and Europe, but their credibility is not yet ensured: the bankruptcy of a major international bank is still likely to send a disastrous shockwave on the markets.

Reducing systemic risk in the derivatives market The majority of standard derivatives are now negotiated in organized venues. Central clearing reduces counterparty risk as well as the risk of default contagion. However, the risk of central counterparties (CCP) defaults must also be taken into account: CCPs have been subject to stress tests since 2016, but a mechanism of controlled resolution of CCPs is yet to be defined. Also, FX derivatives remain a poorly regulated sector.

Regulation of the parallel banking system Reforms are being implemented to regulate 'shadow banks':

1. Mutual funds (LTCM);
2. Non-bank lenders dependent on short-term financing, such as consumer credit companies, etc.;
3. Market intermediaries that rely on short-term financing or secured financing of client assets, such as brokers and dealers;
4. Companies facilitating credit creation, such as credit insurance companies, financial guarantors and monoline insurers;
5. Securitization intermediaries.

Improving competition and transparency in the financial markets

- Switch from bilateral to multilateral trading: over 2/3 of swap transactions in the US now take place at MFT with a centralized order book

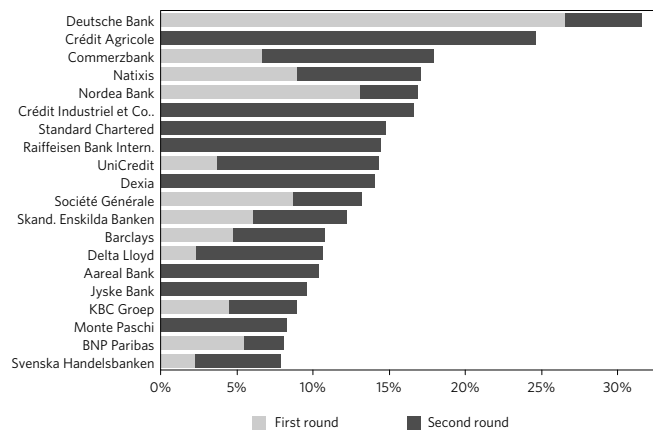


Figure 1.7: A climate stress test of the financial system: capital losses of a sample of 20 banks in the scenario of loss of 100% of the capitalization of the fossil energy / utility sector. Source: [1].

- Regulations such as MIFID-2 have mandated pre- and post-trade transparency.

Emerging risks: the tragedy of the horizons In addition to the endogenous instability of the financial system, future crises may have their origins outside the financial world; among important potential risk sources are pandemics, geopolitical instability, cyber-related events and, perhaps most importantly, climate change.

The *climate change* creates unprecedented systemic risks for our economy (see Figure 1.7), which have been categorized [5] into physical risks (destruction of assets) and transition risks (losses due to regulatory action, stranded assets). Actors of the financial system are becoming aware of the need to take into account these risks and to develop instruments for funding the energy transition.

1.5 The principles of derivative pricing

Pricing through hedging The business of an option trading desk is to sell options and implement hedging strategies to minimize the risks for the bank. In the real world it is of course not possible to hedge all risk completely, but provisions / reserves must be allocated for all unhedged risks and the regulator ensures that the pricing method – hedging strategy – provisions/reserves is globally conservative. The “first best” situation is a back-to-back transaction, where two products with identical characteristics

compensate each other in the bank's portfolio (but even in this case counterparty exposure may create unhedged risks). The “second best” setting is that of “simple” derivative products like futures and swaps, which can be hedged with static positions in the underlying, without using a model. Finally, for more complex “nonlinear” products such as options, a probabilistic model of the market is required.

Dynamic hedging and absence of arbitrage In the most simple setting, a probabilistic model consists of (i) a set of possible scenarios for market evolution with associated probabilities and (ii) definition of *admissible portfolios* which represent allowed trading strategies for the agents. In the context of dynamic trading, an important notion is that of *self-financing* strategies: admissible strategies in which the gain arises only from variation of asset prices, with no additional funds added. The pricing methodology is based on the following assumption of absence of arbitrage.

Absence of arbitrage

There is no self-financing portfolio, with zero initial cost, such that its value at a future date T is nonnegative, and strictly positive with a non-zero probability.

If the class of admissible portfolios is large enough, one can “sometimes” find one which replicates exactly the pay-off of the option we want to price. By absence of arbitrage, the price of the option is then equal to the value of the replicating portfolio. The possibility of perfect replication is therefore model-dependent: in more realistic models with many risk factors perfect hedging is unlikely to be possible.

Example: one-period binomial model Consider a simple market where trading happens at two dates, $t = 0$ and $t = 1$. The market consists of two assets, whose prices at $t = 0$ are S_0^0 and S_0^1 and at date $t = 1$ are S_1^0 and S_1^1 . The asset S^0 is riskless: $S_1^0 = (1+r)S_0^0$ where $r > 0$ is the interest rate. The asset S^1 is risky: S_1^1 is not known at $t = 0$ and takes two possible values $S_1^1 = uS_0^1$ with probability p and $S_1^1 = dS_0^1$ with probability $1 - p$.

Let ϕ^0 be the quantity of risk-free asset and ϕ^1 be the quantity of risky asset. The condition of perfect replication under two scenarios writes:

$$\phi^0 S_0^0(1+r) + \phi^1 S_0^1 d = h(S_1^1 d), \quad \phi^0 S_0^0(1+r) + \phi^1 S_0^1 u = h(S_1^1 u).$$

Solving these equations we find the optimal strategy:

$$\begin{pmatrix} \phi^0 \\ \phi^1 \end{pmatrix} = \begin{pmatrix} (1+r)S_0^0 & dS_0^1 \\ (1+r)S_0^0 & uS_0^1 \end{pmatrix}^{-1} \begin{pmatrix} h(dS_0^1) \\ h(uS_0^1) \end{pmatrix} = \begin{pmatrix} \frac{uh(dS_0^1) - dh(uS_0^1)}{S_0^0(1+r)(u-d)} \\ \frac{h(uS_0^1) - h(dS_0^1)}{S_0^1(u-d)} \end{pmatrix}.$$

Consequently, the value of the replicating portfolio at $t = 0$ and the option's price are

$$\begin{aligned} V_0 &= \phi^0 S_0^0 + \phi^1 S_0^1 = \frac{uh(dS_0^1) - dh(uS_0^1)}{(1+r)(u-d)} + \frac{h(uS_0^1) - h(dS_0^1)}{(u-d)} \\ &= \frac{qh(uS_0^1) + (1-q)h(dS_0^1)}{1+r}, \end{aligned}$$

where $q := \frac{1+r-d}{u-d} \in (0, 1)$.

The option price thus takes form of expectation under the probability $\mathbb{Q} = \{q, 1 - q\}$ which is not directly related to the historical one $\mathbb{P} = \{p, 1 - p\}$. $\mathbb{Q}$ is called *risk-neutral probability* because the prices of risky assets are given by (discounted) expectations under $\mathbb{Q}$, while under the real-world measure risky assets command a lower price because of the presence of the risk premium.

The insurance approach The financial approach of pricing by replication is fundamentally different from the insurance approach based on aggregation of independent risks. To illustrate this idea, consider an insurance company with N clients. The payment to n -th client is denoted by S^n , and the variables $(S^1, \dots, S^N)$ are assumed to be i.i.d.

Let $m = \mathbb{E}[S^1]$ and $\sigma^2 = \text{Var}[S^1]$. By the central limit theorem, as $N \rightarrow \infty$,

$$\mathbb{P} \left[\sum_{n=1}^N \left(-S^n + m - \frac{x\sigma}{\sqrt{N}} \right) \leq 0 \right] \rightarrow N(x),$$

where $N(x)$ is the standard normal distribution function. When the number of clients is large, to ensure solvency with probability α the premium must be equal to

$$\mathbb{E}[S^1] - \frac{N^{-1}(\alpha)\sigma}{\sqrt{N}}.$$

Therefore, modulo a small correction, the insurance premium equals the expectation under the historical measure of the terminal cashflow.

Before the subprime crisis, the functioning of CDOs was based on the logic of insurance: it was believed that the diversification of the risks of non-repayment of the various borrowers reduced the risk of these products to almost zero. In reality,

however, this is not the case: real estate borrowers are far from independent, as they are all sensitive to the systemic risk of a downturn in the real estate market.

Chapter 2

Black-Scholes option pricing model: the martingale approach

Key concepts of this chapter

- *Pricing and hedging financial derivatives in a specific probabilistic model of the financial market*
- *Classical setting: Brownian filtration, Itô processes*
- *Standard assumptions: perfect markets, everything can be hedged*
- *Dynamics of a self-financing portfolio containing several risky assets. Representation of portfolio strategies in terms of the number of units of each asset, the amount invested in each asset, and the proportion of total wealth invested in each asset.*
- *Admissible strategies, absence of arbitrage opportunities, risk premium.*
- *Risk neutral valuation formula for contingent claims.*

2.1 Description of the model

Key assumptions and modeling choices The following table illustrates the key assumption behind the Black-Scholes model. As one can see, most of them are not satisfied by the actual financial markets. Nevertheless, this model provides a useful

approximation to the reality – but one of course needs to be aware of its shortcomings and deploy appropriate risk management tools.

There are no riskless arbitrage opportunities (impossible to outperform risk-free asset or earn money from zero initial investment without taking any risk) ⇒ True in normal market conditions (but remember transaction costs)	True
Security trading is continuous ⇒ Markets close at night; portfolio readjustment is discrete; there are “circuit breakers” which stop trading in emergency situations	False
No transactions costs or taxes ⇒ There are multiple transaction costs: bid-ask spreads, exchange fees, order book liquidity costs For liquid assets and low hedging frequencies costs are small	False
All securities are perfectly divisible ⇒ Minimum trade: one share, often more on futures For large positions, residual P&L due to this is usually small	False
No dividends during the life of the derivative ⇒ false in most cases as stocks distribute dividends <u>Most models are easy to adapt for dividends</u>	False
The short selling of securities is permitted ⇒ Short selling permitted via repo but requires posting a margin; may be impossible under stressed conditions	True
There exists a risk-free asset ⇒ Some government bonds and short-term collateralized interbank loans may be considered risk-free (Remember subprime crisis and Eurozone crisis)	True
The stock price dynamics is driven by a multidimensional Brownian motion ⇒ in practice stock prices may exhibit jumps	False
The market is complete: all risk may be hedging by investing in risky assets ⇒ In practice there are unhedged risks (stochastic volatility)	False

The model for asset prices Let $W = (W_t^1, \dots, W_t^d)_{t \geq 0}^T$ be a standard Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$, and $(\mathcal{F}_t)_{t \geq 0}$ its completed natural filtration. Consider a financial market with a risk-free asset

$$S_t^0 = \exp \left(\int_0^t r(u) du \right),$$

and d risky assets with the dynamics

$$S_t^i = S_0^i \exp \left(\int_0^t \left(b^i(u) - \frac{1}{2} \sum_{j=1}^d |\sigma^{ij}(t)|^2 \right) dt + \int_0^t \sum_{j=1}^d \sigma^{ij}(t) dW_t^j \right), \quad i = 1 \dots d,$$

where r, μ and σ are adapted process satisfying

$$\int_0^T (\|r(t)\| + \|b(t)\| + \|\sigma(t)\|^2) dt < \infty, \quad \text{p.s.,}$$

We assume that $\sigma(t)$ is invertible and $r(t) \geq 0$ for all t .

By Itô's formula,

$$dS_t^i = S_t^i \left(b^i(t) dt + \sum_{j=1}^d \sigma^{ij}(t) dW_t^j \right),$$

or with matrix notation

$$dS_t = \text{diag}[S_t](b(t)dt + \sigma(t)dW_t),$$

where $S_t = (S_t^1, \dots, S_t^d)^T$, $b(t) = (b^1(t), \dots, b^d(t))^T$.

Introduce the *risk premium vector*: $\lambda(t) = \sigma(t)^{-1}(b(t) - r(t)\mathbf{1})$, so that

$$dS_t = \text{diag}[S_t] \{r(t)\mathbf{1}dt + \sigma(t)(\lambda(t)dt + dW_t)\}.$$

To simplify notation, we use discounting

$$\tilde{S}_t = \frac{S_t}{S_t^0} = S_t e^{-\int_0^t r(s) ds} \Rightarrow d\tilde{S}_t = \text{diag}[\tilde{S}_t] \sigma(t)(\lambda(t)dt + dW_t).$$

Portfolio strategies A strategy can be expressed

- in terms of *quantities* of each risky asset $\delta_t = (\delta_t^1, \dots, \delta_t^d) \in \mathbb{R}^d$,
- in terms of *amounts* to invest in each risky asset $\pi_t = (\pi_t^1, \dots, \pi_t^d) \in \mathbb{R}^d$
- if the portfolio value X_t is positive at all times, in terms of *proportions* of wealth invested in each risky asset $\omega_t = (\omega_t^1, \dots, \omega_t^d) \in \mathbb{R}^d$.

These coefficients are related as follows:

$$\pi_t^i = \delta_t^i S_t^i \quad \text{et} \quad \omega_t^i = \frac{\delta_t^i S_t^i}{X_t}, \quad i = 1, \dots, d.$$

The amount invested into risk-free asset is given by,

$$X_t - \sum_{i=1}^d \delta_t^i S_t^i = X_t - \sum_{i=1}^d \pi_t^i = X_t \left(1 - \sum_{i=1}^d \omega_t^i \right)$$

Self-financing strategies The self-financing equation expresses the fact that no amount is withdrawn or injected into the portfolio. The dynamics of a self-financing portfolio is given by the following formulas:

- In terms of quantities:

$$dX_t = \sum_{i=1}^d \delta_t^i dS_t^i + \left(X_t - \sum_{i=1}^d \delta_t^i S_t^i \right) \frac{dS_t^0}{S_t^0}.$$

- In terms of amounts:

$$dX_t = \sum_{i=1}^d \pi_t^i \frac{dS_t^i}{S_t^i} + \left(X_t - \sum_{i=1}^d \pi_t^i \right) \frac{dS_t^0}{S_t^0}.$$

- In terms of proportions:

$$\frac{dX_t}{X_t} = \sum_{i=1}^d \omega_t^i \frac{dS_t^i}{S_t^i} + \left(1 - \sum_{i=1}^d \omega_t^i \right) \frac{dS_t^0}{S_t^0}.$$

From now on we focus portfolio strategies expressed in terms of *amounts*. A strategy is a pair $(x, (\pi_t)_{0 \leq t \leq T})$, where x is the initial value and $\pi_t = (\pi_t^1, \dots, \pi_t^d) \in \mathbb{R}^d$ are the amounts to invest in risky assets. The *discounted* self-financing portfolio value then writes:

$$d\tilde{X}_t = \tilde{\pi}_t \text{diag}[\tilde{S}_t]^{-1} d\tilde{S}_t = \tilde{\pi}_t \sigma(t) \{ \lambda(t) dt + dW_t \}.$$

We denote by $X_t^{x, \pi}$ the value of portfolio with initial value x and strategy π .

Risk-neutral probability Assume Novikov's condition:

$$\mathbb{E} \left[\exp \left(\frac{1}{2} \int_0^T \|\lambda(t)\|^2 dt \right) \right] < \infty,$$

then under the measure $\mathbb{Q}$ defined by

$$\frac{d\mathbb{Q}}{d\mathbb{P}} \Big|_{\mathcal{F}_T} = \exp \left(- \int_0^T \lambda(t) dW_t - \frac{1}{2} \int_0^T \|\lambda(t)\|^2 dt \right),$$

the process $\widehat{W}_t = \int_0^t \lambda(s) ds + W_t$ is a Brownian motion and we write

$$\tilde{X}_t^{x,\pi} = x + \int_0^t \tilde{\pi}_u \sigma(u) d\widehat{W}_u. \quad (2.1)$$

Under $\mathbb{Q}$, the discounted values of assets and portfolios are *local martingales*, but not always true martingales. The measure $\mathbb{Q}$ is called *risk-neutral probability* because under $\mathbb{Q}$ the expected returns of assets do not depend on their risk level.

Admissible strategies We now describe the class of strategies the agents are allowed to use in our market.

Definition 1. The strategy $(\pi_t)_{0 \leq t \leq T}$ is said to be admissible if

- i. the stochastic integral in (2.1) is well defined, in other words,

$$\int_0^T \|\tilde{\pi}_u \sigma(u)\|^2 du < \infty \quad \text{p.s.,}$$

- ii. the portfolio value satisfies

$$\tilde{X}_t^{0,\pi} \geq M_t, \quad 0 \leq t \leq T, \quad (2.2)$$

where M is an a.s. continuous $\mathbb{Q}$ -martingale (finite credit line condition).

The set of admissible strategies is denoted $\mathcal{A}$.

2.2 Absence of arbitrage

The theory of contingent claim pricing is largely based on the following property of financial markets.

Definition 2 (Absence of arbitrage opportunities). A financial market does not admit arbitrage opportunity if for any admissible strategy π ,

$$X_T^{0,\pi} \geq 0 \quad \text{p.s.} \quad \Rightarrow \quad X_T^{0,\pi} = 0 \quad \text{p.s.}$$

Theorem 1. *The financial market defined above in this section does not admit arbitrage opportunity*

Proof. Let π be an admissible strategy with $X_T^{0,\pi} \geq 0$ a.s., and define

$$\tau_n = \inf \left\{ t : \int_0^t \|\tilde{\pi}_u \sigma(u)\|^2 du \geq n \right\} \wedge T.$$

Then, (τ_n) converges a.s. to T and

$$\mathbb{E}^{\mathbb{Q}} \left[\int_0^{\tau_n} \|\tilde{\pi}_u \sigma(u)\|^2 du \right] \leq n < \infty,$$

so that

$$\mathbb{E}^{\mathbb{Q}}[\tilde{X}_{\tau_n}^{0,\pi}] = 0.$$

for all n . By Fatou's lemma, since $\tilde{X}_{\tau_n}^{0,\pi} - M_{\tau_n} \geq 0$, we have,

$$\mathbb{E}^{\mathbb{Q}}[\tilde{X}_T^{0,\pi}] = \mathbb{E}^{\mathbb{Q}}[\lim_{n \rightarrow \infty} (\tilde{X}_{\tau_n}^{0,\pi} - M_{\tau_n})] + \mathbb{E}^{\mathbb{Q}}[M_0] \leq \lim_{n \rightarrow \infty} \mathbb{E}^{\mathbb{Q}}[(\tilde{X}_{\tau_n}^{0,\pi} - M_{\tau_n})] + \mathbb{E}^{\mathbb{Q}}[M_0] = 0,$$

so that $X_T^{0,\pi} = 0$ a.s. □

Doubling strategies The “finite credit line condition” (2.2) is needed to avoid “doubling strategies”, an example of which is given below.

Assume that the interest rate is null and the single risky asset value follows $S_t = W_t$, where W is a standard Brownian motion. The strategy consists in holding $2^{n/2}$ units of risky asset on the interval $(t_{k-1}, t_k]$, for $k = 0, 1, \dots$, where $t_k = 1 - \frac{1}{2^k}$. The portfolio value at t_n is then given by

$$X_{t_n} = \sum_{k=1}^n 2^{k/2} (W_{t_k} - W_{t_{k-1}}) = \sum_{k=1}^n Z_k,$$

where $(Z_k)_{k \geq 1}$ is a sequence of identically distributed standard normal random variables. In other words, $(X_{t_n})_{n \geq 1}$ is a Gaussian random walk which is known to satisfy

$$\inf\{n : X_{t_n} \geq K\} < \infty \quad \text{p.s.} \quad \forall K.$$

This strategy is thus an arbitrage opportunity (see below) and since the finite credit line condition (2.2) excludes arbitrage opportunities, we conclude that this strategy does not satisfy (2.2).

2.3 Pricing contingent claims

Superhedging price We start our discussion of contingent claim pricing by introducing the notion of superhedging price, that is, the minimum cost of a dominating portfolio for a given contingent claim.

Definition 3 (Superhedging / subhedging price). Let $G \in \mathcal{F}_T$. The superhedging price of G at $t = 0$ and $t \in [0, T]$ is defined by

$$\begin{aligned}\bar{V}(G) &= \inf\{x \in \mathbb{R} : \exists \pi \text{ with } \pi \in \mathcal{A} \text{ and } X_T^{x,\pi} \geq G\}, \\ \bar{V}_t(G) &= \text{essinf}\{X \in \mathcal{F}_t : \exists \pi \in \mathcal{A} \text{ with } X_t^{x,\pi} = X \text{ and } X_T^{x,\pi} \geq G\}.\end{aligned}$$

Similarly, the subhedging price is defined by

$$\begin{aligned}\underline{V}(G) &= \sup\{x \in \mathbb{R} : \exists \pi \text{ with } -\pi \in \mathcal{A} \text{ and } X_T^{x,\pi} \leq G\}, \\ \underline{V}_t(G) &= \text{esssup}\{X \in \mathcal{F}_t : \exists \pi \text{ with } -\pi \in \mathcal{A}, X_t^{x,\pi} = X \text{ and } X_T^{x,\pi} \leq G\}.\end{aligned}$$

The presence of $-\pi$ in the definition of the subhedging price makes sense: if the asset is available in the market at a price below its subhedging price, the arbitrage strategy is to buy the asset and sell the subhedging portfolio, and for this to happen, the strategy $-\pi$ must be admissible. The following proposition is formulated for the time-zero prices $\bar{V}(G)$ and $\underline{V}(G)$, but the results also hold for the time t prices $\bar{V}_t(G)$ and $\underline{V}_t(G)$.

Proposition 1 (Properties of the superhedging price). *Let G, G_1, G_2 belong to $\mathcal{F}_T$. The following properties hold true.*

1. *Monotonicity: if $G_1 \geq G_2$ a.s., then $\bar{V}(G_1) \geq \bar{V}(G_2)$.*
2. *Sub-additivity: for all G_1, G_2 , $\bar{V}(G_1 + G_2) \leq \bar{V}(G_1) + \bar{V}(G_2)$.*
3. *Relationship with the subhedging price: for all G , $\underline{V}(G) = -\bar{V}(-G)$ and $\underline{V}(G) \leq \bar{V}(G)$.*
4. *Bounds for the market price: In the absence of arbitrage, the market price of a pay-off G , denoted by $V(G)$, satisfies $\underline{V}(G) \leq V(G) \leq \bar{V}(G)$. In particular, if a claim G is replicable with a portfolio (x, π) such that both π and $-\pi$ are admissible, then $V(G) = x$.*

Proof.

1. Monotonicity. From the definition of the superhedging price, for any $\varepsilon > 0$, one can find $x \in \mathbb{R}$ and $\pi \in \mathcal{A}$ such that $x \leq \bar{V}(G_1) + \varepsilon$ and

$$X_T^{x,\pi} \geq G_1.$$

Thus also, $X_T^{x,\pi} \geq G_2$, so that $\bar{V}(G_2) \leq x \leq \bar{V}(G_1) + \varepsilon$. Since ε is arbitrary, we conclude that $\bar{V}(G_2) \leq \bar{V}(G_1)$.

2. Sub-additivity. From the definition of the superhedging price, for any $\varepsilon > 0$ we can find $x_1, x_2 \in \mathbb{R}$ and $\pi_1, \pi_2 \in \mathcal{A}$ such that

$$x_1 \leq \bar{V}(G_1) + \varepsilon, \quad x_2 \leq \bar{V}(G_2) + \varepsilon, \quad X_T^{x_1, \pi_1} \geq G_1 \text{ and } X_T^{x_2, \pi_2} \geq G_2.$$

By linearity of portfolio,

$$X_T^{x_1+x_2, \pi_1+\pi_2} \geq G_1 + G_2,$$

so that

$$\bar{V}(G_1 + G_2) \leq x_1 + x_2 \leq \bar{V}(G_1) + \bar{V}(G_2) + 2\varepsilon.$$

Since ε is arbitrary, we conclude that

$$\bar{V}(G_1 + G_2) \leq x_1 + x_2 \leq \bar{V}(G_1) + \bar{V}(G_2).$$

3. Relationship with the superhedging price. By definition of the subhedging price,

$$\begin{aligned} \underline{V}(G) &= \sup\{x \in \mathbb{R} : \exists \pi \text{ with } -\pi \in \mathcal{A} \text{ and } X_T^{x,\pi} \leq G\} \\ &= -\inf\{x \in \mathbb{R} : \exists \pi \text{ with } \pi \in \mathcal{A} \text{ and } X_T^{-x, -\pi} \leq G\} \\ &= -\inf\{x \in \mathbb{R} : \exists \pi \text{ with } \pi \in \mathcal{A} \text{ and } X_T^{x,\pi} \geq -G\} = -\bar{V}(-G). \end{aligned}$$

By sub-additivity,

$$\bar{V}(G) - \underline{V}(G) = \bar{V}(G) + \bar{V}(-G) \geq \bar{V}(G - G) = \bar{V}(0) = 0.$$

where the last equality is due to AOA

4. Bounds for the market price and pricing replicable contingent claims. If a contingent claim can be both bought and sold at price $V(G)$, clearly arbitrage can be constructed if $V(G) \notin [\underline{V}(G), \bar{V}(G)]$. If a claim G is replicable with a portfolio (x, π) , then $\bar{V}(G) \leq x$. If $-\pi$ is admissible, then also $\underline{V}(G) \geq x$. Since $\bar{V}(G) \geq \underline{V}(G)$, we conclude that $\underline{V}(G) = \bar{V}(G) = V(G) = x$.

□

The following theorem shows that in the market of this section, every contingent claim (subject to an integrability condition) may be priced by replication.

Theorem 2 (Pricing contingent claims). *Let $G \in \mathcal{F}_T$ be a contingent claim with $\tilde{G} \in L^2(\mathcal{F}_T, \mathbb{Q})$. Then there exists a replicating portfolio for G and its no arbitrage price is given by*

$$V_t(G) = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T r(s)ds} G \middle| \mathcal{F}_t \right].$$

Proof. By martingale representation, there is a unique adapted process $H \in \mathcal{H}^2$ with

$$\tilde{G} = \mathbb{E}[\tilde{G}] + \int_0^T H_s d\widehat{W}_s.$$

Let $\tilde{\pi}_t = \sigma(t)^{-1} H_t$. Then

$$\tilde{X}_t^{x,\pi} := \mathbb{E}[\tilde{G}] + \int_0^t \tilde{\pi}_s \sigma(s) d\widehat{W}_s$$

is the discounted value of a self-financing portfolio which replicates $\tilde{G}$. Moreover $\tilde{X}_t$ is a $\mathbb{Q}$ -martingale hence it is admissible. □

Pricing contingent claims

In the model of this lecture, the unique no-arbitrage price of an option with pay-off $G \in L^2(\mathcal{F}_T, \mathbb{Q})$ at time t is given by

$$V_t(G) = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T r(s)ds} G \middle| \mathcal{F}_t \right],$$

where $\mathbb{Q}$ is the risk-neutral probability, under which the risky assets satisfy

$$dS_t = \text{diag}[S_t](r(t)\mathbf{1}dt + \sigma(t)d\widehat{W}_t),$$

where $\widehat{W}$ is a $\mathbb{Q}$ -brownian motion.

Black-Scholes formula Assume that there is a single risky asset, the pay-off is $G = g(S_T)$, and r and σ are deterministic. The pricing formula becomes

$$\begin{aligned} V_t(G) &= \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T r(s)ds} g(S_T) \middle| \mathcal{F}_t \right] \\ &= \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T r(s)ds} g \left(S_t e^{\int_t^T \left(r(s) - \frac{\sigma^2(s)}{2} \right) ds + \int_t^T \sigma(s) d\widehat{W}_s} \right) \middle| \mathcal{F}_t \right] := v(t, S_t), \end{aligned}$$

where

$$\begin{aligned} v(t, S) &= \mathbb{E}^Q \left[e^{-\int_t^T r(s)ds} g \left(S e^{\int_t^T \left(r(s) - \frac{\sigma^2(s)}{2} \right) ds + \int_t^T \sigma(s) d\widehat{W}_s} \right) \right] \\ &= \mathbb{E}^Q \left[e^{-(T-t)\bar{r}} g \left(S e^{\left(\bar{r} - \frac{\bar{\sigma}^2}{2} \right) (T-t) + \bar{\sigma} \widehat{W}_{T-t}} \right) \right], \end{aligned}$$

with

$$\bar{r} := \frac{1}{T-t} \int_t^T r(s)ds \quad \text{et} \quad \bar{\sigma}^2 = \frac{1}{T-t} \int_t^T \sigma^2(s)ds.$$

For the call option $g(S) = (S - K)^+$ and

$$\begin{aligned} V_t(G) &= \mathbb{E}^Q \left[e^{-\int_t^T r(s)ds} (S_T - K)^+ \middle| \mathcal{F}_t \right] \\ &= e^{\int_0^t r(s)ds} \mathbb{E}^Q \left[Z_T \mathbf{1}_{S_T \geq K} \middle| \mathcal{F}_t \right] - K e^{-\int_t^T r(s)ds} \mathbb{Q}[S_T \geq K | \mathcal{F}_t] \end{aligned} \quad (2.3)$$

with

$$Z_T = e^{-\int_0^T \frac{\sigma^2(s)}{2} ds + \int_0^T \sigma(s) d\widehat{W}_s}.$$

Introduce a new probability (share measure) $\tilde{\mathbb{Q}}$ given by

$$\frac{d\tilde{\mathbb{Q}}}{d\mathbb{Q}} \Big|_{\mathcal{F}_T} = Z_T, \quad \text{such that} \quad \widetilde{W}_t = \widehat{W}_t - \int_0^t \sigma(s)ds$$

is a $\tilde{\mathbb{Q}}$ -Brownian motion. Then,

$$\begin{aligned} V_t(G) &= S_t \tilde{\mathbb{Q}}[S_T \geq K | \mathcal{F}_t] - K e^{-\int_t^T r(s)ds} \mathbb{Q}[S_T \geq K | \mathcal{F}_t] \\ &= S_t \tilde{\mathbb{Q}} \left[S_t e^{\int_t^T \left(r(s) + \frac{\sigma^2(s)}{2} \right) ds + \int_t^T \sigma(s) d\widetilde{W}_s} \geq K \middle| \mathcal{F}_t \right] \\ &\quad - K e^{-\int_t^T r(s)ds} \mathbb{Q} \left[S_t e^{\int_t^T \left(r(s) - \frac{\sigma^2(s)}{2} \right) ds + \int_t^T \sigma(s) d\widehat{W}_s} \geq K \middle| \mathcal{F}_t \right] \end{aligned} \quad (2.4)$$

$$\begin{aligned} &= S_t \tilde{\mathbb{Q}} \left[S_t e^{\left(\bar{r} + \frac{\bar{\sigma}^2}{2} \right) (T-t) + \bar{\sigma} (\widetilde{W}_T - \widetilde{W}_t)} \geq K \middle| \mathcal{F}_t \right] \\ &\quad - K e^{-\bar{r}(T-t)} \mathbb{Q} \left[S_t e^{\left(\bar{r} - \frac{\bar{\sigma}^2}{2} \right) (T-t) + \bar{\sigma} (\widehat{W}_T - \widehat{W}_t)} \geq K \middle| \mathcal{F}_t \right] \end{aligned} \quad (2.5)$$

$$= S_t N(d_1) - K e^{-\bar{r}(T-t)} N(d_2) \quad (2.6)$$

with

$$d_{12} = \frac{\log \frac{S_t}{K e^{-\bar{r}(T-t)}} \pm \frac{\bar{\sigma}^2(T-t)}{2}}{\bar{\sigma} \sqrt{T-t}}.$$

The formula (6.1) is known as the Black-Scholes formula for the call option price. We leave the detailed study of this formula for the next section.

Chapter 3

Black-Scholes model: the PDE approach

Bird's eye view and key concepts of this chapter

- *A Markov diffusion model for the financial market*
- *The pricing equation; explicit form of the hedging portfolio*
- *Black-Scholes formula and the greeks*
- *Beyond Black-Scholes: discretization error and gamma hedging; transaction costs via Leland's approach; volatility risk and robustness of the Black-Scholes formula; implied volatility*

[Click here to access the Jupyter notebook for this chapter](#)

3.1 The pricing equation

The risk-neutral pricing formula

$$V_t(G) = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T r(s)ds} G | \mathcal{F}_t \right]$$

does not allow to compute the hedging portfolio. The “standard” way to obtain the hedge ratio is through Itô formula in the Markovian setting. The Markovian setting also allows to express the option price as solution to a PDE. We first recall the very

important notion of stochastic differential equation, which we need to build a model for asset prices.

Reminder: Stochastic differential equations

We consider an equation:

$$X_t = x + \int_0^t b(s, X_s)ds + \int_0^t a(s, X_s)dW_s,$$

or equivalently

$$dX_t = b(t, X_t)dt + a(t, X_t)dW_t, \quad X_0 = x \quad (*)$$

where W is a d -dimensional BM on $(\Omega, \mathcal{F}, \mathbb{P})$ with filtration $(\mathcal{F}_t)_{t \geq 0}$, and b and a are deterministic coefficients: $a : [0, T] \times \mathbb{R}^n \mapsto \mathbb{R}^{n \times d}$ and $b : [0, T] \times \mathbb{R}^n \mapsto \mathbb{R}^n$.

A *strong solution* of $(*)$ is a $(\mathcal{F}_t)_{t \geq 0}$ -adapted process X which satisfies $(*)$ and

$$\int_0^T |b(s, X_s)|ds + \int_0^T |a(s, X_s)|^2 ds < \infty.$$

The following basic existence result holds true: assume that $\exists K > 0$ such that $\forall t \in [0, T], \forall x, y \in \mathbb{R}$,

$$|a(t, x)| + |b(t, x)| \leq K(1 + |x|) \text{ and } |a(t, x) - a(t, y)| + |b(t, x) - b(t, y)| \leq K|x - y|.$$

Then, $(*)$ admits a unique strong solution, which is a Markov process.

The diffusion model for a financial market Let the prices of risky assets be given by the solution of the system of SDE:

$$dS_t = \text{diag}[S_t](\mu(t, S_t)dt + \sigma(t, S_t)dW_t),$$

and the risk-free asset be given by

$$S_t^0 = \exp\left(\int_0^t r(u, S_u)du\right).$$

We assume that $\mu \in \mathbb{R}^d$, $\sigma \in \mathbb{R}^{d \times d}$, σ^{-1} and $r \geq 0$ are bounded and $\text{diag}[S]\mu$ and $\text{diag}[S]\sigma$ are Lipschitz continuous in S uniformly in t . Under these assumptions, the SDE admits a unique strong solution, and the risk premium $\lambda(t, S) := \sigma(t, S)^{-1}(\mu(t, S) - r(t, S)\mathbf{1})$ is bounded.

Risk-neutral probability Introduce the risk-neutral probability $\mathbb{Q}$ via

$$\frac{d\mathbb{Q}}{d\mathbb{P}} \Big|_{\mathcal{F}_T} = \exp \left(- \int_0^T \lambda(t, S_t) dW_t - \frac{1}{2} \int_0^T \|\lambda(t, S_t)\|^2 dt \right).$$

Then,

$$\widehat{W}_t = W_t + \int_0^t \lambda(s, S_s) ds$$

is a $\mathbb{Q}$ -Brownian motion, the prices of risky assets satisfy

$$dS_t = \text{diag}[S_t](r(t, S_t)\mathbf{1}dt + \sigma(t, S_t)d\widehat{W}_t), \quad (3.1)$$

and the discounted prices satisfy

$$d\widetilde{S}_t = \text{diag}[\widetilde{S}_t]\sigma(t, S_t)d\widehat{W}_t.$$

We now recall two important technical results which will be needed to study the behavior of prices in this model and establish pricing formulas for options.

Reminder: Burkholder-Davis-Gundy inequalities and Gronwall inequality

- For any $p > 0$, there exist universal constants c_p and C_p such that for every continuous local martingale X ,

$$c_p \mathbb{E}[\langle X \rangle_T^{p/2}] \leq \mathbb{E}[(X_T^*)^p] \leq C_p \mathbb{E}[\langle X \rangle_T^{p/2}],$$

where $X_T^* := \max_{0 \leq t \leq T} |X_t|$.

- If u is a finite function and for some constants $\alpha \in \mathbb{R}$ and $\beta \geq 0$,

$$u(t) \leq \alpha + \beta \int_a^t u(s) ds, \quad t \in [a, b],$$

then

$$u(t) \leq \alpha e^{\beta(t-a)}, \quad t \in [a, b].$$

Together the above results allow to establish the boundedness of moments of asset prices. We present this result with proofs because it allows to illustrate the important technique of localisation, which will be useful in the sequel.

Proposition 2 (Bounded moments). *For all $n \geq 1$,*

$$\mathbb{E}^{\mathbb{Q}} \left[\max_{0 \leq t \leq T} \|\widetilde{S}_t\|^n \right] < \infty.$$

Proof. To simplify notation, let $d = 1$ and $n \geq 2$. Let $\tau_M := \inf\{t > 0 : \tilde{S}_t > M\}$ and $\tilde{S}_t^M := \tilde{S}_{t \wedge \tau_M}$, where we recall that $\tilde{S}$ is the discounted version of S , which is a $\mathbb{Q}$ -local martingale. By Burkholder-Davis-Gundy, the bound on σ , and Jensen's inequality

$$\begin{aligned} \mathbb{E}^{\mathbb{Q}}[\max_{0 \leq t \leq T} |\tilde{S}_t^M - \tilde{S}_0|^n] &\leq C \mathbb{E}^{\mathbb{Q}}[\langle \tilde{S} \rangle_{\tau_M}^{n/2}] = C \mathbb{E}^{\mathbb{Q}} \left[\left(\int_0^{\tau_M} \tilde{S}_t^2 \sigma^2(t, S_t) dt \right)^{n/2} \right] \\ &\leq C \mathbb{E}^{\mathbb{Q}} \left[\left(\int_0^{\tau_M} (\tilde{S}_t^M)^2 dt \right)^{n/2} \right] \leq C \mathbb{E}^{\mathbb{Q}} \left[\int_0^T (\tilde{S}_t^M)^n dt \right] \end{aligned}$$

where the constant C may vary from line to line.

Using the inequality $|a + b|^n \leq 2^n(|a|^n + |b|^n)$,

$$\mathbb{E}^{\mathbb{Q}}[\max_{0 \leq t \leq T} |\tilde{S}_t^M|^n] \leq C \tilde{S}_0^n + C \int_0^T \mathbb{E}^{\mathbb{Q}}[(\tilde{S}_t^M)^n] dt$$

and thus, by Gronwall's inequality,

$$\mathbb{E}^{\mathbb{Q}}[\max_{0 \leq t \leq T} |\tilde{S}_t^M|^n] \leq C \tilde{S}_0^n e^{C'T}$$

Finally, monotone convergence allows to conclude that

$$\mathbb{E}^{\mathbb{Q}}[\max_{0 \leq t \leq T} |\tilde{S}_t|^n] < \infty.$$

□

To the stochastic differential equation 3.1 we associate a differential operator, called the infinitesimal generator of the SDE:

$$\begin{aligned} \mathcal{L}_s u(t, S) &= r(t, S) \sum_{i=1}^d S^i \frac{\partial u}{\partial S^i} + \frac{1}{2} \sum_{i,j=1}^d a_{ij}(t, S) S^i S^j \frac{\partial^2 u}{\partial S^i \partial S^j} \\ a_{ij}(t, S) &= \sum_{k=1}^d \sigma_{ik}(t, S) \sigma_{jk}(t, S). \end{aligned}$$

Then, for any $C^{1,2}$ function u , the process

$$\begin{aligned} M_t^u &= e^{-\int_0^t r(s, S_s) ds} u(t, S_t) - \int_0^t e^{-\int_0^s r(v, S_v) dv} \left(\frac{\partial u}{\partial t} + L_s u - r(s, S_s) u \right) ds \\ &= u(0, S_0) + \int_0^t e^{-\int_0^s r(v, S_v) dv} \sum_{i,j=1}^d \frac{\partial u}{\partial S^i} S_t^i \sigma_{ij}(s, S_s) dW_s^j \end{aligned}$$

is a local martingale.

Theorem 3 (The pricing equation). *Consider a contingent claim with pay-off at time T given by $G = g(S_T)$, where g is a function of class $C^0((0, \infty))$ with polynomial growth, and consider the pricing equation:*

$$\frac{\partial u}{\partial t} + L_s u - r(t, S)u = 0, \quad (t, x) \in [0, T) \times (0, \infty),$$

with terminal condition $u(T, x) = g(x)$ for $x \in (0, \infty)$.

Assume that this equation admits a solution u of class $C^0([0, T] \times (0, \infty)) \cap C^{1,2}([0, T) \times (0, \infty))$ with polynomial growth in x uniformly on $t \in [0, T]$.

Then the unique price of the claim is given by

$$V_t = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T r(s, S_s) ds} g(S_T) \middle| \mathcal{F}_t \right] = u(t, S_t).$$

In the one-dimensional case, the pricing equation for European options takes the familiar form:

$$\frac{\partial u}{\partial t} + \frac{1}{2} \sigma^2(t, S) S^2 \frac{\partial^2 u}{\partial S^2} = r(t, S) \left(u - S \frac{\partial u}{\partial S} \right), \quad (t, x) \in [0, T) \times (0, \infty).$$

This is known as Black-Scholes PDE.

Proof. Let u be the solution of the pricing equation, and define the process M^u as above:

$$M_t^u = e^{-\int_0^t r(s, S_s) ds} u(t, S_t) = u(0, S_0) + \int_0^t e^{-\int_0^s r(v, S_v) dv} \sum_{i,j=1}^d \frac{\partial u}{\partial S_i} S_t^i \sigma_{ij}(s, S_s) dW_s^j$$

Define a sequence of stopping times

$$\tau_n = \inf \{ t > 0 : |\log S_t / S_0| > n \} \wedge T - \frac{1}{n}$$

Then, $\tau_n \rightarrow T$ as $n \rightarrow \infty$ and for every n , M^u stopped at τ_n is a square integrable martingale so that $M_{\tau_n \wedge t}^u = \mathbb{E}^{\mathbb{Q}} \left[M_{\tau_n}^u \middle| \mathcal{F}_t \right]$ and

$$e^{-\int_0^{t \wedge \tau_n} r(s, S_s) ds} u(t \wedge \tau_n, S_{t \wedge \tau_n}) = \mathbb{E}^{\mathbb{Q}} [e^{-\int_0^{\tau_n} r(s, S_s) ds} u(\tau_n, S_{\tau_n}) | \mathcal{F}_t]$$

Taking the limit $n \rightarrow \infty$, using the dominated convergence theorem with the moment bound of Proposition 2, we get:

$$u(t, S_t) = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T r(s, S_s) ds} g(S_T) \middle| \mathcal{F}_t \right].$$

□

The hedging portfolio The PDE theory of this section allows to express the hedging portfolio in explicit form, in terms of the derivatives of the function u . Consider a portfolio with initial value $x = u(0, S_0)$ and strategy

$$\pi_t^i = \frac{\partial u}{\partial S^i} S_t^i, \quad t < T \quad \text{and} \quad \pi_T^i = 0.$$

For $t < T$, since $\frac{\partial u}{\partial S}$ is continuous, $\int_0^t \|\tilde{\pi}_s \sigma(s, S_s)\|^2 ds < \infty$ p.s. and, by Itô formula,

$$X_t^x = u(t, S_t) \iff \tilde{X}_t^x = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_0^T r(s, S_s) ds} g(S_T) \middle| \mathcal{F}_t \right],$$

so that

$$\mathbb{E}^{\mathbb{Q}}[(\tilde{X}^x)_t] = \mathbb{E}^{\mathbb{Q}} \left[\int_0^t \|\tilde{\pi}_s \sigma(s, S_s)\|^2 ds \right] = \mathbb{E}^{\mathbb{Q}}[(\tilde{X}_t^x)^2 - x^2] < C < \infty.$$

By monotone convergence, we then conclude that $\mathbb{E}^{\mathbb{Q}} \left[\int_0^T \|\tilde{\pi}_s \sigma(s, S_s)\|^2 ds \right] < \infty$, so that the portfolio is admissible on $[0, T]$.

Finally, by continuity, $X_T^x = g(S_T)$, the portfolio replicates the option.

Black-Scholes formula and the greeks The dynamic hedging strategy allowing to eliminate the risk associated to a Call option in the Black-Scholes model is given by

$$\delta_t = \frac{\partial \text{Call}}{\partial S}(t, S_t) = N(d_1), \quad d_1 = \frac{\log \frac{S}{K e^{-r(T-t)}} + \frac{\sigma^2(T-t)}{2}}{\sigma \sqrt{T-t}}.$$

To see this, one can exchange the derivative and the expectation (using dominated convergence) in Equation (2.3), obtaining

$$\begin{aligned} \frac{\partial \text{Call}(t, S)}{\partial S} &= \frac{\partial}{\partial S} \mathbb{E}[e^{-r(T-t)} (S e^{(r-\frac{\sigma^2}{2})(T-t) + \sigma W_{T-t}} - K)^+] \\ &= \mathbb{E}[e^{-\frac{\sigma^2}{2}(T-t) + \sigma W_{T-t}} \mathbf{1}_{S e^{(r-\frac{\sigma^2}{2})(T-t) + \sigma W_{T-t}} \geq K}] \\ &= \tilde{\mathbb{Q}}[S e^{(r+\frac{\sigma^2}{2})(T-t) + \sigma \tilde{W}_{T-t}} \geq K] = N(d_1). \end{aligned}$$

More generally, to understand the behavior of options as function of the model parameters / risk factors and quantify / manage risks associated to parameter variation, the traders and risk managers compute the *greeks or sensitivities* of Black-Scholes formula to these parameters.

- **Delta.** The *delta* (Figure 3.1) is the sensitivity of the option price to the present value of the underlying:

$$\frac{\partial C_{BS}}{\partial S} = N(d_1), \quad \frac{\partial P_{BS}}{\partial S} = N(d_1) - 1.$$

It is by far the most important risk factor affecting the option price.

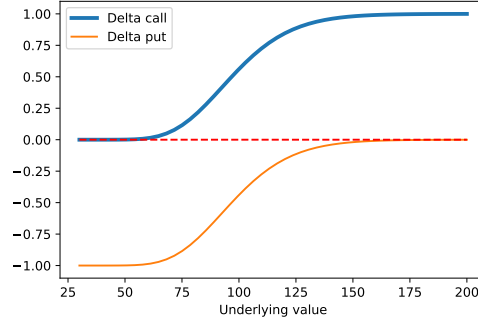


Figure 3.1: Delta of call and put option as function of the underlying value.

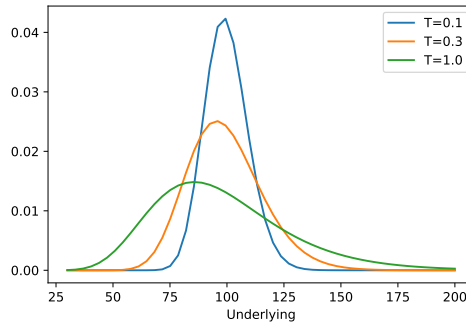


Figure 3.2: Gamma of the call option as function of the underlying value for different times to maturity.

- **Gamma.** The *gamma* (Figure 3.2) is defined as the second derivative of the option price or first derivative of its delta.

$$\begin{aligned}\frac{\partial^2 C_{BS}}{\partial S^2} &= \frac{\partial^2 P_{BS}}{\partial S^2} = \frac{\partial}{\partial S} N(d_1) \\ &= \frac{n(d_1)}{S\sigma\sqrt{T-t}}, \quad n(x) = \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}} = N'(x).\end{aligned}$$

- The gammas for call and put option with the same strike and maturity are equal.
- The gamma has the double interpretation of price sensitivity to large variations of underlying, and the rate of change in delta.

- Since gamma of a call/put option is always positive, call/put prices are convex in S .
- The gamma of a European option is large when the option is close to expiry and the value of the underlying is close to the option's strike.

- **Vega.** The *vega* is the sensitivity of the option price to the volatility parameter:

$$\frac{\partial C_{BS}}{\partial \sigma} = \frac{\partial P_{BS}}{\partial \sigma} = Sn(d_1)\sqrt{T-t}.$$

- Since the vega is always positive, call and put prices are increasing in volatility.
- Vega is large at the money, but decreases for options close to expiry.
- A portfolio of n options with prices $C_1, \dots, C_n$ containing w_i units of i -th option is gamma-neutral (resp., vega-neutral) if

$$\sum_{i=1}^n w_i \frac{\partial^2 C_i}{\partial S^2} = 0 \quad \text{resp.} \quad \sum_{i=1}^n w_i \frac{\partial C_i}{\partial \sigma} = 0.$$

- In Black-Scholes, it is easy to see that *if all options in a portfolio have the same expiry date*, a portfolio is vega-neutral if and only if it is gamma-neutral.

- **Other greeks.** The *rho* is the sensitivity to interest rate:

$$\begin{aligned} \frac{\partial C_{BS}}{\partial r} &= K(T-t)e^{-r(T-t)}N(d_2) > 0, \\ \frac{\partial P_{BS}}{\partial r} &= -K(T-t)e^{-r(T-t)}N(-d_2) < 0. \end{aligned}$$

The call is increasing and the put is decreasing as function of interest rate.

The *theta* measures the time dependence when all other variables are constant.

$$\begin{aligned} \frac{\partial C_{BS}}{\partial t} &= -\frac{Sn(d_1)\sigma}{2\sqrt{T-t}} - rKe^{-r(T-t)}N(d_2) < 0 \\ \frac{\partial P_{BS}}{\partial t} &= -\frac{Sn(d_1)\sigma}{2\sqrt{T-t}} + rKe^{-r(T-t)}N(-d_2) \end{aligned}$$

Call theta is always negative but put theta may be both positive and negative

3.2 Imperfections of the Black-Scholes model

Gamma hedging and discretization error Hedging portfolios are readjusted at discrete dates: either fixed (e.g., daily, at market close) or determined from option parameters. The remaining risk is small in normal conditions but may be important if something unusual happens in the market. If the trader is long gamma (convexity), the discretization risk is limited.

To understand the behavior of the discretization error, let us perform a Taylor expansion of the option price of order 1 in t and order 2 in S , assuming $r = 0$:

$$\begin{aligned} u(T, S_T) - u(0, S_0) &= \sum_{i=1}^n \{u(t_i, S_{t_i}) - u(t_{i-1}, S_{t_{i-1}})\} \\ &\approx \sum_{i=1}^n \left\{ \frac{\partial u}{\partial t}(t_{i-1}, S_{t_{i-1}}) \Delta t_i + \frac{\partial u}{\partial S}(t_{i-1}, S_{t_{i-1}}) \Delta S_{t_i} + \frac{1}{2} \frac{\partial^2 u}{\partial S^2}(t_{i-1}, S_{t_{i-1}}) \Delta S_{t_i}^2 \right\} \\ &\approx \sum_{i=1}^n \left\{ \frac{\partial u}{\partial t} \Delta t_i + \frac{\partial u}{\partial S} \Delta S_{t_i} + \frac{1}{2} \frac{\partial^2 u}{\partial S^2} S_{t_{i-1}}^2 \sigma^2 \Delta t_i + \frac{1}{2} \frac{\partial^2 u}{\partial S^2} S_{t_{i-1}}^2 \sigma^2 (\Delta W_{t_i}^2 - \Delta t_i) \right\}. \end{aligned}$$

Using the Black-Scholes PDE and subtracting the hedging portfolio, we get the following form of the discretization error:

$$\text{P\&L}_T = u(T, S_T) - X_T \approx \frac{1}{2} \sum_{i=1}^n \frac{\partial^2 u}{\partial S^2} S_{t_{i-1}}^2 \sigma^2 (\Delta W_{t_i}^2 - \Delta t_i),$$

or, with nonzero interest rate:

$$u(T, S_T) - X_T \approx \frac{e^{rT}}{2} \sum_{i=1}^n e^{-rt_{i-1}} \frac{\partial^2 u}{\partial S^2} S_{t_{i-1}}^2 \sigma^2 (\Delta W_{t_i}^2 - \Delta t_i),$$

The increment $\Delta W_{t_i}^2 - \Delta t_i$ has zero expectation and variance

$$\text{Var} [\Delta W_{t_i}^2 - \Delta t_i] = 2\Delta t_i^2,$$

and it can be further shown that by a functional central limit theorem, the process

$$\sum_{t_i \leq t} \frac{1}{\sqrt{2\Delta t_i}} (\Delta W_{t_i}^2 - \Delta t_i),$$

converges in law to a standard Brownian motion W^* independent from W .

Finally, the discretization error satisfies:

$$\text{P\&L}_T \approx \sqrt{\frac{\Delta t}{2}} e^{rT} \int_0^T e^{-rt} \frac{\partial^2 u}{\partial S^2} S_t^2 \sigma_t^2 dW_t^*.$$

From this formula, the following conclusions can be drawn:

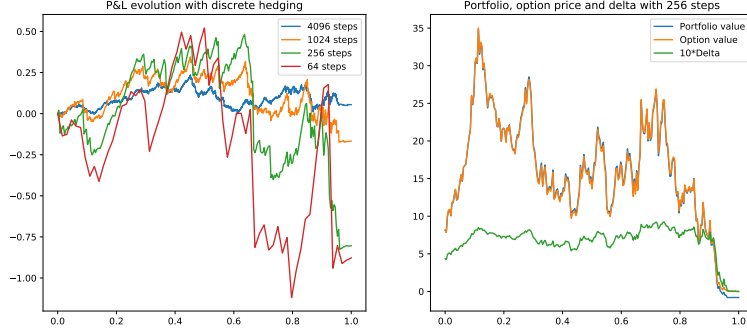


Figure 3.3: Discretization error. Left: portfolio evolution at different discretization frequency. Right: portfolio evolution, option price and delta with 256 steps (approximately daily frequency).

- The discretization error is proportional to option's gamma and square root of time step.
- It is larger when gamma is higher: close to the money and to the expiry date.
- To reduce discretization error: increase readjustment frequency, or reduce gamma via gamma hedging, possible for large portfolios.

$$\gamma_t = \frac{\frac{\partial^2 u}{\partial S^2}}{\frac{\partial^2 u_0}{\partial S^2}}.$$

Figure 3.3 plots the evolution of the discretization error for different rebalancing frequencies.

Optimizing the readjustment frequency Assuming a fixed cost δ per transaction and rebalancing frequency ω_t , on a small interval $[t, t + \Delta t]$ (still containing many readjustment dates) we get:

$$\text{Cost: } \delta \omega_t \Delta t \quad \text{Error: } \sqrt{\frac{\Delta t}{2\omega_t}} \int_t^{t+\Delta t} \frac{\partial^2 u}{\partial S^2}(u, S_u) \sigma^2 S_u^2 dW_u^*$$

which leads to the optimal frequency

$$\omega_t = \left(\frac{\Gamma_t^2 S_t^4 \sigma^4}{4\delta^2} \right)^{\frac{1}{3}}.$$

The discretization error can be further reduced by readjusting the portfolio at stochastic dates (which are determined by the deviation of the portfolio from its initial value).

Transaction costs: Leland's approach The assumption of no transaction costs is in general false. There are multiple transaction costs: bid-ask spreads, exchange fees. Selling/buying large quantities of low liquid assets may be problematic. In practice, for liquid assets and low rebalancing frequencies transaction costs are small. In the presence of proportional transaction costs, it is impossible to replicate an option exactly [22]: the superhedging price is the price of the stock. Leland [18] proposes a simple strategy allowing to compensate transaction costs *on average*.

The main idea of the approach is to use Black-Scholes hedging with volatility $\bar{\sigma}$ higher than the true one. This creates a positive drift in the P&L which compensates transaction costs on average.

Assume the trader uses the Black-Scholes model with volatility $\bar{\sigma}$ and let $r = 0$. The value of trader's portfolio writes:

$$\begin{aligned} X_T &= u_{\bar{\sigma}}(0, S_0) + \sum_{i=1}^n \frac{\partial u_{\bar{\sigma}}}{\partial S}(t_{i-1}, S_{t_{i-1}}) \Delta S_{t_i} \\ &\quad - k \sum_{i=1}^n \left| \frac{\partial u_{\bar{\sigma}}}{\partial S}(t_{i-1}, S_{t_{i-1}}) - \frac{\partial u_{\bar{\sigma}}}{\partial S}(t_i, S_{t_i}) \right| S_{t_i} \\ &\approx u_{\bar{\sigma}}(0, S_0) + \sum_{i=1}^n \frac{\partial u_{\bar{\sigma}}}{\partial S}(t_{i-1}, S_{t_{i-1}}) \Delta S_{t_i} \\ &\quad - k \sum_{i=1}^n \frac{\partial^2 u_{\bar{\sigma}}}{\partial S^2}(t_{i-1}, S_{t_{i-1}}) S_{t_{i-1}} |\Delta S_{t_i}| \end{aligned}$$

Using the pricing equation, the approximate P&L evolution writes:

$$\begin{aligned} \text{P\&L}_T &= X_T - g(S_T) \approx u(0, S_0) - u(T, S_T) + \sum_{i=1}^n \frac{\partial u}{\partial S} \Delta S_{t_i} - k \sum_{i=1}^n \frac{\partial^2 u}{\partial S^2} S_{t_{i-1}} |\Delta S_{t_i}| \\ &\approx \frac{1}{2} \sum S_t^2 \frac{\partial^2 u}{\partial S^2} \sigma^2 (\Delta W_t^2 - \Delta t) + \sum S_t^2 \frac{\partial^2 u}{\partial S^2} \left\{ \frac{(\bar{\sigma}^2 - \sigma^2) \Delta t}{2} - k \sigma |\Delta W_t| \right\} \end{aligned}$$

To cancel the P&L on average, take

$$\bar{\sigma}^2 = \sigma^2 + \frac{2k\sigma\mathbb{E}[|\Delta W_t|]}{\Delta t} \quad \Rightarrow \quad \bar{\sigma} = \sigma \sqrt{1 + \sqrt{\frac{8}{\pi}} \frac{k}{\sigma \sqrt{\Delta t}}}$$

Under Leland's choice of volatility, the residual error satisfies

$$\text{P\&L}_T \approx \sqrt{\frac{\Delta t}{2}} \int_0^T S_t^2 \sigma^2 \Gamma_t dW_t^* + k \sqrt{1 - \frac{2}{\pi}} \int_0^T S_t^2 \sigma \Gamma_t dW_t',$$

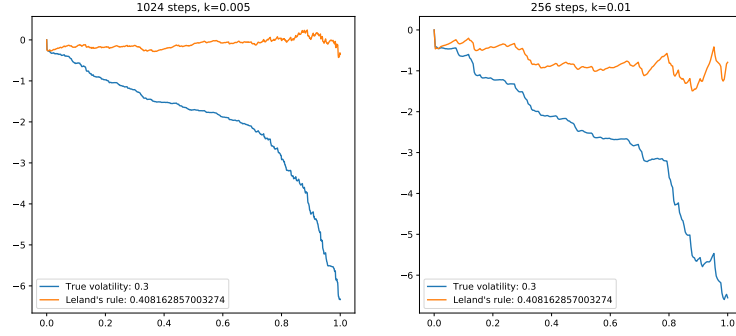


Figure 3.4: Portfolio P&L evolution in the presence of transaction costs, for different cost levels and discretization step values.

where W^* and W' are independent Brownian motions.

Figure 3.4 shows the portfolio P&L evolution for two different values of proportional transaction cost and discretization step. The values have been chosen to obtain the same value of Leland's volatility (and thus the same cost of hedging) in both cases. We see that on average the P&L is close to zero in both cases, but the variance of the P&L is higher for higher transaction costs, because the discretization step must be larger.

Discretization error and transaction costs In the absence of transaction costs:

$$\text{P\&L}_T \approx \sqrt{\frac{\Delta t}{2}} e^{rT} \int_0^T e^{-rt} \frac{\partial^2 u}{\partial S^2} S_t^2 \sigma_t^2 dW_t^*.$$

In the presence of proportional transaction costs k , Leland's rule demands to use the hedging volatility

$$\bar{\sigma} = \sigma \sqrt{1 + \sqrt{\frac{8}{\pi}} \frac{k}{\sigma \sqrt{\Delta t}}}.$$

Then,

$$\text{P\&L}_T \approx \sqrt{\frac{\Delta t}{2}} \int_0^T S_t^2 \sigma^2 \Gamma_t dW_t^* + k \sqrt{1 - \frac{2}{\pi}} \int_0^T S_t^2 \sigma \Gamma_t dW_t',$$

where W^* and W' are independent Brownian motions. To make the additional error negligible as $k \rightarrow 0$, we therefore need to choose the discretization step such that $\lim_{k \rightarrow 0} \frac{k}{\sqrt{\Delta t}} = 0$. With this choice of discretization step, up to the leading order term, the

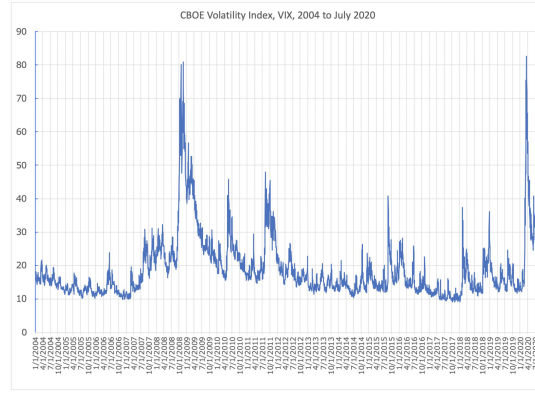


Figure 3.5: Evolution of the VIX volatility index.

total error is dominated by the discretization error:

$$\sqrt{\frac{\Delta t}{2}} \int_0^T S_t^2 \sigma^2 \Gamma_t dW_t^*$$

This error corresponds to the following cost of hedging (because we use a higher volatility):

$$u(\bar{\sigma}) - u(\sigma) \approx \frac{\partial u}{\partial \sigma} (\bar{\sigma} - \sigma) \approx \frac{\partial u}{\partial \sigma} \sqrt{\frac{2}{\pi}} \frac{k}{\sigma \sqrt{\Delta t}}$$

Minimizing $\mathbb{E}[\text{Cost}^2 + \text{Error}^2]$ yields

$$\frac{\Delta t}{2} \int_0^T S_t^4 \sigma^4 \Gamma_t^2 dt = \left(\frac{\partial u}{\partial \sigma} \right)^2 \frac{2}{\pi} \frac{k^2}{\Delta t}$$

The optimal discretization step is thus $\Delta t \sim k$.

Volatility risk and robustness of the BS formula The second most important risk after delta risk is the vega risk arising from changes in underlying volatility. In practice volatility is stochastic: between 14/02/20 and 16/03/20, the VIX index went up from 13% to 82%; in normal times variations are also observed (see Figure 8.1).

Assume that the true volatility σ_t follows a stochastic dynamics:

$$\frac{dS_t}{S_t} = \mu dt + \sigma_t dW_t,$$

but the trader uses Black-Scholes model with constant volatility $\bar{\sigma}$, and let $r = 0$.

Portfolio dynamics writes:

$$dX_t = \frac{\partial u_{\bar{\sigma}}}{\partial S}(t, S_t) dS_t, \quad X_0 = u_{\bar{\sigma}}(0, S_0).$$

The hedging error is then given by:

$$\varepsilon_T = X_T - g(S_T) = X_T - u_{\bar{\sigma}}(T, S_T).$$

By Itô formula,

$$du_{\bar{\sigma}}(t, S_t) = \left(\frac{\partial u_{\bar{\sigma}}}{\partial t} + \frac{1}{2} \frac{\partial^2 u_{\bar{\sigma}}}{\partial S^2} \sigma_t^2 S_t^2 \right) dt + \frac{\partial u_{\bar{\sigma}}}{\partial S}(t, S_t) dS_t,$$

but $u_{\bar{\sigma}}(t, S)$ as a function satisfies Black-Scholes PDE:

$$\frac{\partial u_{\bar{\sigma}}}{\partial t} + \frac{1}{2} \bar{\sigma}^2 S^2 \frac{\partial^2 u_{\bar{\sigma}}}{\partial S^2} = 0.$$

Gathering three equations, we get:

$$\varepsilon_T = \frac{1}{2} \int_0^T \frac{\partial^2 u_{\bar{\sigma}}}{\partial S^2} (\bar{\sigma}^2 - \sigma_t^2) S_t^2 dt.$$

or with nonzero interest rate,

$$\varepsilon_T = X_T - g(S_T) = \frac{1}{2} e^{rT} \int_0^T e^{-rt} \frac{\partial^2 u_{\bar{\sigma}}}{\partial S^2} (\bar{\sigma}^2 - \sigma_t^2) S_t^2 dt.$$

- If the true volatility is higher than $\bar{\sigma}$, P&L of option's buyer is positive;
- If $\bar{\sigma}$ dominates the true volatility, the seller has a positive pay-off.

Figure 3.6 plots the evolution of delta-hedged portfolio and option price under stochastic volatility bounded above by 40%. It is clear that when the trader uses a volatility of 40% for delta hedging, the P&L is increasing.

3.3 Implied volatility

In the BS model, unique unobservable parameter is the volatility. The function $\sigma \mapsto C_{BS}(\sigma)$ satisfies

$$\begin{aligned} \lim_{\sigma \downarrow 0} C_{BS}(\sigma) &= (S - Ke^{-r(T-t)})^+ & \lim_{\sigma \uparrow \infty} C_{BS}(\sigma) &= S \\ \frac{\partial C_{BS}}{\partial \sigma} &= Sn(d_1) \sqrt{T-t} > 0. \end{aligned}$$

Thus $C_{BS}(\sigma) = C$ has a unique solution for all C satisfying no arbitrage constraint

$$(S - Ke^{-r(T-t)})^+ < C < S$$

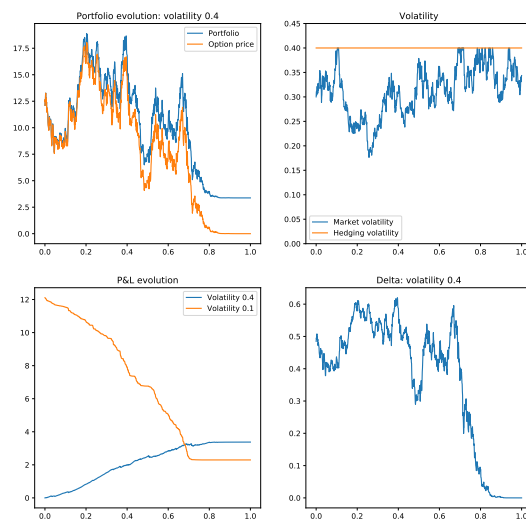


Figure 3.6: Evolution of delta-hedged portfolio and option price under stochastic volatility.

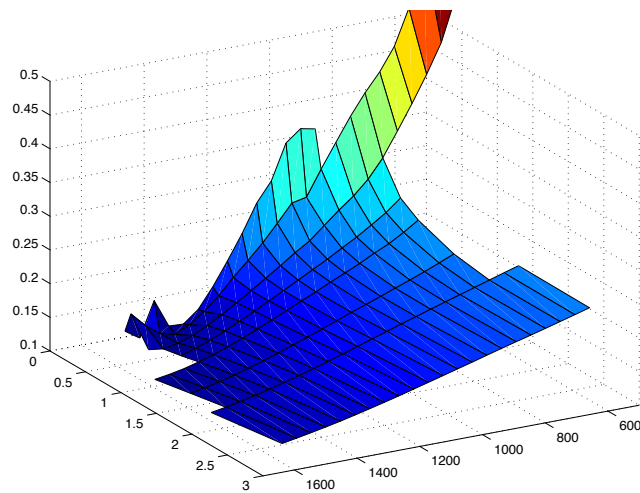


Figure 3.7: Implied volatility smile of S&P index option. Data source: CBOE.

Implied volatility

The solution $I(C)$ of equation $C_{BS}(\sigma) = C$, with C the observed price of a European call option is called *implied volatility* of this option.

The implied volatility observed in the market:

- Is often greater than the historical one;
- Depends on strike and maturity of the option;
- For almost all strikes, decreases as function of strike (*implied volatility skew*);
- For very large strikes, sometimes an increase is observed (*implied volatility smile*);
- Skew and smile are more pronounced for short maturity options; the smile flattens for longer maturities.

Figure 3.7 shows a typical shape of the implied volatility smile observed in the S&P index options market.

Several explanations for the presence of the implied volatility smile may be given:

- Hedging an option is more expensive in reality than in the model because of transaction costs and the need to cover sources of risk not taken into account by the model (e.g. volatility risk).

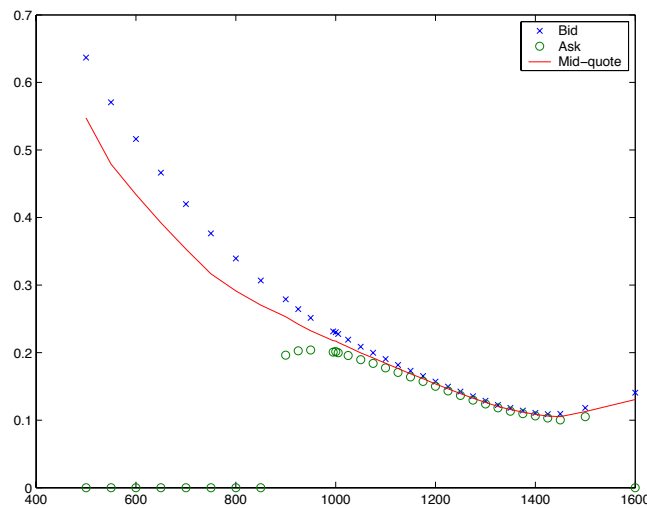


Figure 3.8: Implied volatility computed from bid and ask prices. Data source: CBOE.

- The skew is due to the fact that the Black-Scholes model underestimates the probability of a stock market crash or a large price movement.
- The smile can be explained by liquidity premia, higher for options away from the money. This is in particular supported by Figure 3.8 which shows that the implied volatility smile for out-of-the-money options is mainly explained by bid quotes.

Chapter 4

Pricing contingent claims by change of numéraire

Bird's eye view and key concepts of this chapter

- *Change of numéraire: a classical method in financial mathematics, allowing to simplify complex option pricing problems by changing the unit used to express asset prices.*
- *Method of change of numéraire. Transformation of volatilities under change of numéraire.*
- *Martingale measure associated to a numéraire.*
- *Generalized Black-Scholes formula.*
- *Pricing quanto options.*
- *Forward-neutral probability.*

4.1 Change of numéraire: theory

Definition 4. A *numéraire* is an asset or self-financing portfolio, whose value (Y_t) satisfies:

- Strict positivity: $Y_t > 0$ for all $t \in [0, T]$ a.s.

- Martingale property: the discounted value $(\tilde{Y}_t)_{0 \leq t \leq T}$ is an a.s. continuous $\mathbb{Q}$ -martingale.

For a more symmetric notation, in this section we write $S_t = (S_t^0, \dots, S_t^d)^\top$ where

$$dS_t^i = S_t^i \left(b^i(t) dt + \sum_{j=1}^d \sigma^{ij}(t) dW_t^j \right), \quad i = 0, \dots, d,$$

with $b^0(t) = r(t)$ and $\sigma^{0j}(t) = 0$. The self-financing portfolio equation then writes

$$X_t = \sum_{i=0}^d \delta_t^i S_t^i, \quad dX_t = \sum_{i=0}^d \delta_t^i dS_t^i.$$

The value of portfolio/asset X expressed with numéraire Y writes $X_t^Y := \frac{X_t}{Y_t}$.

Change of numéraire and volatility Let the dynamics of asset X be

$$dX_t = X_t \{ r(t) dt + \gamma^X(t) (\lambda(t) dt + dW_t) \}$$

and the dynamic of numéraire Y be

$$dY_t = Y_t \{ r(t) dt + \gamma^Y(t) (\lambda(t) dt + dW_t) \}.$$

By Itô formula, express the dynamics of X in the numéraire Y :

$$\begin{aligned} dX_t^Y &= \frac{dX_t}{Y_t} - \frac{X_t}{Y_t^2} dY_t + \frac{X_t}{Y_t^3} d\langle Y \rangle_t - \frac{d\langle X, Y \rangle_t}{Y_t^2} \\ &= X_t^Y (\gamma^X(t) - \gamma^Y(t)) \{ (\lambda(t) - \gamma^Y(t)^\top) dt + dW_t \}. \end{aligned}$$

Change of numéraire and volatility

In the new numéraire, the vector of volatilities of X is $\gamma^X(t) - \gamma^Y(t)$, and the vector or risk premia is $\lambda(t) - \gamma^Y(t)^\top$.

The dynamics of risky assets under new numéraire is

$$dS_t^Y = \text{diag}[S_t^Y] (\sigma(t) - \mathbf{1} \gamma^Y(t)) ((\lambda(t) - \gamma^Y(t)^\top) dt + dW_t)$$

and under the risk-neutral measure,

$$dS_t^Y = \text{diag}[S_t^Y] (\sigma(t) - \mathbf{1} \gamma^Y(t)) (-\gamma^Y(t)^\top dt + d\widehat{W}_t).$$

Example Let X et Y follow the dynamics

$$dX_t = X_t \{b^X(t)dt + \sigma^X(t)dW_t^X\} \quad \text{et} \quad dY_t = Y_t \{b^Y(t)dt + \sigma^Y(t)dW_t^Y\},$$

where W^X and W^Y are one-dimensional Brownian motions such that $d\langle W^X, W^Y \rangle = \rho dt$. Express the dynamics of X and Y with independent Brownian motions $\widehat{W}^1$ and $\widehat{W}^2$):

$$\begin{aligned} dX_t &= X_t \left\{ b^X(t)dt + \sigma^X(t)d\widehat{W}_t^1 \right\} \\ dY_t &= Y_t \left\{ b^Y(t)dt + \sigma^Y(t)(\rho d\widehat{W}_t^1 + \sqrt{1-\rho^2}d\widehat{W}_t^2) \right\}. \end{aligned}$$

or in vector form:

$$\begin{aligned} dX_t &= X_t \left\{ b^X(t)dt + \gamma^X(t)d\widehat{W}_t \right\}, & \gamma^X(t) &= (\sigma^X(t), 0), \\ dY_t &= Y_t \left\{ b^Y(t)dt + \gamma^Y(t)d\widehat{W}_t \right\}, & \gamma^Y(t) &= (\sigma^Y(t)\rho, \sigma^Y(t)\sqrt{1-\rho^2}). \end{aligned}$$

The scalar volatility of X in the numéraire Y is thus:

$$\|\gamma^X(t) - \gamma^Y(t)\| = \sqrt{\sigma^X(t)^2 + \sigma^Y(t)^2 - 2\rho\sigma^X(t)\sigma^Y(t)}.$$

Change of numéraire and change of probability In the numéraire Y , discounted asset prices are not local martingales under risk-neutral measure $\mathbb{Q}$. Risk-neutral measure is replaced with *martingale measure associated to numéraire Y* .

Let

$$Z_T = e^{\int_0^T \gamma^Y(t)d\widehat{W}_t - \frac{1}{2} \int_0^T \|\gamma^Y(t)\|^2 dt} = \frac{Y_T}{Y_0 e^{\int_0^T r(s)ds}}.$$

By assumption, $\mathbb{E}^{\mathbb{Q}}[Z_T] = 1$ and we introduce a new probability $\mathbb{Q}^Y$

$$\frac{d\mathbb{Q}^Y}{d\mathbb{Q}} = Z_T.$$

Under $\mathbb{Q}^Y$, the process $W_t^Y = \widehat{W}_t - \int_0^t \gamma^Y(s)^T ds$ is a Brownian motion, and the asset prices expressed using the numéraire Y are local martingales:

$$dS_t^Y = \text{diag}[S_t^Y](\sigma(t) - \mathbf{1}\gamma^Y(t))dW_t^Y$$

Change of numéraire and portfolios

Proposition 3. *Self-financing portfolios are invariant with respect to numéraire change.*

Proof. Let the numéraire dynamics be given by

$$dY_t = Y_t (b^Y(t)dt + \gamma^Y(t)dW_t),$$

and let X be a self-financing portfolio with dynamics $dX_t = \sum_{i=0}^d \delta_t^i dS_t^i$.

Applying Itô formula to function $f(x, y) = \frac{x}{y}$,

$$\begin{aligned} d\left(\frac{X_t}{Y_t}\right) &= \frac{dX_t}{Y_t} - \frac{X_t}{Y_t^2} dY_t + \frac{X_t}{Y_t^3} d\langle Y \rangle_t - \frac{d\langle X, Y \rangle_t}{Y_t^2} \\ &= \sum_{i=0}^d \delta_i \left\{ \frac{dS_t^i}{Y_t} - \frac{S_t^i}{Y_t^2} dY_t + \frac{S_t^i}{Y_t^3} d\langle Y \rangle_t - \frac{d\langle S^i, Y \rangle_t}{Y_t^2} \right\} = \sum_{i=0}^d \delta_t^i d\left(\frac{S_t^i}{Y_t}\right). \end{aligned}$$

□

Change of numéraire and arbitrage When the portfolio is expressed in numéraire Y , we say that it is admissible if

$$X_t^Y \geq M_t^Y, \quad 0 \leq t \leq T$$

almost surely, where M^Y is a $\mathbb{Q}^Y$ -martingale.

Proposition 4. *Let X be a self-financing portfolio and Y a numéraire. Then X is admissible for the numéraire Y if and only if it is admissible for the risk-free asset as numéraire.*

Proof.

$$\tilde{X}_t \geq M_t, \quad 0 \leq t \leq T \quad \Longleftrightarrow \quad X_t^Y \geq \frac{M_t e^{\int_0^t r(s)ds}}{Y_t} := M_t^Y,$$

and it is easy to see that M^Y is a $\mathbb{Q}^Y$ -martingale. □

This proposition shows in particular that if a strategy is an arbitrage under one numéraire, it is an arbitrage under any other numéraire.

Theorem 4. *Pricing formula under numéraire change Let $G \in \mathcal{F}_T$ be a contingent claim with $\tilde{G} \in L^2(\mathcal{F}_T, \mathbb{Q})$ which admits a replicating portfolio, and let Y be a numéraire.*

Then, the no arbitrage price of G is given by

$$V_t(G) = Y_t \mathbb{E}^{\mathbb{Q}^Y} \left[\frac{G}{Y_T} \middle| \mathcal{F}_t \right].$$

Proof. By the risk-neutral pricing formula,

$$\begin{aligned} V_t(G) &= \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T r(s)ds} G | \mathcal{F}_t \right] = e^{\int_0^t r(s)ds} Y_0 \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_0^T r(s)ds} \frac{Y_T}{Y_0} \frac{G}{Y_T} | \mathcal{F}_t \right] \\ &= \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T r(s)ds} G | \mathcal{F}_t \right] = Y_t \mathbb{E}^{\mathbb{Q}^Y} \left[\frac{G}{Y_T} | \mathcal{F}_t \right] \end{aligned}$$

□

4.2 Change of numéraire: applications

Example: exchange option Consider an option to exchange cash flows F^1 and F^2 at date T . We assume that F^1 and F^2 admit replicating portfolios with values (F_t^1) and (F_t^2) , which are *lognormal* with volatility vectors $\gamma^1(t)$ and $\gamma^2(t)$, i.e., they follow the dynamics:

$$\frac{dF_t^1}{F_t^1} = b^1(t)dt + \gamma^1(t)dW_t, \quad \frac{dF_t^2}{F_t^2} = b^2(t)dt + \gamma^2(t)dW_t,$$

where $b^1, b^2, \gamma^1, \gamma^2$ and the interest rate r are deterministic functions.

Let $\mathbb{Q}^1$ and $\mathbb{Q}^2$ be martingale measures associated to numéraires (F_t^1) and (F_t^2) .

Then,

$$\frac{d\mathbb{Q}^1}{d\mathbb{Q}} = \frac{F_T^1}{F_0^1} e^{-\int_0^T r(s)ds} \quad \text{et} \quad \frac{d\mathbb{Q}^2}{d\mathbb{Q}} = \frac{F_T^2}{F_0^2} e^{-\int_0^T r(s)ds}.$$

The exchange option price satisfies:

$$\begin{aligned} V_t &= \mathbb{E}^{\mathbb{Q}} [e^{-\int_t^T r(s)ds} (F_T^1 - F_T^2)^+ | \mathcal{F}_t] \\ &= \mathbb{E}^{\mathbb{Q}} [e^{-\int_t^T r(s)ds} F_T^1 \mathbf{1}_{F_T^1 \geq F_T^2} | \mathcal{F}_t] - \mathbb{E}^{\mathbb{Q}} [e^{-\int_t^T r(s)ds} F_T^2 \mathbf{1}_{F_T^1 \geq F_T^2} | \mathcal{F}_t] \\ &= F_t^1 \mathbb{Q}^1 [F_T^1 \geq F_T^2 | \mathcal{F}_t] - F_t^2 \mathbb{Q}^2 [F_T^1 \geq F_T^2 | \mathcal{F}_t] \\ &= F_t^1 \mathbb{Q}^1 \left[\frac{F_T^2}{F_T^1} \leq 1 | \mathcal{F}_t \right] - F_t^2 \mathbb{Q}^2 \left[\frac{F_T^1}{F_T^2} \geq 1 | \mathcal{F}_t \right] \\ &= F_t^1 N \left(\frac{\log \frac{F_t^1}{F_t^2} + \frac{1}{2} \text{Var}_t^T}{\sqrt{\text{Var}_t^T}} \right) - F_t^2 N \left(\frac{\log \frac{F_t^1}{F_t^2} - \frac{1}{2} \text{Var}_t^T}{\sqrt{\text{Var}_t^T}} \right), \end{aligned} \quad (4.1)$$

where $\text{Var}_t^T = \int_t^T \|\gamma^1(s) - \gamma^2(s)\|^2 ds$, since $\frac{F_t^2}{F_t^1}$ is a lognormal martingale with volatility $\gamma^1(t) - \gamma^2(t)$ under $\mathbb{Q}^1$ and $\frac{F_t^1}{F_t^2}$ is a lognormal martingale with volatility $\gamma^1(t) - \gamma^2(t)$ under $\mathbb{Q}^2$.

To determine the price of option to exchange F^1 and F^2 , use the following algorithm:

- Identify replicating portfolios F_t^1 and F_t^2 (for instance with risk-neutral pricing), and compute volatilities $\gamma^1(t)$ and $\gamma^2(t)$ *with respect to the same Brownian motion*.
- Compute the price of the exchange option with the generalized BS formula

$$V_t = F_t^1 N \left(\frac{\log \frac{F_t^1}{F_t^2} + \frac{1}{2} \text{Var}_t^T}{\sqrt{\text{Var}_t^T}} \right) - F_t^2 N \left(\frac{\log \frac{F_t^1}{F_t^2} - \frac{1}{2} \text{Var}_t^T}{\sqrt{\text{Var}_t^T}} \right),$$

$$\text{where } \text{Var}_t^T = \int_t^T \|\gamma^1(s) - \gamma^2(s)\|^2 ds$$

To determine the hedging strategy, remark that by Itô formula, the dynamics of V_t is

$$dV_t = N \left(\frac{\log \frac{F_t^1}{F_t^2} + \frac{1}{2} \text{Var}_t^T}{\sqrt{\text{Var}_t^T}} \right) dF_t^1 - N \left(\frac{\log \frac{F_t^1}{F_t^2} - \frac{1}{2} \text{Var}_t^T}{\sqrt{\text{Var}_t^T}} \right) dF_t^2.$$

The option can thus be replicated with a portfolio containing

$$N \left(\frac{\log \frac{F_t^1}{F_t^2} + \frac{1}{2} \text{Var}_t^T}{\sqrt{\text{Var}_t^T}} \right)$$

units of F^1 and

$$-N \left(\frac{\log \frac{F_t^1}{F_t^2} - \frac{1}{2} \text{Var}_t^T}{\sqrt{\text{Var}_t^T}} \right)$$

units of F^2 .

Foreign exchange options Often, several currencies are involved in the pay-off of an option. In this paragraph, we shall discuss the general approach to pricing such options, in the log-normal context, before considering a specific example, the quanto option, in the next paragraph. We assume that there are two market, the *domestic market* where the prices are expressed in the domestic currency (e.g., euro) and the *foreign market*, where the prices are expressed in foreign currency. Let $r(t)$ resp. $r^f(t)$ be the domestic resp. foreign interest rate, assumed deterministic. The foreign asset S is assumed to have the following dynamics (in foreign currency, under the real world measure)

$$\frac{d\hat{S}_t}{\hat{S}_t} = b(t)dt + \gamma(t)dW_t,$$

and the exchange rate X (value of one unit of foreign currency) has the dynamics

$$\frac{dX_t}{X_t} = b^X(t)dt + \gamma^X(t)dW_t,$$

where W is a 2-dimensional Brownian motion and $\gamma(t)$ and $\gamma^X(t)$ are linearly independent for all t .

Pricing foreign exchange options can be considered from the point of view of the domestic market as well as from that of the foreign market. The *domestic market*, where asset prices are denominated in domestic currency is composed of the domestic risk-free asset $S_t^0 = e^{\int_0^t r(s)ds}$, the foreign risk-free asset $S_t^{0,f} = X_t e^{\int_0^t r^f(s)ds}$ and the foreign risky asset $S_t = X_t \hat{S}_t$, with dynamics

$$\begin{aligned}\frac{dS_t^0}{S_t^0} &= r(t)dt, & \frac{dS_t^{0,f}}{S_t^{0,f}} &= (r^f(t) + b^X(t))dt + \gamma^X(t)dW_t \\ \frac{dS_t}{S_t} &= (b(t) + b^X(t) + \gamma(t)^\top \gamma^X(t))dt + (\gamma(t) + \gamma^X(t))dW_t\end{aligned}$$

The *domestic risk-neutral probability* is thus defined by

$$\begin{aligned}\frac{d\mathbb{Q}}{d\mathbb{P}}\Big|_{\mathcal{F}_T} &= \exp\left(-\int_0^T \lambda(t)dW_t - \frac{1}{2}\int_0^T \|\lambda(t)\|^2 dt\right), \\ \lambda(t) &= (\gamma^X(t), \gamma(t) + \gamma^X(t))^{-1} \begin{pmatrix} r^f(t) + b^X(t) - r(t) \\ b(t) + b^X(t) + \gamma(t)^\top \gamma^X(t) - r(t) \end{pmatrix}\end{aligned}$$

Under the domestic risk-neutral probability, the dynamics of the assets in the domestic market is then given by

$$\begin{aligned}\frac{dS_t^0}{S_t^0} &= r(t)dt, & \frac{dS_t^{0,f}}{S_t^{0,f}} &= r(t)dt + \gamma^X(t)dW_t^\mathbb{Q} \\ \frac{dS_t}{S_t} &= r(t)dt + (\gamma(t) + \gamma^X(t))dW_t^\mathbb{Q},\end{aligned}$$

and the exchange rate follows the dynamics

$$\frac{dX_t}{X_t} = (r(t) - r^f(t))dt + \gamma^X(t)dW_t^\mathbb{Q}.$$

Alternatively, one can consider the *foreign market*, where prices are denominated in foreign currency, and which is composed of the foreign risk-free asset $\hat{S}_t^{0,f} = e^{\int_0^t r^f(s)ds}$, the domestic risk-free asset $\hat{S}_t^0 = X_t^{-1} e^{\int_0^t r(s)ds}$ and the foreign risky asset $\hat{S}_t$, with dynamics

$$\begin{aligned}\frac{d\hat{S}_t^{0,f}}{\hat{S}_t^{0,f}} &= r^f(t)dt, & \frac{d\hat{S}_t^0}{\hat{S}_t^0} &= (r(t) - b^X(t) + \gamma^X(t)^\top \gamma^X(t))dt - \gamma^X(t)dW_t \\ \frac{d\hat{S}_t}{\hat{S}_t} &= b^X(t)dt + \gamma(t)dW_t\end{aligned}$$

The *foreign risk-neutral probability* is different from the domestic risk-neutral probability, because it is associated to taking the foreign risk-free asset as numéraire. It is thus defined by

$$\frac{d\hat{\mathbb{Q}}}{d\mathbb{P}} \Big|_{\mathcal{F}_T} = \exp \left(- \int_0^T \hat{\lambda}(t) dW_t - \frac{1}{2} \int_0^T \|\hat{\lambda}(t)\|^2 dt \right),$$

$$\hat{\lambda}(t) = (-\gamma^X(t), \gamma(t))^{-1} \begin{pmatrix} r(t) - b^X(t) + \gamma^X(t)^\top \gamma^X(t) - r^f(t) \\ b^X(t) - r^f(t) \end{pmatrix}$$

Under the foreign risk-neutral probability, the dynamics of the assets in the foreign market is given by

$$\frac{d\hat{S}_t^{0,f}}{\hat{S}_t^{0,f}} = r^f(t)dt, \quad \frac{dS_t^0}{S_t^0} = r^f(t)dt - \gamma^X(t)dW_t^{\hat{\mathbb{Q}}}$$

$$\frac{d\hat{S}_t}{\hat{S}_t} = r^f(t)dt + \gamma(t)dW_t^{\hat{\mathbb{Q}}}$$

Example: quanto option A quanto option is an option on a foreign asset with strike in foreign currency, whose pay-off is converted to domestic currency with a fixed exchange rate $\bar{X}$:

$$\bar{X}(\hat{S}_T - K),$$

To simplify, we let $\bar{X} = 1$.

To price the quanto option, we aim to apply the formula for the exchange option of the previous example with $F^1 = \hat{S}_T$ et $F^2 = K$ (both in domestic currency). The replicating portfolio for F^2 is clearly $F_t^2 = e^{-\int_t^T r(s)ds} K$. To compute the cost of replicating portfolio for F^1 , we use the domestic market viewpoint. The replication cost is then:

$$F_t^1 = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T r(s)ds} \hat{S}_T \Big| \mathcal{F}_t \right] = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T r(s)ds} \frac{S_T}{X_T} \Big| \mathcal{F}_t \right]$$

To evaluate this expression, let us apply Itô's formula to

$$e^{\int_0^t \bar{\gamma}(s)ds} \frac{S_t}{X_t}.$$

This gives:

$$de^{\gamma t \int_0^t \bar{\gamma}(s)ds} \frac{\hat{S}_t}{X_t} = e^{\gamma t \int_0^t \bar{\gamma}(s)ds} \frac{S_t}{X_t} \left\{ \gamma(t) dW_t^{\mathbb{Q}} + (\bar{\gamma}(t) + r^f(t) - \gamma^X(t)^\top \gamma(t)) dt \right\}$$

Thus, taking $\bar{\gamma}(t) = -r^f(t) + \gamma^X(t)^\top \gamma(t)$, $e^{\gamma^t \int_0^t \bar{\gamma}(s) ds} \frac{S_t}{X_t}$ becomes a $\mathbb{Q}$ -martingale, and therefore, with this choice of $\bar{\gamma}$,

$$\begin{aligned} \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T r(s) ds} \frac{S_T}{X_T} \middle| \mathcal{F}_t \right] &= e^{-\gamma \int_t^T \bar{\gamma}(s) ds - \int_t^T r(s) ds} \frac{S_t}{X_t} \\ &= \frac{S_t}{X_t} e^{\int_t^T (r^f(s) - r(s)) ds} e^{-\int_t^T \gamma(s)^\top \gamma^X(s) ds}. \end{aligned}$$

The option price is given by formula (4.1) with $Var_t^T = \int_t^T \|\gamma(s)\|^2 ds$.

- The replicating strategy for the quanto option contains $N(d_t^1)$ units of portfolio F_t^1 and $-N(d_t^2)$ units of portfolio F_t^2 .
- Portfolio F_t^2 contains domestic risk-free asset.
- Portfolio F_t^1 writes

$$\begin{aligned} F_t^1 &= \widehat{S}_t e^{\int_t^T (r^f(s) - r(s)) ds} e^{-\int_t^T \gamma(s)^\top \gamma^X(s) ds} \\ &= \frac{S_t}{S_t^{0,f}} e^{\int_t^T (2r^f(s) - r(s)) ds} e^{-\int_t^T \gamma(s)^\top \gamma^X(s) ds} \end{aligned}$$

By Itô formula,

$$\begin{aligned} dF_t^1 &= \frac{dS_t}{S_t^{0,f}} e^{\int_t^T (2r^f(s) - r(s)) ds} e^{-\int_t^T \gamma(s)^\top \gamma^X(s) ds} \\ &\quad - \frac{S_t dS_t^{0,f}}{(S_t^{0,f})^2} e^{\int_t^T (2r^f(s) - r(s)) ds} e^{-\int_t^T \gamma(s)^\top \gamma^X(s) ds} \end{aligned}$$

Example: stochastic interest rates and forward neutral probability If the interest rate varies stochastically, the zero coupons are risky assets. We assume the zero-coupon with maturity T follows the log-normal dynamics:

$$dB_t(T) = B_t(T)(\mu^B(t)dt + \gamma^B(t)dW_t),$$

while the stock price follows

$$dS_t = S_t(\mu(t)dt + \gamma(t)dW_t),$$

where W is a 2-dimensional Brownian motion and γ and γ^B are linearly independent. The price of a European call with pay-off $(S_T - K)^+$ can then be computed as the price of an exchange option with $F^1 = S_T$ and $F^2 = K$. The replicating portfolios are $F_t^1 = S_t$ et $F_t^2 = B_t(T)$ and the variance is $Var_t^T = \int_t^T \|\gamma(s) - \gamma^B(s)\|^2 ds$. In

the following, we present an alternative method for pricing this option, which is based on the very important notion of forward-neutral probability.

By the risk-neutral valuation formula,

$$V_t = \mathbb{E}^Q \left[e^{-\int_t^T r(s)ds} (S_T - K)^+ | \mathcal{F}_T \right].$$

Let us choose the zero-coupon with maturity T as numéraire. The associated martingale measure is called *forward neutral probability* and given by

$$\frac{d\mathbb{Q}^T}{d\mathbb{Q}} = \frac{e^{-\int_0^T r(s)ds}}{B_0(T)}, \quad Z_t = \frac{e^{-\int_0^T r(s)ds} B_t(T)}{B_0(T)}.$$

Using this measure, the option price can therefore be computed as follows:

$$V_t = \mathbb{E}^Q \left[e^{-\int_t^T r(s)ds} (S_T - K)^+ | \mathcal{F}_T \right] = B_t(T) \mathbb{E}^{\mathbb{Q}^T} \left[(S_T - K)^+ | \mathcal{F}_T \right],$$

In other words, the effect of the stochastic interest rate disappears under $\mathbb{Q}^T$.

Since, $\frac{S_t}{B_t(T)}$ is a $\mathbb{Q}^T$ -martingale, the option price is computed using the standard Black-Scholes formula applied to $\frac{S_t}{B_t(T)}$:

$$V_t = S_t N(d_1) - K B_t(T) N(d_2), \quad d_{1,2} = \frac{\log \frac{S_t}{K B_t(T)} \pm \frac{1}{2} \text{Var}_t^T}{\sqrt{\text{Var}_t^T}}.$$

Chapter 5

Numerical methods

Bird's eye view and key concepts of this chapter

Outside the constant volatility world, explicit formulas are rare. The standard approaches to compute option prices are

- *Discretization of the pricing PDE (trees, finite differences, finite elements)*
⇒ Fast for low-dimensional problems, better error control
- *Approximation of the expectation risk-neutral pricing formula with an empirical mean (Monte Carlo methods)*
⇒ Slow ($N^{-1/2}$ convergence rate), stochastic errors, but generic and works for very high dimensions

We discuss both approaches in the context of one-dimensional diffusion model (local volatility).

Key concepts (PDE):

- *Finite difference method;*
- *Explicit/implicit scheme;*
- *Boundary conditions.*

Key concepts (Monte Carlo):

- *Basic formulation of the Monte Carlo method and error analysis*
- *Variance reduction with control variates and importance sampling*
- *Discretization of stochastic differential equations: Euler scheme, weak error, strong error, overall approximation error*

[Click here to access the Jupyter notebook for this chapter](#)

5.1 PDE methods

We consider the Markov diffusion model of the previous chapter in the one-dimensional setting, with constant interest rate r :

$$dS_t = S_t(b(t, S_t)dt + \sigma(t, S_t)dW_t), \quad S_t^0 = e^{rt}.$$

The pricing PDE for the European option with pay-off $h(S_T)$ at time T takes the form:

$$\frac{\partial v}{\partial t} + \frac{1}{2}\sigma(t, S)^2 S^2 \frac{\partial^2 v}{\partial S^2} = rv(t, S) - rS \frac{\partial v}{\partial S}, \quad v(T, S) = h(S), \quad (PDE)$$

where

$$\begin{aligned} v(t, S) &= \mathbb{E}^{\mathbb{Q}}[h(S_T^{(t, S)})], \\ S_T^{(t, S)} &= S + \int_t^T S_s^{(t, S)} r ds + \int_t^T S_s^{(t, S)} \sigma(s, S_s^{(t, S)}) dW_s^{\mathbb{Q}}. \end{aligned} \quad (SDE)$$

PDE methods discretize the equation (PDE), while Monte Carlo methods discretize the process (SDE).

Change of variables To simplify the equation, introduce a new variable

$$x = \log \frac{S}{S_0} + r(T - t)$$

and a new unknown function

$$u(t, x) = e^{r(T-t)} v(t, S_0 e^{x-r(T-t)})$$

so that

$$v(t, S) = e^{-r(T-t)} u(t, r(T-t) + \log \frac{S}{S_0}).$$

The derivatives of v then write:

$$\begin{aligned} \frac{\partial v}{\partial t} &= e^{-r(T-t)} \left\{ ru + \frac{\partial u}{\partial t} - r \frac{\partial u}{\partial x} \right\}, \quad \frac{\partial v}{\partial S} = e^{-r(T-t)} \frac{1}{S} \frac{\partial u}{\partial x}, \\ \frac{\partial^2 v}{\partial S^2} &= e^{-r(T-t)} \frac{1}{S^2} \left\{ \frac{\partial^2 u}{\partial x^2} - \frac{\partial u}{\partial x} \right\} \end{aligned}$$

The pricing equation in terms of u and x becomes

$$0 = \frac{\partial u}{\partial t} + \frac{1}{2} \tilde{\sigma}^2(t, x) \left\{ \frac{\partial^2 u}{\partial x^2} - \frac{\partial u}{\partial x} \right\}, \quad t \in [0, T], x \in \mathbb{R},$$

where $\tilde{\sigma}(t, x) = \sigma(t, S_0 e^{x-r(T-t)})$.

Domain truncation and space discretization We discretize the equation on a bounded rectangular domain $[0, T] \times [-A, A]$ with an equidistant grid in time $t_n = n\Delta t$, $\Delta t = \frac{T}{N}$, $n = 0, 1, \dots, N$ and in space $x_m = -A + m\Delta x$, $\Delta x = \frac{2A}{M}$, $m = 0, 1, \dots, M$. The approximate solution at (t_n, x_m) is denoted $\hat{u}_{n,m}$. For $m = 1, \dots, M-1$, the space derivatives are approximated by centered differences:

$$\begin{aligned}\frac{\partial u(t_n, x_m)}{\partial x} &\approx \frac{u(t_n, x_{m+1}) - u(t_n, x_{m-1})}{2\Delta x} \\ \frac{\partial^2 u(t_n, x_m)}{\partial x^2} &\approx \frac{u(t_n, x_{m+1}) - 2u(t_n, x_m) + u(t_n, x_{m-1}))}{\Delta x^2}.\end{aligned}$$

Boundary conditions At the boundaries $m = 0$ and $m = M$, boundary conditions must be specified. Here we use Dirichlet conditions, that is, $u(t, x) = g(t, x)$ for $x = -A$ and $x = A$. Therefore, instead of pricing equation, we solve the equation on the truncated domain:

$$0 = \frac{\partial \hat{u}}{\partial t} + \frac{1}{2}\tilde{\sigma}^2(t, x) \left\{ \frac{\partial^2 \hat{u}}{\partial x^2} - \frac{\partial \hat{u}}{\partial x} \right\}, \quad t \in [0, T], x \in (-A, A)$$

with boundary conditions $\hat{u}(t, -A) = g(t, -A)$, $\hat{u}(t, A) = g(t, A)$, $t \in [0, T]$.

Time discretization: explicit scheme The simplest approach is to approximate the time derivative with a “one step back” finite difference:

$$\frac{\partial u(t_n, x_m)}{\partial t} \approx \frac{u(t_n, x_m) - u(t_{n-1}, x_m)}{\Delta t}, \quad 1 \leq n \leq N.$$

Starting with the terminal condition given by the option’s pay-off: $u(t_N, x_m) = \tilde{h}(x_m)$, $\tilde{h}(x) = h(s_0 e^x)$, we get a finite difference equation for $\hat{u}_{n,m}$ obtained by substituting discrete derivatives into the pricing equation:

$$\begin{aligned}\hat{u}_{n-1,m} &= \hat{u}_{n,m} \left\{ 1 - \frac{\tilde{\sigma}^2(t_n, x_m)\Delta t}{\Delta x^2} \right\} + \hat{u}_{n,m+1} \frac{\tilde{\sigma}^2(t_n, x_m)\Delta t}{2\Delta x^2} \left(1 - \frac{\Delta x}{2} \right) \\ &\quad + \hat{u}_{n,m-1} \frac{\tilde{\sigma}^2(t_n, x_m)\Delta t}{2\Delta x^2} \left(1 + \frac{\Delta x}{2} \right), \quad m = 1, \dots, M-1 \\ \hat{u}_{n,0} &= g(t_n, x_0), \quad \hat{u}_{n,M} = g(t_n, x_M), \quad \forall n; \quad \hat{u}_{N,m} = \tilde{h}(x_m), \quad \forall m.\end{aligned}$$

This equation is solved backwards from $n = N$ to $n = 0$. Each step of the scheme gives $\hat{u}_{n-1,\cdot}$ explicitly in terms of $\hat{u}_{n,\cdot}$. The scheme is therefore called *explicit*. It is simple to implement but may suffer from stability problems discussed below.

Stability analysis and the CFL condition To avoid explosion of numerical errors, discretization schemes must be *stable*. One of the possible definitions of stability is the following: if terminal condition is bounded by C , the solution at any date must be bounded by C .

The explicit scheme may be written as

$$\hat{u}_{n-1,m} = p_{n,m}\hat{u}_{n,m+1} + q_{n,m}\hat{u}_{n,m-1} + (1 - p_{n,m} - q_{n,m})\hat{u}_{n,m}$$

for some coefficients $p_{n,m}$ and $q_{n,m}$. It is clear that the scheme is stable in the sense of the above definition if $p_{n,m} \geq 0$, $q_{n,m} \geq 0$ and $1 - p_{n,m} - q_{n,m} \geq 0$ for all n, m . From the above formulas, we can see that this is the case if $\Delta x < 2$ and the Courant-Friedrichs-Lewy (CFL) condition is satisfied for all n, m :

$$\tilde{\sigma}^2(t_n, x_m)\Delta t < \Delta x^2.$$

In other words, the explicit scheme is *conditionally stable* and in practice a very small time step may be required to ensure that the CFL condition is satisfied and the scheme is stable.

Discretization: implicit scheme Another possibility is to approximate the time derivative by the “one step ahead” finite difference:

$$\frac{\partial u(t_n, x_m)}{\partial t} \approx \frac{u(t_{n+1}, x_m) - u(t_n, x_m)}{\Delta t}, \quad 0 \leq n \leq N - 1$$

Substituting this, together with the approximation formulas for the space derivatives, into the equation, we obtain the *implicit scheme*:

$$\begin{aligned} \hat{u}_{n,m} &= \hat{u}_{n-1,m} \left\{ 1 + \frac{\tilde{\sigma}^2(t_{n-1}, x_m)\Delta t}{\Delta x^2} \right\} - \hat{u}_{n-1,m+1} \frac{\tilde{\sigma}^2(t_{n-1}, x_m)\Delta t}{2\Delta x^2} \left(1 - \frac{\Delta x}{2} \right) \\ &\quad - \hat{u}_{n-1,m-1} \frac{\tilde{\sigma}^2(t_{n-1}, x_m)\Delta t}{2\Delta x^2} \left(1 + \frac{\Delta x}{2} \right) \\ &= \hat{u}_{n-1,m}(1 + p_{n-1,m} + q_{n-1,m}) - \hat{u}_{n-1,m+1}p_{n-1,m} - \hat{u}_{n-1,m-1}q_{n-1,m}, \end{aligned}$$

for $m = 1, \dots, M - 1$. In this scheme, the unknown function $\hat{u}_{n-1,\cdot}$ is computed from $\hat{u}_{n,\cdot}$ by matrix inversion.

Let us introduce the matrix A^{n-1} of size $M - 1 \times M - 1$:

$$\begin{aligned} A_{m,m}^{n-1} &= p_{n-1,m} + q_{n-1,m}, \quad m = 1, \dots, M - 1 \\ A_{m,m+1}^{n-1} &= -p_{n-1,m}, \quad m = 1, \dots, M - 2 \\ A_{m,m-1}^{n-1} &= -q_{n-1,m}, \quad m = 2, \dots, M - 1, \end{aligned}$$

and let $\hat{u}_n = (\hat{u}_{nm})_{m=1,\dots,M-1}$. Then, in matrix notation, the implicit scheme writes:

$$\hat{u}_n = (I + A_{n-1})\hat{u}_{n-1} + v^{n-1},$$

where $v_1^{n-1} = -q_{n-1,0}g(t_{n-1}, x_0)$, $v_{M-1}^{n-1} = -p_{n-1,M}g(t_{n-1}, x_M)$ and all other elements are zero.

The implicit scheme is unconditionnally stable. Indeed, for simplicity, assume that $g \equiv 0$ so that $v \equiv 0$. Since the matrix A_{n-1} is diagonally dominant hence positive definite,

$$u_n = \prod_{i=n}^{N-1} (I + A_i)^{-1} u_N,$$

and

$$\|u_n\| \leq \prod_{i=n}^{N-1} \|(I + A_i)^{-1}\| \|u_N\| \leq \|u_N\|.$$

Truncation error analysis In this paragraph, we analyze the error from truncating the equation to a bounded domain. Assume the volatility is bounded by a constant $\bar{\sigma}$ and the pay-off h is convex. Then by the robustness of the Black-Scholes formula,

$$\tilde{h}(x) \leq u(t, x) \leq u^{\bar{\sigma}}(t, x),$$

where $u^{\bar{\sigma}}(t, x)$ is the solution of pricing equation with constant volatility $\bar{\sigma}$.

The truncated solution $\hat{u}$ satisfies:

$$\hat{u}(t, x) = \mathbb{E}[\tilde{h}(X_T^{(t,x)})\mathbf{1}_{\tau(t,x) > T} + g(\tau^{(t,x)}, X_{\tau(t,x)}^{(t,x)})\mathbf{1}_{\tau(t,x) \leq T}],$$

where

$$X_s^{(t,x)} = x - \frac{1}{2} \int_t^s \sigma^2(r, X_r^{(t,x)}) dr + \int_t^s \sigma(r, X_r^{(t,x)}) dW_r, \quad s \geq t,$$

and $\tau^{(t,x)} = \inf\{s \geq t : X_s^{(t,x)} \notin [-A, A]\}$. This means in particular that $\hat{u}(t, x)$ is increasing in the boundary condition. On the other hand, clearly,

$$u(t, x) = \mathbb{E}[\tilde{h}(X_T^{(t,x)})\mathbf{1}_{\tau(t,x) > T} + u(\tau^{(t,x)}, X_{\tau(t,x)}^{(t,x)})\mathbf{1}_{\tau(t,x) \leq T}].$$

so that This means that, on the one hand, $u(t, x)$ is bounded from above by the solution to the truncated equation with boundary condition $g(t, x) = u^{\bar{\sigma}}(t, x)$, and on the other hand, $u(t, x)$ is bounded from below by the solution to the truncated equation with boundary condition $g(t, x) = \tilde{h}(x)$. Thus, to solve the original (non-truncated) equation to required precision, we need to solve the truncated equation with these two boundary conditions, and increase the domain until the difference between the two becomes smaller than the maximum allowed error.

Convergence analysis The order of convergence of a discretization scheme refers to the speed of convergence of the discretized solution to the true solution, when the time step Δt and the space step Δx go to zero. When the pricing equation admits a smooth solution, both schemes are order 1 in time and order 2 in space, meaning that

$$|\hat{u}(t_n, x_m) - \hat{u}_{n,m}| = O(\Delta t + \Delta x^2).$$

Let us prove this result for the explicit scheme assuming that the CFL condition is satisfied. Under this assumption, we may introduce the discretized process: a Markov chain $(\hat{X}_n)$ such that

$$\begin{aligned}\mathbb{P}[\hat{X}_{n+1} = x_{m+1} | \hat{X}_n = x_m] &= \frac{\tilde{\sigma}^2(t_{n+1}, x_m) \Delta t}{2\Delta x^2} \left(1 - \frac{\Delta x}{2}\right) := p(t_{n+1}, x_m) \\ \mathbb{P}[\hat{X}_{n+1} = x_{m-1} | \hat{X}_n = x_m] &= \frac{\tilde{\sigma}^2(t_{n+1}, x_m) \Delta t}{2\Delta x^2} \left(1 + \frac{\Delta x}{2}\right) := q(t_{n+1}, x_m) \\ \mathbb{P}[\hat{X}_{n+1} = x_m | \hat{X}_n = x_m] &= 1 - \frac{\tilde{\sigma}^2(t_{n+1}, x_m) \Delta t}{\Delta x^2}.\end{aligned}$$

Assuming no truncation, the explicit scheme writes

$$\hat{u}_{n,m} = \mathbb{E}[\tilde{h}(\hat{X}_N) | \hat{X}_n = x_m].$$

The difference between the explicit scheme and the true solution can then be evaluated as follows:

$$\begin{aligned}\hat{u}_{n,m} - u(t_n, x_m) &= \mathbb{E}[u(t_N, \hat{X}_N) - u(t_n, \hat{X}_n) | \hat{X}_n = x_m] \\ &= \sum_{i=n}^{N-1} \mathbb{E}[u(t_{i+1}, \hat{X}_{i+1}) - u(t_i, \hat{X}_i) | \hat{X}_n = x_m] \\ &= \sum_{i=n}^{N-1} \mathbb{E}[\{u(t_{i+1}, \hat{X}_i + \Delta x) - u(t_{i+1}, \hat{X}_i)\} p(t_{i+1}, \hat{X}_i) | \hat{X}_n = x_m] \\ &\quad + \sum_{i=n}^{N-1} \mathbb{E}[\{u(t_{i+1}, \hat{X}_i - \Delta x) - u(t_{i+1}, \hat{X}_i)\} q(t_{i+1}, \hat{X}_i) | \hat{X}_n = x_m] \\ &\quad + \sum_{i=n}^{N-1} \mathbb{E}[u(t_{i+1}, \hat{X}_i) - u(t_i, \hat{X}_i) | \hat{X}_n = x_m]\end{aligned}$$

By Taylor expansion and pricing equation,

$$\begin{aligned}
\hat{u}_{n,m} - u(t_n, x_m) &= \sum_{i=n}^{N-1} \mathbb{E}[\{u'_x(t_{i+1}, \hat{X}_i) \Delta x + \frac{1}{2} u''_{xx}(t_{i+1}, \hat{X}_i) \Delta x^2\} p(t_{i+1}, \hat{X}_i) | \hat{X}_n = x_m] \\
&+ \sum_{i=n}^{N-1} \mathbb{E}[\{-u'_x(t_{i+1}, \hat{X}_i) \Delta x + \frac{1}{2} u''_{xx}(t_{i+1}, \hat{X}_i) \Delta x^2\} q(t_{i+1}, \hat{X}_i) | \hat{X}_n = x_m] \\
&+ \sum_{i=n}^{N-1} \mathbb{E}[u'_t(t_{i+1}, \hat{X}_i) | \hat{X}_n = x_m] \Delta t + O(\Delta t) + O(\Delta x^2) \\
&= \sum_{i=n}^{N-1} \Delta t \mathbb{E}[\frac{1}{2} (-u'_x(t_{i+1}, \hat{X}_i) + u''_{xx}(t_{i+1}, \hat{X}_i)) \tilde{\sigma}^2(t_{i+1}, \hat{X}_i) + u'_t(t_{i+1}, \hat{X}_i) | \hat{X}_n = x_m] \\
&+ O(\Delta t) + O(\Delta x^2) = O(\Delta t) + O(\Delta x^2).
\end{aligned}$$

5.2 Monte Carlo methods

To introduce the basic Monte Carlo method, consider the pricing of a European call option in the Black-Scholes model

$$P = e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+] = e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_0 e^{\sigma W_T + (r - \sigma^2/2)T} - K)^+]$$

Reminder: strong law of large numbers

Assume $(\xi^{(l)})_{l \geq 1}$ i.i.d. with $\mathbb{E}[|\xi^{(1)}|] < \infty$ and let $\hat{S}_N = \frac{1}{N} \sum_{l=1}^N \xi^{(l)}$ be the empirical mean. Then,

$$\lim_{N \rightarrow \infty} \hat{S}_N = \mathbb{E}[\xi^{(1)}] \quad \text{a.s.}$$

This suggests the Monte Carlo estimator

$$\hat{P}_N = \frac{1}{N} \sum_{l=1}^N P^{(l)}, \quad P^{(l)} = e^{-rT} \left(S_0 e^{\sigma \sqrt{T} N^{(l)} + (r - \sigma^2/2)T} - K \right)^+$$

with $(N^{(l)})_{l \geq 1}$ i.i.d. standard normal.

The basic error estimate of the Monte Carlo method is given by the central limit theorem.

Reminder: central limit theorem

Let $(\xi^{(l)})_{l \geq 1}$ i.i.d. with $\mathbb{E}[|\xi^{(1)}|] < \infty$ and $\sigma^2 := \text{Var} \xi^{(1)}$. Then,

$$Y_N = \frac{\sqrt{N}}{\sigma} (\hat{S}_N - \mathbb{E}[\xi^{(1)}])$$

satisfies $Y_N \xrightarrow{d} N(0, 1)$ as $N \rightarrow \infty$.

Since N is large, $\hat{P}_N$ is close to normal with mean P and variance $\frac{1}{N} \text{Var } P^{(1)}$. The asymptotic centered confidence interval for P at the level α is

$$\left[P + \frac{\sqrt{\text{Var } P^{(1)}}}{\sqrt{N}} \Phi^{-1} \left(\frac{1 - \alpha}{2} \right), P + \frac{\sqrt{\text{Var } P^{(1)}}}{\sqrt{N}} \Phi^{-1} \left(\frac{1 + \alpha}{2} \right) \right]$$

The error of the basic Monte Carlo estimator therefore decays like $\sqrt{N}$ independently of dimension. This is a relatively slow rate, and achieving the desired precision simply by increasing the number of samples is not always possible. The variance reduction methods, discussed below, therefore attempt to improve the precision of the Monte Carlo estimator by decreasing the constant in the error estimate.

5.2.1 Variance reduction: control variates

The main idea of the method of control variates is to find a random variable, strongly correlated with the one of interest, and whose expectation is known explicitly. Assume that we want to estimate $\mathbb{E}[Y]$ and let X be a random variable correlated with Y , such that $\mathbb{E}[X]$ is known. The estimator

$$\hat{Y}_b = \frac{1}{N} \sum_{l=1}^N \left\{ Y^{(l)} + b(X^{(l)} - \mathbb{E}[X]) \right\}$$

is an unbiased estimator of $\mathbb{E}[Y]$ but its variance depends on b :

$$\text{Var } \hat{Y}_b = \frac{1}{N} \left\{ \text{Var } Y + b^2 \text{Var } X + 2b \text{Cov}(Y, X) \right\}.$$

The variance of $\hat{Y}_b$ can therefore be made smaller than the variance of the basic estimator $\hat{Y}_0$, by choosing b optimally. The optimal value of b is

$$b^* = -\frac{\text{Cov}(Y, X)}{\text{Var } X} = -\rho_{XY} \frac{\sigma_Y}{\sigma_X},$$

where we denote $\sigma_Y^2 = \text{Var } Y$, $\sigma_X^2 = \text{Var } X$ and $\rho_{XY} = \frac{\text{Cov}(Y, X)}{\sigma_X \sigma_Y}$. The optimal variance is therefore given by

$$\text{Var } \hat{Y}_{b^*} = \frac{\sigma_Y^2}{N} \{1 - \rho_{XY}^2\} = \text{Var } \hat{Y}_0 \{1 - \rho_{XY}^2\}.$$

It is therefore important to choose control variates which are strongly correlated with the variable of interest.

In practice, $\text{Cov}(X, Y)$ and often $\text{Var } X$ are not known and are replaced with their Monte Carlo estimators in the formula for the optimal parameter b :

$$\hat{b} = -\frac{\sum_{l=1}^N (X^{(l)} - \bar{X})(Y^{(l)} - \bar{Y})}{\sum_{l=1}^N (X^{(l)} - \bar{X})^2},$$

where $\bar{X}$ denotes the componentwise sample average of X . If several control variates $X_1, \dots, X_m$ are available, the formulas can be easily generalized using matrix notation:

$$\begin{aligned}\hat{Y}_b &= \bar{Y} + \sum_{k=1}^m b_k (\bar{X}_k - \mathbb{E}[X_k]) \\ \text{Var } \hat{Y}_b &= \frac{1}{N} \{ \text{Var } Y + 2b^\top \Sigma^{XY} + b^\top \Sigma^X b \}\end{aligned}$$

with $\Sigma_i^{XY} = \text{Cov}(Y, X_i)$ and $\Sigma_{ij}^X = \text{Cov}(X_i, X_j)$. The optimal value is therefore $b^* = -(\Sigma^X)^{-1} \Sigma^{XY}$, where the covariances can also be approximated with their sample estimators.

Example: basket option in the Black-Scholes model Consider the pricing of a basket call option in the multidimensional Black-Scholes model:

$$\begin{aligned}C(K) &= e^{-rT} \mathbb{E} \left[\left(\sum_{i=1}^d \alpha_i S_T^{(i)} - K \right)^+ \right], \quad \sum_{i=1}^d \alpha_i = 1, \quad \alpha_i \geq 0 \\ S_T^i &= S_0^i \exp \left(\sigma_i W_T^{(i)} + \left(r - \frac{\sigma_i^2}{2} \right) T \right),\end{aligned}$$

where $(W^{(1)}, \dots, W^{(d)})$ is a Brownian motion with correlation matrix R . Since no explicit formula is available for this product, and using PDE is difficult due to potentially high dimension of the problem, we need to resort to the Monte Carlo method, whose precision may be improved by control variates as we shall see below.

The first control variate (which incidentally gives an upper bound for the price) can be obtained by Jensen's inequality:

$$C(K) \leq \sum_{i=1}^d \alpha_i e^{-rT} \mathbb{E} \left[(S_T^{(i)} - K)^+ \right] = \sum_{i=1}^d \alpha_i C^{BS}(K, S_0^{(i)}, \sigma_i) := C^+(K).$$

We therefore choose the first control variate to be given by

$$CV^1 = \sum_{i=1}^d \alpha_i e^{-rT} (S_T^{(i)} - K)^+.$$

N	K	Price control variate	Standard error	Price Monte Carlo	Standard error	Reduction ratio
1000	100	6.02	0.0177	5.57	0.28	15.8
1000	120	0.786	0.00903	0.712	0.102	11.3
1000	80	21.0	0.0185	21.5	0.436	23.5

Table 5.1: Monte Carlo basket option prices with and without control variates.

The second control variate (which gives a lower bound for the price) is also obtained by Jensen's inequality, used at a different point in the computation:

$$\begin{aligned}
C(K) &\geq e^{-rT} \mathbb{E} \left[\left(e^{\sum_{i=1}^d \alpha_i X_T^{(i)}} - K \right)^+ \right] := C^-(K) = C^{BS}(K, \tilde{S}_0, \tilde{\sigma}), \\
X_T^{(i)} &= \sigma_i W_T^{(i)} + \left(r - \frac{1}{2} \sigma_i^2 \right) T + \log S_0^i \\
\tilde{S}_0 &= \exp \left(\sum_{i=1}^d \alpha_i \log S_0^i + \frac{T}{2} \left(\sum_{i,j=1}^d \alpha_i \alpha_j \sigma_i \sigma_j R_{ij} - \sum_{i=1}^d \alpha_i \sigma_i^2 \right) \right) \\
\tilde{\sigma}^2 &= \sum_{i,j=1}^d \alpha_i \alpha_j \sigma_i \sigma_j R_{ij}.
\end{aligned}$$

We therefore choose the second control variate to be given by

$$CV^2 = e^{-rT} \left(e^{\sum_{i=1}^d \alpha_i X_T^{(i)}} - K \right)^+.$$

Table 5.1 shows the prices of a basket option computed by Monte Carlo with both control variates and without control variates, with 1000 simulations and for different strike values. The model parameters are maturity $T = 0.5$, interest rate $r = 0.02$,

$$S_0 = \begin{pmatrix} 100 \\ 100 \\ 100 \end{pmatrix} \quad \alpha = \begin{pmatrix} 1/3 \\ 1/3 \\ 1/3 \end{pmatrix} \quad \sigma = \begin{pmatrix} 0.3 \\ 0.3 \\ 0.2 \end{pmatrix} \quad R = \begin{pmatrix} 1 & 0.3 & 0.3 \\ 0.3 & 1 & 0.3 \\ 0.3 & 0.3 & 1 \end{pmatrix}$$

It is clear that control variates provide a substantial reduction of the standard error: with control variates one can obtain a precise estimate of the price with as little as 1000 simulations.

5.2.2 Variance reduction: importance sampling

Importance sampling is another variance reduction technique, particularly useful in the context of rare events (e.g., price of an option with low exercise probability). The main

idea of the importance sampling method: is to find a new probability measure under which the estimator has a smaller variance, in particular because the rare event is no longer rare under the new probability.

As a motivating example, consider the pricing of an out of the money binary option in the Black-Scholes model. Assuming that the interest rate is zero, its price is given by the (risk-neutral) exercise probability:

$$p = \mathbb{P}[S_T \leq K], \quad S_T = Se^{-\frac{\sigma^2}{2}T + \sigma W_T},$$

The standard Monte Carlo estimator then writes:

$$\hat{p}_N = \frac{1}{N} \sum_{l=1}^N \mathbf{1}_{S_T^{(l)} \leq K}.$$

Its variance equals

$$\text{Var } \hat{p}_N = \frac{1}{N} \text{Var } \mathbf{1}_{S_T \leq K} = \frac{p - p^2}{N} \approx \frac{p}{N},$$

and the relative error, defined by the ratio of the standard error to the value of the quantity being estimated is given by

$$\frac{\sqrt{\text{Var } \hat{p}_N}}{p} \approx \frac{1}{\sqrt{pN}}.$$

If the probability in question is $p = 1\%$, this shows that $N = 10^6$ trajectories are needed for a 2-digit precision: Monte Carlo estimation in the context of rare events is very slow. This happens because for a vast majority of the trajectories the indicator function is equal to zero, and provides no information on the probability of the event of interest.

Importance sampling estimator The unknown quantity p can alternatively be computed as

$$p = \mathbb{E}^{\mathbb{Q}} \left[\frac{d\mathbb{P}}{d\mathbb{Q}} \mathbf{1}_{S_T \leq K} \right]$$

for any other probability measure $\mathbb{Q}$. One may thus construct an alternative estimator

$$\hat{p}_N^{\mathbb{Q}} = \frac{1}{N} \sum_{l=1}^N \left[\frac{d\mathbb{P}}{d\mathbb{Q}} \right]^{(l)} \mathbf{1}_{S_T^{\mathbb{Q},(l)} \leq K}.$$

In practice, especially in the Gaussian setting, one often uses Girsanov transform:

$$\frac{d\mathbb{P}}{d\mathbb{Q}} = e^{\lambda W_T^{\mathbb{Q}} - \frac{\lambda^2}{2}T}, \quad \Rightarrow \quad W_T = W_T^{\mathbb{Q}} - \lambda T$$

leading to the estimator

$$\hat{p}_N^\lambda = \frac{1}{N} \sum_{l=1}^N e^{\lambda W_T^{(l)} - \frac{\lambda^2}{2} T} \mathbf{1}_{S e^{-\frac{\sigma^2}{2} T + \sigma(W_T^{(l)} - \lambda T)} \leq K}.$$

The question is now how to optimize λ to obtain an estimator with a smaller variance than the original one.

With the Girsanov measure change,

$$\begin{aligned} \text{Var}^{\mathbb{Q}} \left[\frac{d\mathbb{P}}{d\mathbb{Q}} \mathbf{1}_{S_T \leq K} \right] &= \mathbb{E} \left[e^{\lambda W_T + \frac{\lambda^2}{2} T} \mathbf{1}_{S_T \leq K} \right] - p^2 \\ &= e^{\lambda^2 T} \tilde{\mathbb{P}} \left[-\frac{\sigma^2 T}{2} + \sigma W_T \leq \log \frac{K}{S} \right] \\ &= e^{\lambda^2 T} \tilde{\mathbb{P}} \left[-\frac{\sigma^2 T}{2} + \sigma(\tilde{W}_T + \lambda T) \leq \log \frac{K}{S} \right] \\ &= e^{\lambda^2 T} N(d - \lambda\sqrt{T}) - p^2, \quad d = \frac{\log \frac{K}{S} + \frac{\sigma^2 T}{2}}{\sigma\sqrt{T}}, \end{aligned}$$

where we have introduced a new probability $\tilde{\mathbb{P}}$ given by

$$\frac{d\tilde{\mathbb{P}}}{d\mathbb{P}} = e^{\lambda W_T - \frac{\lambda^2}{2} T}.$$

Therefore, to find the optimal λ , we need to minimize $f(\lambda) = e^{\lambda^2 T} N(d - \lambda\sqrt{T})$. The first order condition for this minimization writes $2\lambda\sqrt{T} = \frac{n(d - \lambda\sqrt{T})}{N(d - \lambda\sqrt{T})}$.

This equation does not admit an explicit solution in general, but let us assume that we are dealing with a low-probability event, that is, $K \ll S$ so that $d \ll 0$. Since $f'(\lambda) < 0$ for $\lambda < 0$, at the optimum $\lambda > 0$ and $d - \lambda\sqrt{T} \ll 0$, so that using the well-known asymptotics for the Gaussian distribution function,

$$\frac{n(d - \lambda\sqrt{T})}{N(d - \lambda\sqrt{T})} \approx \lambda\sqrt{T} - d \quad \text{and so} \quad \lambda^{opt} \approx -\frac{d}{\sqrt{T}}.$$

Under the candidate optimal probability we have found,

$$S_T = S e^{\sigma(W_T^{\mathbb{Q}} - \lambda^{opt} T) - \frac{\sigma^2}{2} T} = K e^{\sigma W_T^{\mathbb{Q}}}$$

which means that exactly 50% of trajectories end up in the exercise region.

The variance satisfies

$$\text{Var}^{\mathbb{Q}} \left[\frac{d\mathbb{P}}{d\mathbb{Q}} \mathbf{1}_{S_T \leq K} \right] = e^{\lambda^2 T} N(-2\lambda\sqrt{T}) \approx -\frac{1}{d\sqrt{8\pi}} e^{-d^2},$$

whereas

$$p = \mathbb{P}[S_T \leq K] = N(d) \approx -\frac{1}{d\sqrt{2\pi}}e^{-\frac{d^2}{2}}.$$

Thus,

$$\text{Var}^{\mathbb{Q}} \left[\frac{d\mathbb{P}}{d\mathbb{Q}} \mathbf{1}_{S_T \leq K} \right] \approx p^2 \sqrt{2\pi \log \frac{1}{p}}.$$

For small p this is much smaller than the variance $\text{Var} [\mathbf{1}_{S_T \leq K}] \approx p$ obtained without importance sampling. In particular, the relative error of the importance sampling estimator, approximately equal to

$$\frac{\sqrt{\sqrt{2\pi \log \frac{1}{p}}}}{\sqrt{N}},$$

grows only logarithmically with $\frac{1}{p}$ when $p \rightarrow 0$.

Importance sampling: numerical results $S_0 = 100$, $T = 0.5$, $\sigma = 0.3$, $N = 10000$.

K	$\hat{p}_N$	Std. dev	Rel. error	$\hat{p}_N^{\mathbb{Q}}$	Std. dev	Rel. error
90	0.342	0.00475	0.0139	0.342	0.00363	0.0106
80	0.176	0.00381	0.0217	0.17	0.00204	0.012
70	0.0578	0.00233	0.0404	0.0582	0.0008	0.0138
60	0.0104	0.00101	0.0975	0.0108	0.000175	0.0162
50	0.0011	0.000331	0.301	0.000763	1.46e-05	0.0191

5.2.3 Simulation of stochastic differential equations

Consider an stochastic differential equation:

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dW_t, \quad X_0 = x$$

where W is a d -dimensional Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$ with filtration $(\mathcal{F}_t)_{t \geq 0}$, $\sigma : [0, T] \times \mathbb{R}^n \mapsto \mathbb{R}^{n \times d}$ and $b : [0, T] \times \mathbb{R}^n \mapsto \mathbb{R}^n$. Assume that there exists $K > 0$ such that for all $t \in [0, T]$ and for all $x, y \in \mathbb{R}$,

$$|\sigma(t, x)| + |b(t, x)| \leq K(1 + |x|) \text{ and } |\sigma(t, x) - \sigma(t, y)| + |b(t, x) - b(t, y)| \leq K|x - y|.$$

Then, the stochastic differential equation admits a unique strong solution with finite moments. Except in some particular cases, the equation cannot be solved explicitly therefore one needs to use numerical approximations. These are usually carried out in two different contexts:

- Context of *scenario simulation*: for $(W_t)_{t \geq 0}$ given, simulate scenarios of $(X_t)_{t \geq 0}$ on the same probability space.
- *Approximation in law*: evaluate expectations $\mathbb{E}[f(X_t, t \leq T)]$ or $\mathbb{E}[f(X_T)]$.

Let X_T^n be a computable approximation with complexity n . Then,

- A scenario simulation scheme has *strong order of convergence* α if

$$\mathbb{E}[\max_{0 \leq t \leq T} |X_t - X_t^n|^2] \leq \frac{c}{n^{2\alpha}}, \quad n \rightarrow \infty$$

- An approximation in law has *weak order of convergence* α if

$$|\mathbb{E}[f(X_T)] - \mathbb{E}[f(X_T^n)]| \leq \frac{c}{n^\alpha}$$

for f regular enough.

Euler scheme By introducing a new process $\tilde{X}_t = (X_t, t)$ we may assume without loss of generality that b and σ do not depend on t . Let $t_k = \frac{kT}{n}$. The Euler discretization scheme for the process X is defined by

$$X_0^n = x, \quad X_{t_{k+1}}^n = X_{t_k} + b(X_{t_k}^n)(t_{k+1} - t_k) + \sigma(X_{t_k}^n)(W_{t_{k+1}} - W_{t_k}).$$

or in continuous time

$$X_t^n = x + \int_0^t b(X_{\tau_s}^n) ds + \int_0^t \sigma(X_{\tau_s}^n) dW_s,$$

where τ_s is the last discretization time before s : $\tau_s = \max\{t_k : t_k \leq s\}$.

The following results show that the Euler scheme has *strong order* $\frac{1}{2}$ and *weak order* 1.

Proposition 5 (Strong convergence rate). *Assume that b and σ are Lipschitz-continuous. Then the Euler scheme has strong convergence order $\frac{1}{2}$.*

Proof. First, by using localization, Gronwall's lemma and Fatou's lemma, it is easy to show, similarly to the proof of Proposition 2 that the second moment of both the SDE and the Euler scheme is bounded:

$$\max_{0 \leq t \leq T} \mathbb{E}[X_t^2] < \infty \quad \text{and} \quad \max_{0 \leq t \leq T} \mathbb{E}[(X_t^n)^2] < \infty \quad \text{uniformly on } n \quad (5.1)$$

Now, by Lipschitz property,

$$\begin{aligned}
\mathbb{E}[(X_t - X_t^n)^2] &\leq 2\mathbb{E}\left[\left(\int_0^t (b(X_s) - b(X_{\tau_s}))ds\right)^2\right] + 2\mathbb{E}\left[\left(\int_0^t (\sigma(X_s) - \sigma(X_{\tau_s}))dW_s\right)^2\right] \\
&\leq 2T\mathbb{E}\left[\int_0^t (b(X_s) - b(X_{\tau_s}))^2 ds\right] + 2\mathbb{E}\left[\int_0^t (\sigma(X_s) - \sigma(X_{\tau_s}))^2 ds\right] \\
&\leq 2(T+1)K^2\mathbb{E}\left[\int_0^t |X_s - X_{\tau_s}^n|^2 ds\right] \\
&\leq 4(T+1)K^2\mathbb{E}\left[\int_0^t |X_s - X_s^n|^2 ds\right] + 4(T+1)K^2\mathbb{E}\left[\int_0^t |X_s^n - X_{\tau_s}^n|^2 ds\right]
\end{aligned}$$

On the other hand

$$\begin{aligned}
\mathbb{E}\left[\int_0^t |X_s^n - X_{\tau_s}^n|^2 ds\right] &= \int_0^t \mathbb{E}\left[|b(X_{\tau_s}^n)(s - \tau_s) + \sigma(X_{\tau_s}^n)(W_s - W_{\tau_s})|^2\right] ds \\
&\leq 2 \int_0^t \mathbb{E}\left[\frac{T^2}{n^2} b(X_{\tau_s}^n)^2 + \sigma(X_{\tau_s}^n)^2 \frac{T}{n}\right] ds.
\end{aligned}$$

Now, using the Lipschitz property of b and σ , and the uniform bound (5.1), we deduce that

$$\mathbb{E}\left[\int_0^t |X_s^n - X_{\tau_s}^n|^2 ds\right] \leq \frac{C}{n}$$

for some constant C which does not depend on n . Then, by Gronwall's lemma, $\mathbb{E}[(X_t - X_t^n)^2] \leq \frac{c}{n}$, $t \in [0, T]$ for another constant c . Using BDG inequality, the estimate for the maximum can be shown as well. $\square$

Proposition 6 (Weak convergence rate). *Let $u(t, x) = \mathbb{E}^{(t, x)}[f(X_T)]$, and let b, σ, f be such that u satisfies the equation*

$$\frac{\partial u}{\partial t} + b(x) \frac{\partial u}{\partial x} + \frac{1}{2} \sigma^2(x) \frac{\partial^2 u}{\partial x^2} = 0,$$

with derivatives $\frac{\partial u}{\partial x}$, $\frac{\partial^2 u}{\partial x^2}$, $\frac{\partial^3 u}{\partial x^3}$, $\frac{\partial^4 u}{\partial x^4}$, $\frac{\partial^2 u}{\partial t \partial x}$, $\frac{\partial^3 u}{\partial t \partial x^2}$, uniformly bounded and the moments of X and X^n up to order 4 uniformly bounded.

Then,

$$\mathbb{E}[f(X_T^n)] - \mathbb{E}[f(X_T)] = \frac{c}{n} + o\left(\frac{1}{n}\right).$$

Proof. By Itô formula,

$$\begin{aligned}
\mathbb{E}[f(X_T^n)] - \mathbb{E}[f(X_T)] &= \mathbb{E}[u(T, X_T^n) - u(0, X_0^n)] \\
&= \int_0^T dt \mathbb{E} \left\{ \frac{\partial u}{\partial t}(t, X_t^n) + b(X_{\tau_t}^n) \frac{\partial u}{\partial x}(t, X_t^n) + \frac{1}{2} \sigma^2(X_{\tau_t}^n) \frac{\partial^2 u}{\partial x^2}(t, X_t^n) \right\} \\
&= \int_0^T dt \mathbb{E} \left\{ \frac{\partial u}{\partial t}(t, X_t^n) - \frac{\partial u}{\partial t}(t, X_{\tau_t}^n) \right\} dt \\
&\quad + \int_0^T \mathbb{E} \left[b(X_{\tau_t}^n) \left\{ \frac{\partial u}{\partial x}(t, X_t^n) - \frac{\partial u}{\partial x}(t, X_{\tau_t}^n) \right\} \right] dt \\
&\quad + \frac{1}{2} \int_0^T \mathbb{E} \left[\sigma^2(X_{\tau_t}^n) \left\{ \frac{\partial^2 u}{\partial x^2}(t, X_t^n) - \frac{\partial^2 u}{\partial x^2}(t, X_{\tau_t}^n) \right\} \right] dt \\
&:= \sum_{i=1}^3 \int_0^T \mathbb{E} [f_i(X_{\tau_t}^n) \{F_i(t, X_t^n) - F(t, X_{\tau_t}^n)\}] dt.
\end{aligned}$$

By Taylor expansion, each term may be estimated as follows:

$$\begin{aligned}
&f_i(X_{\tau_t}^n) \{F_i(t, X_t^n) - F(t, X_{\tau_t}^n)\} \\
&= f_i(X_{\tau_t}^n) \left\{ F'_i(t, X_t^n)(X_t^n - X_{\tau_t}^n) + \frac{1}{2} F''_i(t, X_t^n)(X_t^n - X_{\tau_t}^n)^2 \right\} + R_t^n
\end{aligned}$$

with

$$|R_t^n| \leq C f_i(X_{\tau_t}^n) |X_t^n - X_{\tau_t}^n|^3.$$

so that

$$\begin{aligned}
\mathbb{E} [f_i(X_{\tau_t}^n) \{F_i(t, X_t^n) - F(t, X_{\tau_t}^n)\} | \mathcal{F}_{\tau_t}] &= f_i(X_{\tau_t}^n) F'_i(t, X_{\tau_t}^n) b(X_{\tau_t}^n) (\tau_t - t) \\
&\quad + \frac{1}{2} f_i(X_{\tau_t}^n) F''_i(t, X_{\tau_t}^n) (b(X_{\tau_t}^n) (\tau_t - t)^2 + \sigma(X_{\tau_t}^n)^2 (\tau_t - t)) + R_t^n
\end{aligned}$$

and

$$\mathbb{E}[|R_t^n| | \mathcal{F}_{\tau_t}] \leq C f_i(X_{\tau_t}^n) n^{-\frac{3}{2}}.$$

Summing up the different terms,

$$\begin{aligned}
\mathbb{E}[f(X_T^n)] - \mathbb{E}[f(X_T)] &= \sum_{i=1}^3 \int_0^T (\tau_t - t) \mathbb{E} \left[f_i(X_{\tau_t}^n) \left(F'_i(t, X_t^n) b(X_{\tau_t}^n) + \frac{1}{2} F''_i(t, X_t^n) \sigma(X_{\tau_t}^n)^2 \right) \right] dt + o\left(\frac{1}{n}\right) \\
&= \frac{T}{2n} \sum_{i=1}^3 \int_0^T \mathbb{E} \left[f_i(X_t) \left(F'_i(t, X_t) b(X_t) + \frac{1}{2} F''_i(t, X_t) \sigma(X_t)^2 \right) \right] dt + o\left(\frac{1}{n}\right).
\end{aligned}$$

□

The expansion of the previous theorem allows to use the so called *Romberg extrapolation* method to improve convergence. The idea is to exploit simultaneously the Euler scheme with n steps and the one with $2n$ steps to remove the leading term of the error:

$$\mathbb{E}[f(X_T)] = -\mathbb{E}[f(X_T^n)] + 2\mathbb{E}[f(X_T^{2n})] + o\left(\frac{1}{n}\right).$$

Monte Carlo method for PDE The weak approximation using the Euler scheme can be employed to compute the solutions to partial differential equations by Monte Carlo. Let u be the solution to backward parabolic PDE:

$$\frac{\partial u}{\partial t} + \frac{1}{2}\sigma^2(x)\frac{\partial^2 u}{\partial x^2} + b(x)\frac{\partial u}{\partial x} = 0, \quad u(T, x) = g(x)$$

From the Feynman-Kac formula, it is known that under appropriate conditions, $u(t, x) = \mathbb{E}^{(t,x)}[g(X_T)]$. This allows to use a combination of Euler scheme / Monte Carlo to approximate u :

$$\hat{u}_{N,n}(t, x) = \frac{1}{N} \sum_{l=1}^N g(X_T^{n,l})$$

This approximation suffers from two types of error: the discretization error of the Euler scheme and the Monte Carlo error:

$$\hat{u}_{N,n}(t, x) - u(t, x) = \frac{1}{N} \sum_{l=1}^N g(X_T^{n,l}) - \mathbb{E}^{(t,x)}[g(X_T^n)] + \mathbb{E}^{(t,x)}[g(X_T^n)] - \mathbb{E}^{(t,x)}[g(X_T)]$$

Since the Monte Carlo method converges at the rate $\frac{1}{\sqrt{N}}$, and the Euler scheme has weak order 1, for sufficiently large N and n , the total error may be estimated as follows:

$$|\hat{u}_{N,n}(t, x) - u(t, x)| \leq \frac{C_1}{\sqrt{N}} + \frac{C_2}{n},$$

where C_1 and C_2 are positive constants. On the other hand, the computational effort is proportional to nN . To minimize the error for a given computational effort, we therefore need to choose $N \sim n^2$ for global error decay of order (computational effort) $^{-1/3}$.

Chapter 6

Dupire's equation and local volatility

Bird's eye view and key concepts of this chapter

- *Local volatility: a one-dimensional Markov diffusion model for a financial market*
- *The calibration problem and Dupire's formula at the level of prices*
- *Difficulties of implementation of Dupire's formula*
- *Dupire's formula at the level of implied volatilities*
- *The SVI parameterization*
- *No arbitrage conditions for option prices and implied volatilities*

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6.1 Local volatility model

The Black-Scholes model cannot reproduce all the option prices observed in the market for a given underlying because their *implied volatility varies with strike and maturity*. To take into account the market implied volatility smile within a Markovian and complete model, one possibility is to model the volatility as a *deterministic function of time*

and the value of the underlying:

$$\frac{dS_t}{S_t} = rdt + \sigma(t, S_t)dB_t,$$

where r is the interest rate, assumed to be constant, and B is the Brownian motion under the risk-neutral measure $\mathbb{Q}$.

We shall see that the local volatility model allows for *exact calibration of the option price surface* on any single day.

Option pricing Recall that the price of an option with payoff $h(S_T)$ at date T is given by

$$C(t, s) = \mathbb{E}^{\mathbb{Q}} \left[e^{-r(T-t)} h(S_T) | S_t = s \right],$$

and is characterized by the *partial differential equation*:

$$rC = \frac{\partial C}{\partial t} + rs \frac{\partial C}{\partial s} + \frac{1}{2} \sigma(t, s)^2 s^2 \frac{\partial^2 C}{\partial s^2}, \quad C(T, s) = h(s). \quad (6.1)$$

The self-financing hedging portfolio contains $\delta_t = (\partial C / \partial s)(t, S_t)$ shares and the amount $\delta_t^0 = C(t, S_t) - \delta_t S_t$ in cash. The pricing equation has the *same form as in the Black-Scholes model*, but *one can no longer deduce an explicit pricing formula*, because the volatility is now a function of the underlying.

6.2 The calibration problem and Dupire's formula

The calibration problem consists in determining the local volatility surface $\sigma(\cdot, \cdot)$ from the observed prices of European Call/Put options. The Black-Scholes pricing equation cannot be used for this task, since it requires the knowledge of option prices for all values of t and S , and on any given date, we only know the prices for one fixed value of t and S . The following theorem derives another equation satisfied by the Call option prices, where the variables are the time to maturity T and the strike K . Since on any given date one can observe option prices for many strikes and maturities, this equation allows to develop a calibration algorithm.

Theorem 5 (Dupire's equation). *Let $(S_t)_{t \geq 0}$ be the solution of SDE*

$$\frac{dS_t}{S_t} = rdt + \sigma(t, S_t)dW_t,$$

such that

1. S_t is square integrable: $E \left[\int_0^T S_t^2 dt \right] < \infty, \quad \forall T$
2. For every $t > 0$, S_t has density $p(t, x)$, continuous on $(0, \infty) \times (0, \infty)$.
3. $\sigma(t, x)$ is continuous on $(0, \infty) \times (0, \infty)$.

Then the call price $C(T, K) = e^{-rT} E[(S_T - K)^+]$ satisfies Dupire's equation

$$\frac{\partial C}{\partial T} = \frac{1}{2} \sigma^2(T, K) K^2 \frac{\partial^2 C}{\partial K^2} - rK \frac{\partial C}{\partial K}, \quad C(0, K) = (S_0 - K)^+$$

Comparison of Dupire and Black-Scholes PDE

The Black-Scholes PDE

$$rC = \frac{\partial C}{\partial t} + rs \frac{\partial C}{\partial s} + \frac{1}{2} \sigma(t, s)^2 s^2 \frac{\partial^2 C}{\partial s^2}, \quad C(T, s) = h(s).$$

- Is a backward equation;
- Is satisfied by all European options as functions of t and S
- Can be used to compute the option price for a single strike and maturity

Dupire's PDE

$$\frac{\partial C}{\partial T} = \frac{1}{2} \sigma^2(T, K) K^2 \frac{\partial^2 C}{\partial K^2} - rK \frac{\partial C}{\partial K}, \quad C(0, K) = (S_0 - K)^+$$

- Is a forward equation;
- Is satisfied by Calls (and Puts) only, as functions of T and K
- Can be used to compute the option prices for all strikes and maturities simultaneously

Proof. We would like to apply Itô's formula to the process $e^{-rt}(S_t - K)^+$. Since the function $f(x) = x^+$ is not C^2 , we regularize the function f :

$$f_\varepsilon(x) = \frac{(x + \varepsilon/2)^2}{2\varepsilon} \mathbf{1}_{-\varepsilon/2 \leq x \leq \varepsilon/2} + x \mathbf{1}_{x > \varepsilon/2}$$

$$f'_\varepsilon(x) = \frac{x + \varepsilon/2}{\varepsilon} \mathbf{1}_{-\varepsilon/2 \leq x \leq \varepsilon/2} + \mathbf{1}_{x > \varepsilon/2}, \quad f''_\varepsilon(x) = \frac{1}{\varepsilon} \mathbf{1}_{-\varepsilon/2 \leq x \leq \varepsilon/2}.$$

Then, we apply Itô's formula to $e^{-rt}f_\varepsilon(S_t - K)$ between T and $T + \theta$:

$$\begin{aligned} e^{-r(T+\theta)}f_\varepsilon(S_{T+\theta} - K) - e^{-rT}f_\varepsilon(S_T - K) &= -r \int_T^{T+\theta} e^{-rt}f_\varepsilon(S_t - K)dt \\ &+ \int_T^{T+\theta} e^{-rt}f'_\varepsilon(S_t - K)dS_t + \frac{1}{2} \int_T^{T+\theta} e^{-rt}f''_\varepsilon(S_t - K)\sigma^2(t, S_t)S_t^2dt. \end{aligned}$$

Taking the expectation of each term,

$$\begin{aligned} e^{-r(T+\theta)}\mathbb{E}[f_\varepsilon(S_{T+\theta} - K)] - e^{-rT}\mathbb{E}[f_\varepsilon(S_T - K)] \\ = -r \int_T^{T+\theta} e^{-rt}\mathbb{E}[f_\varepsilon(S_t - K)]dt + \int_T^{T+\theta} e^{-rt}\mathbb{E}[f'_\varepsilon(S_t - K)S_t]r dt \\ + \frac{1}{2} \int_T^{T+\theta} dt e^{-rt} \int_{K-\varepsilon/2}^{K+\varepsilon/2} x^2 \sigma^2(t, x) \frac{1}{\varepsilon} p(t, x) dx. \end{aligned}$$

By square integrability of S , and continuity of σ and p , take the limit $\varepsilon \rightarrow 0$:

$$\begin{aligned} C(T + \theta, K) - C(T, K) &= -r \int_T^{T+\theta} \mathbb{E}[(S_t - K)^+] dt \\ &+ r \int_T^{T+\theta} e^{-rt}\mathbb{E}[S_t \mathbf{1}_{S_t \geq K}] dt + \frac{1}{2} \int_T^{T+\theta} e^{-rt}\sigma^2(t, K)K^2 p(t, K) dt \\ &= rK \int_T^{T+\theta} e^{-rt}\mathbb{P}[S_t \geq K] dt + \frac{1}{2} \int_T^{T+\theta} e^{-rt}\sigma^2(t, K)K^2 p(t, K) dt. \end{aligned}$$

Dividing both sides by θ and passing to the limit $\theta \rightarrow 0$, this gives

$$\frac{\partial C}{\partial T} = rK e^{-rT}\mathbb{P}[S_T \geq K] + \frac{1}{2} e^{-rT}\sigma^2(T, K)K^2 p(T, K).$$

Finally, observing that

$$e^{-rT}\mathbb{P}[S_T \geq K] = -\frac{\partial C}{\partial K} \quad \text{and} \quad e^{-rT}p(T, K) = \frac{\partial^2 C}{\partial K^2},$$

the proof of Dupire's equation is completed. $\square$

Implied diffusion The volatility function is given by Dupire's formula

$$\sigma^2(T, K) = 2 \frac{\frac{\partial C}{\partial T} + rK \frac{\partial C}{\partial K}}{K^2 \frac{\partial^2 C}{\partial K^2}}.$$

This means that we can recover the volatility from a complete set of call prices at a given date *if these prices were produced by a local volatility model*. Given a system of call option prices $(C(T, K))_{T \geq 0, K \geq 0}$, does there exist a continuous diffusion model

reproducing these prices? To apply Dupire's formula, we need to at least assume that $\frac{\partial^2 C}{\partial K^2} > 0$ and $\frac{\partial C}{\partial T} + rK \frac{\partial C}{\partial K} \geq 0$: this is related to the positivity of *butterfly spreads* and *calendar spreads*. It may happen that the volatility $\sigma(t, s)$ exists but does not lead to a Markov process satisfying the assumptions of theorem. When it exists, the local volatility model is *one among many models* which calibrate a given option price surface. This idea is developed in the next paragraph.

Markovian projection Let $(S_t)_{t \geq 0}$ be a stochastic volatility model:

$$\frac{dS_t}{S_t} = rdt + \sigma_t dW_t,$$

and let $\sigma_{loc}^2(t, x) := E[\sigma_t^2 | S_t = x]$. It can be shown that option prices in the model associated to (S_t) satisfy Dupire's equation with volatility $\sigma_{loc}(t, x)$. The function $\sigma_{loc}^2(t, x) = E[\sigma_t^2 | S_t = x]$ thus defines a *Markovian projection* of (S_t) : a Markovian diffusion with the same marginals as (S_t) . All models with Markovian projection σ_{loc}^2 are perfectly calibrated to the smile, but may lead to very different prices of exotics. As an example, consider a (fictitious) option on *instantaneous variance* in a stochastic volatility model. The price of this option satisfies

$$\begin{aligned} \mathbb{E}[(\sigma_t^2 - K)^+] &= \mathbb{E}[\mathbb{E}[(\sigma_t^2 - K)^+ | S_t]] \geq \mathbb{E}[(\mathbb{E}[\sigma_t^2 | S_t] - K)^+] \\ &= \mathbb{E}[(\sigma_{loc}^2(t, S_t) - K)^+] = \mathbb{E}^{loc}[(\sigma_{loc}^2(t, S_t) - K)^+], \end{aligned}$$

where the inequality is from Jensen's inequality and $\mathbb{E}^{loc}$ denotes the expectation in a model with local volatility σ_{loc} . We see that the local volatility model price for an exotic option may be different from its price in the original stochastic volatility model. In this particular example, the local volatility price is the *lower bound* of prices in all stochastic volatility models calibrated to the same option prices, however this result does not hold in general as we explain in the following paragraph.

Model risk and model-free bounds As we have seen, even with *perfect calibration* to the market, there are *many compatible models* which give different prices to non-quoted options. If a derivative product is priced with an arbitrage-free model calibrated to quoted option prices, it is therefore still possible that this price will not be sufficient to put in place a satisfactory hedging strategy. To analyze this model risk, one can compute price bounds given by calibration constraints *without making assumptions* on the underlying.

Let $\mathcal{A}$ be the set of quoted products with pay-offs $\{H_a\}_{a \in \mathcal{A}}$ and prices $\{P_a\}_{a \in \mathcal{A}}$, and let $\mathcal{M}$ be the class of admissible models. We assume the interest rate to be zero. For a pay-off H we define the upper and lower superhedging prices, where the hedging is done using a dynamic strategy in the underlying and a static strategy in quoted European options:

$$P_{min}(H) = \max_{\lambda, \delta, V_0} \{V_0 + \sum_{a \in \mathcal{A}} \lambda_a P_a : V_0 + \int_0^T \delta_t dS_t + \sum_{a \in \mathcal{A}} \lambda_a H_a \leq H\}. \quad (6.2)$$

$$P_{max}(H) = \min_{\lambda, \delta, V_0} \{V_0 + \sum_{a \in \mathcal{A}} \lambda_a P_a : V_0 + \int_0^T \delta_t dS_t + \sum_{a \in \mathcal{A}} \lambda_a H_a \geq H\}, \quad (6.3)$$

where the superhedging property must hold $\mathbb{P}$ -almost surely for every admissible model $\mathbb{P} \in \mathcal{M}$.

Alternatively, the price bounds can be expressed in terms of *calibrated pricing measures*:

$$P_{min}(H) = \inf_{\mathbb{Q} \in \mathcal{M}, \mathbb{E}^{\mathbb{Q}}[H_a] = P_a \forall a \in \mathcal{A}} \mathbb{E}^{\mathbb{Q}}[H]$$

$$P_{max}(H) = \sup_{\mathbb{Q} \in \mathcal{M}, \mathbb{E}^{\mathbb{Q}}[H_a] = P_a \forall a \in \mathcal{A}} \mathbb{E}^{\mathbb{Q}}[H].$$

The model risk of H is quantified by $P_{max}(H) - P_{min}(H)$.

These formulas are abstract but can be made explicit in specific cases or approximated in the general case. They are nonlinear: model risk of a portfolio is smaller than the sum of model risks of its constituents.

6.3 Dupire's formula in practice

Dupire's formula cannot be used directly on market-quoted option prices, because these are available for a finite number of strikes and maturities, and Dupire's formula requires the knowledge of option prices for all strikes and maturities. Market prices can of course be interpolated and the final result is very sensitive to the interpolation method used. This is characteristic of ill-posed problems, where small data errors lead to very large errors in the solution. Figure 6.1 shows that while from model-generated data Dupire's equation allows to recover a perfectly smooth local volatility, when used on market option prices with spline interpolation, the resulting volatility surface is very irregular

To avoid solving an ill-posed problem, one generally uses one of two approaches:

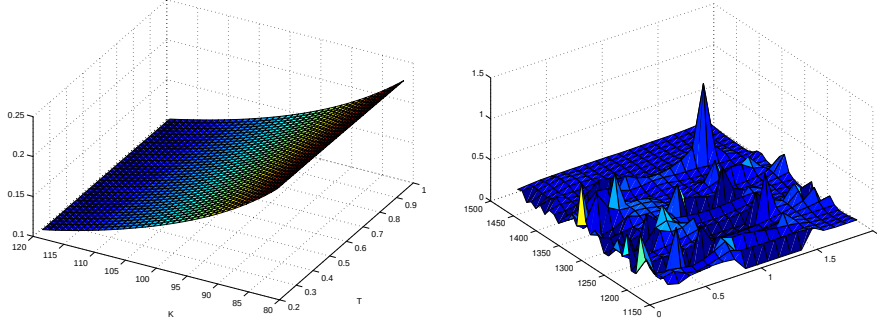


Figure 6.1: Left: local volatility computed from artificial (model-generated) data. Right: local volatility computed from S&P 500 option prices with spline interpolation.

- First, calibrate a parametric functional form to the implied volatility surface. With this smooth parametric function, recalculate option prices for all strikes, which are then used to calculate the local volatility by Dupire's formula.
- Reformulate Dupire's equation as an optimization problem by introducing a penalty term to limit the oscillations of the volatility surface:

$$J(\sigma) \equiv \sum_{i=1}^N w_i (C(T_i, K_i, \sigma) - C_M(T_i, K_i))^2 + \alpha \|\nabla \sigma\|_2^2,$$

$$\|\nabla \sigma\|_2^2 \equiv \int_{K_{\min}}^{K_{\max}} dK \int_{T_{\min}}^{T_{\max}} dT \left\{ \left(\frac{\partial \sigma}{\partial K} \right)^2 + \left(\frac{\partial \sigma}{\partial T} \right)^2 \right\},$$

where $C_M(T_i, K_i)$ are the market prices and $C(T_i, K_i, \sigma)$ are the local vol prices.

We will briefly discuss the first approach. To compute the local volatility from the interpolated implied volatility, one needs a version of Dupire's formula working at the level of implied volatilities. For every option,

$$C(T, K) = C_{BS}(T, K, I(T, K)),$$

where $C_{BS}(T, K, \sigma)$ denotes the Black-Scholes price of a call option with volatility σ and $I(T, K)$ is the implied volatility observed for maturity T and strike K . It will be helpful to express the prices in terms of *log-strike* $k = \log(K/S) - rT$, with $I(T, K) = \tilde{I}(T, k)$ and $\sigma(T, K) = \tilde{\sigma}(T, k)$ as well as the dimensionless implied

volatility $v(T, k) = \tilde{I}(T, k)\sqrt{T}$. We then get:

$$\frac{\partial C}{\partial T} + rK \frac{\partial C}{\partial K} = \frac{\partial \tilde{C}}{\partial T} = Sn(d_1) \frac{\partial v}{\partial T} \quad (6.4)$$

and

$$K^2 \frac{\partial^2 C}{\partial K^2} = \frac{\partial^2 \tilde{C}}{\partial k^2} - \frac{\partial \tilde{C}}{\partial k} = Sn(d_1) \left\{ v''_{kk} - \frac{v}{4}(v'_k)^2 + \frac{1}{v} \left(1 - \frac{kv'_k}{v} \right)^2 \right\}, \quad (6.5)$$

which finally leads to Dupire's formula in terms of implied volatility:

$$\begin{aligned} \tilde{\sigma}^2(T, k) &= \frac{2vv'_T}{vv''_{kk} - \frac{v^2}{4}(v'_k)^2 + \left(1 - \frac{kv'_k}{v} \right)^2} \\ &= \frac{2\tilde{I}T\tilde{I}'_T + \tilde{I}^2}{\tilde{I}T\tilde{I}''_{kk} - \frac{\tilde{I}^2T^2}{4}(\tilde{I}'_k)^2 + \left(1 - \frac{k\tilde{I}'_k}{\tilde{I}} \right)^2}. \end{aligned}$$

To interpolate the implied volatility, we shall use the parameterization of implied volatility as function of log-strike $k = \log \frac{K}{S} - rT$, proposed by J. Gatheral under the name SVI (stochastic volatility inspired).

For the given maturity, the implied variance $V(T, k) = \tilde{I}^2(T, k)T = v^2(T, k)$ satisfies

$$V(k) = a + b\{\rho(k - m) + \sqrt{(k - m)^2 + \sigma^2}\}$$

with $a \geq 0, b \geq 0, \sigma \geq 0, \rho \in [-1, 1]$ et $m \in \mathbb{R}$. Here,

- a controls the level of the smile;
- b controls the slope of the wings;
- ρ controls the asymmetry, “rotation” of the smile;
- m corresponds to a translation;
- σ controls the ATM convexity

The SVI form is calibrated separately for each maturity. The implied variance at intermediate dates is defined by linear interpolation. Local volatility can then be found from Dupire's formula. This parameterization does not guarantee that the resulting implied volatility surface is arbitrage free (i.e., does not necessarily satisfy the conditions given in the next section). See [15] for a method to construct an arbitrage-free SVI parameterization for multiple maturities.

6.4 No arbitrage conditions for option prices and implied volatilities

Arbitrage-free option price surfaces

Definition 5. An option price surface $(T_i, K_i, C_i)_{i \in I}$ is *arbitrage free* if there is a probability space $(\Omega, \mathcal{F}, \mathbb{P} = (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ and a martingale X with $C_i = \mathbb{E}[(X_{T_i} - K_i)^+]$, $i \in I$.

Proposition 7. $C : [0, \infty) \times [0, \infty)$ is an arbitrage-free option price surface if and only if there exists $s > 0$ such that

1. *Convexity in K :* for all $T \geq 0$, $C(T, \cdot)$ is convex.
2. *Monotony in T :* for all $K \geq 0$, $C(K, \cdot)$ is increasing.
3. *Bounds:* $(s - K)^+ \leq C(T, K) \leq s \quad \forall K \geq 0, \forall T \geq 0$.
4. *Value at expiry:* $C(0, K) = (s - K)^+ \quad \forall K \geq 0$.
5. *Large strike limit:* $\lim_{K \rightarrow \infty} C(T, K) = 0 \quad \forall T \geq 0$.

Proof Let $T \geq 0$ and $F_T(x) = 1 + C'_+(T, x)$ for $x \geq 0$ and $F_T(x) = 0$ for $x < 0$, where $C'_+(T, x)$ is the right derivative of C wrt x .

By convexity, $F_T(x)$ is right-continuous increasing in x .

By property 5, $\lim_{x \rightarrow \infty} F_T(x) = 1$, and by property 3, $F_T(x) \geq 0$.

$\Rightarrow F_T(x)$ is a distribution function of probability measure μ_T and for all $K, T \geq 0$,

$$C(K, T) = s - \int_{[0, \infty)} (x \wedge K) \mu_T(dx),$$

which implies, by dominated convergence, that

$$C(K, T) = \int_{[0, \infty)} (x - K)^+ \mu_T(dx).$$

By property 2, the family $(\mu_T)_{T \geq 0}$ is increasing in convex order, and by Kellerer's theorem there exists a martingale with marginals $(\mu_T)_{T \geq 0}$.

Arbitrage-free implied volatility surfaces We use the *log-strike* $k = \log \frac{Ke^{-rT}}{S}$ and let $v(T, k) = I(T, Se^{k+rT})\sqrt{T}$

Proposition 8. Let $s > 0$ and assume $v(T, k)$ is twice differentiable in k for all $T > 0$ and

1. Durrleman's condition: for $T > 0$ and $k \in \mathbb{R}$,

$$vv_{kk}'' - \frac{v^2}{4}(v_k')^2 + \left(1 - \frac{kv_k'}{v}\right)^2 \geq 0.$$

2. Monotony in T : for all $k \in \mathbb{R}$, $v(\cdot, k)$ is increasing.
3. For all $T > 0$ and $k \in \mathbb{R}$, $v(T, k) > 0$.
4. Expiry value: $k \in \mathbb{R}$, $v(0, k) = 0$.
5. Large strike limit $T > 0$, $\lim_{k \rightarrow +\infty} d_+(k, v(T, k)) = -\infty$, where $d_{\pm}(k, v) := -\frac{k}{v} \pm \frac{v}{2}$.

Then the option price surface $\tilde{C}(T, k) = SN(d_+(k, v(T, k))) - Se^k N(d_-(k, v(T, k)))$ is arbitrage free.

6.5 Dynamics of implied volatility

In this section, we are interested in two related questions:

- How does the ATM implied volatility evolve with S ?
- What are the implications on delta hedging?

To describe possible answers to these questions, practitioners have identified two 'regimes':

- “*Sticky strike*” regime, where the implied volatility does not depend on S , but only on strike, so that $\frac{\partial I(T, K, S_0)}{\partial S_0} = 0$. In this regime, the rate of change of ATM implied volatility with the underlying is therefore equal to ATM implied volatility skew:

$$\frac{\partial I(0, S_0, S_0)}{\partial S_0} = \frac{\partial I(0, K, S_0)}{\partial S_0} \Big|_{K=S_0} + \frac{\partial I(T, K, S_0)}{\partial K} \Big|_{K=S_0} = \frac{\partial I(T, K, S_0)}{\partial K} \Big|_{K=S_0}.$$

The Black-Scholes delta hedge remains optimal in this regime, in spite of the presence of implied volatility smile:

$$\frac{\partial P}{\partial S_0} \Big|_{SS} = \frac{\partial P^{BS}}{\partial S_0} + \frac{\partial P^{BS}}{\partial \sigma} \frac{\partial I(T, K)}{\partial S_0} = \frac{\partial P^{BS}}{\partial S_0}$$

- “*Sticky moneyness*” regime, where the implied volatility only depends on the moneyness ratio $\frac{K}{S_0}$: $\frac{\partial I(T, K, S_0)}{\partial S_0} = \frac{\partial I(T, K/S_0)}{\partial S_0} = -\frac{K}{S_0} \frac{\partial I}{\partial K}$. In this regime the ATM implied volatility remains constant over time (because the moneyness for ATM options is constant and equal to 1):

$$\frac{\partial I(0, S_0, S_0)}{\partial S_0} = 0.$$

The Black-Scholes delta hedge is no longer optimal in this model, and must be corrected to account for implied volatility smile:

$$\frac{\partial P}{\partial S_0} \Big|_{SM} = \frac{\partial P^{BS}}{\partial S_0} + \frac{\partial P^{BS}}{\partial \sigma} \frac{\partial I(T, K)}{\partial S_0} = \frac{\partial P^{BS}}{\partial S} - \frac{K}{S_0} \frac{\partial P^{BS}}{\partial \sigma} \frac{\partial I}{\partial K}$$

These two regimes are by no means the only possible descriptions of the smile dynamics, but they illustrate the fact that knowledge of the smile at any given day provides little information about the evolution of the smile as time goes on. We shall see that the local volatility correspond to a third “regime”. To characterize smile dynamics under local volatility, we shall use an explicit formula for implied volatility in local volatility models, valid in small time.

Let $T \rightarrow 0$ and assume that the implied volatility and its derivatives remain bounded. Then, by Dupire’s formula for implied volatility,

$$\tilde{\sigma}^2(0, k) = \frac{\tilde{I}(0, k)^2}{\left(1 - \frac{k \tilde{I}'_k(0, k)}{\tilde{I}(0, k)}\right)^2} \Rightarrow \tilde{I}(0, k) = \left(\int_0^1 \frac{dy}{\tilde{\sigma}(0, ky)} \right)^{-1}.$$

In other words,

$$I(0, K, S_0) = \left(\int_0^1 \frac{dy}{\sigma(0, S_0 \left(\frac{K}{S_0}\right)^y)} \right)^{-1}$$

How does the implied volatility move as the stock price moves? From the above formula, we have, for fixed strike, and small time

$$\begin{aligned} & \frac{\partial I(0, K, S_0)}{\partial S_0} \Big|_{K=S_0} \\ &= \left(\int_0^1 \frac{dy}{\sigma(0, S_0 \left(\frac{K}{S_0}\right)^y)} \right)^{-2} \int_0^1 \frac{(1-y) \left(\frac{K}{S_0}\right)^y \sigma'(0, S_0 \left(\frac{K}{S_0}\right)^y) dy}{\sigma^2(0, S_0 \left(\frac{K}{S_0}\right)^y)} \Big|_{K=S_0} = \frac{\sigma'(S_0)}{2}. \end{aligned}$$

Similarly,

$$\frac{\partial I(0, K, S_0)}{\partial K} \Big|_{K=S_0} = \frac{\sigma'(S_0)}{2},$$

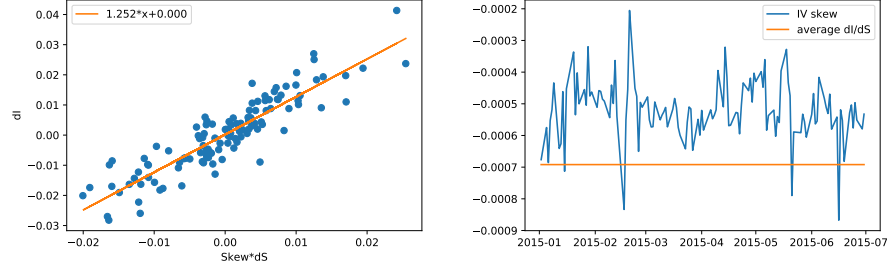


Figure 6.2: Variation of ATM implied volatility and the ATM IV skew of S&P index options with 1 month maturity.

so that

$$\frac{\partial I(0, S_0, S_0)}{\partial S_0} = \sigma'(S_0) = 2 \frac{\partial I(0, K, S_0)}{\partial K} \Big|_{K=S_0} \quad (6.6)$$

The rate of change of ATM implied volatility with the underlying is therefore equal to *twice the ATM implied volatility skew*. This can be compared with *sticky strike* regime with $\frac{\partial I(T, S_0, S_0)}{\partial S_0} = \frac{\partial I(T, K, S_0)}{\partial K} \Big|_{K=S_0}$ and *sticky moneyness* regime with $\frac{\partial I(T, S_0, S_0)}{\partial S_0} = 0$.

The *local volatility delta* at the money is given by

$$\begin{aligned} \frac{\partial P}{\partial S} \Big|_{LV} &= \frac{\partial P^{BS}}{\partial S} + \frac{\partial P^{BS}}{\partial \sigma} \frac{\partial I(T, K)}{\partial S_0} \Big|_{K=S_0} \approx \frac{\partial P^{BS}}{\partial S} + \frac{\partial P^{BS}}{\partial \sigma} \frac{\sigma'(S_0)}{2} \\ &= \frac{\partial P^{BS}}{\partial S} + \frac{\partial P^{BS}}{\partial \sigma} \frac{\partial I(0, K, S_0)}{\partial K} \Big|_{K=S_0} \end{aligned}$$

There is therefore a correction to Black-Scholes delta which has the *opposite sign* to that of the sticky moneyness regime.

Formula (6.6) provides a simple way to test whether observed ATM implied volatility dynamics obeys the local volatility model. Figure 6.2, left graph, plots the variations of implied volatility of one-month ATM options on the S&P index against the values of the implied volatility skew multiplied by the variations of the underlying. The points do seem to fall on a straight line, but the slope of this line is not 2, as would have been the case in a local volatility model, but 1.25. Similarly, the right graph plots the implied volatility skew against the average daily variation of ATM implied volatility over the entire period. Once again, we observe that the ratio of ATM implied volatility variations to the skew is smaller than the local volatility model prediction.

Chapter 7

Stochastic volatility models

Bird's eye view and key concepts of this chapter

- *Stochastic volatility model: market incompleteness and hedging*
- *Heston model pricing via Fourier transform*
- *Impact of parameters on implied volatility smile in Heston's model*
- *Interpretation of ATM implied volatility skew and convexity in short time*

[Click here to access the Jupyter notebook for this chapter](#)

In this chapter we discuss the “classical” stochastic volatility model, where one models the dynamics of the instantaneous volatility σ_t directly with a stochastic process. We first discuss the general features of these models and next consider the Heston model as an archetypical example.

7.1 Stochastic volatility models: general theory

In stochastic volatility models, the volatility is a positive, often mean-reverting stochastic process, correlated with the underlying:

$$\begin{aligned}\frac{dS_t}{S_t} &= \mu_t dt + \sigma_t dW_t \\ d\sigma_t &= a_t dt + b_t dW'_t, \quad d\langle W, W' \rangle_t = \rho dt\end{aligned}$$

Unlike local volatility models, vega risk is here introduced explicitly, which makes it possible to design hedging strategies. The market is incomplete (2 sources of risk),

but may be completed with another liquid asset allowing for perfect hedging of vega risk.

On the other hand, the volatility is not directly observable, which creates *model risk*. The model can be calibrated to options market, but calibration is time-consuming and not perfect; as a result, dynamics of hedging instruments may be wrong. In other words, such a stochastic volatility model is not a market model meaning that the risk factors are not traded assets and cannot be used directly for hedging.

Pricing and hedging in stochastic volatility models We place ourselves in the Markov framework and assume that the coefficients a and b are deterministic functions of time, volatility and the value of the underlying $a_t = a(t, \sigma_t, S_t)$, $b_t = b(t, \sigma_t, S_t)$. Since there are two risk sources in the model, we aim to design a hedging with two risky assets. Suppose therefore that there exists a liquid asset (e.g., variance swap) quoted at price

$$C_t^0 = C^0(t, \sigma_t, S_t),$$

where C^0 is a regular function. Let V_t be the value of a self-financing portfolio containing δ_t units of stock and ω_t units of C^0 . The dynamics of this portfolio is given by

$$\begin{aligned} dV_t &= (V_t - \delta_t S_t - \omega_t C_t^0)rdt + \delta_t dS_t + v_t dC_t^0 \\ &= (V_t - \delta_t S_t - \omega_t C_t^0)rdt \\ &\quad + (\delta_t + \omega_t \frac{\partial C^0}{\partial S})dS_t + \omega_t \frac{\partial C^0}{\partial \sigma} d\sigma_t + \omega_t \mathcal{L}_t C^0 dt \\ \text{with } \mathcal{L}_t &= \frac{\partial}{\partial t} + \frac{1}{2} S_t^2 \sigma_t^2 \frac{\partial^2}{\partial S^2} + \frac{1}{2} b_t^2 \frac{\partial^2}{\partial \sigma^2} + S_t \sigma_t b_t \rho \frac{\partial^2}{\partial S \partial \sigma} \end{aligned}$$

Let us now look for a self-financing portfolio which satisfies $V_t = C(t, \sigma_t, S_t)$ for a given regular function C . By Itô's formula,

$$dV_t = \mathcal{L}_t C dt + \frac{\partial C}{\partial S} dS_t + \frac{\partial C}{\partial \sigma} d\sigma_t$$

Identifying the coefficients in front of dS and $d\sigma$ (because these processes contain independent noises), we get the hedge ratios for delta-vega hedging in a stochastic volatility model:

$$\omega_t = \frac{\partial C / \partial \sigma}{\partial C^0 / \partial \sigma} \quad \delta_t = \frac{\partial C}{\partial S} - \omega_t \frac{\partial C^0}{\partial S},$$

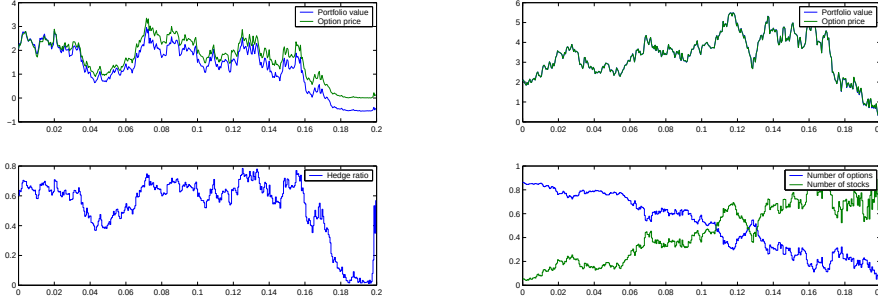


Figure 7.1: Left: delta-only hedging in a stochastic volatility market. Right: delta-vega hedging.

Figure 7.1 illustrates the performance of hedging using the underlying only (which has some residual risk) as well as the one using the underlying and another liquid asset (which leads to perfect hedging on simulated data).

The residual terms show that C must satisfy a partial differential equation:

$$\frac{\mathcal{L}_t C - rC + rS_t \frac{\partial C}{\partial S}}{\partial C / \partial \sigma} = \frac{\mathcal{L}_t C^0 - rC^0 + rS_t \frac{\partial C^0}{\partial S}}{\partial C^0 / \partial \sigma}.$$

Let

$$\lambda(t, \sigma, S) = - \frac{\mathcal{L}_t C^0 - rC^0 + rS \frac{\partial C^0}{\partial S}}{\partial C^0 / \partial \sigma}$$

Then the price of an option with pay-off $h(S_T)$ satisfies:

$$\begin{aligned} \mathcal{L}_t C - rC + rS \frac{\partial C}{\partial S} + \lambda(t, \sigma, S) \frac{\partial C}{\partial \sigma} &= 0, \quad C(T, \sigma, S) = h(S) \\ \text{avec } \mathcal{L}_t &= \frac{\partial}{\partial t} + \frac{1}{2} S^2 \sigma^2 \frac{\partial^2}{\partial S^2} + \frac{1}{2} b_t^2 \frac{\partial^2}{\partial \sigma^2} + S \sigma b_t \rho \frac{\partial^2}{\partial S \partial \sigma} \end{aligned}$$

This equation is an extension of the Black-Scholes PDE to two dimensions. Remark that unlike the one-dimensional Black-Scholes equation, this one depends on an unknown function λ . This translates the market incompleteness of stochastic volatility models. The role of this function will become clear in the next paragraph.

Risk-neutral pricing Using the function λ and Itô's formula, the dynamics of the option price C_t may be written as follows:

$$dC_t = rC_t dt + \frac{\partial C}{\partial S} (dS_t - rS_t dt) + \frac{\partial C}{\partial \sigma} (d\sigma_t - \lambda(t, \sigma_t, S_t) dt).$$

Since, under the risk-neutral probability, the discounted value of C_t must be a local martingale, we deduce that the dynamics of σ and S under $\mathbb{Q}$ must be described by the following equations:

$$\begin{aligned}\frac{dS_t}{S_t} &= rdt + \sigma_t d\widehat{W}_t \\ d\sigma_t &= \lambda_t dt + b_t d\widehat{W}'_t, \quad d\langle \widehat{W}, \widehat{W}' \rangle_t = \rho dt.\end{aligned}$$

The function λ thus appears as the drift of the volatility process under the risk-neutral measure. Note that the volatility of volatility b_t must be the same under $\mathbb{P}$ and under $\mathbb{Q}$ to guarantee equivalence of the two measures.

7.2 The Heston parameterization

The Heston stochastic volatility model [17] under the risk-neutral probability writes:

$$\begin{aligned}\frac{dS_t}{S_t} &= rdt + \sigma_t dW_t \\ d\sigma_t^2 &= \kappa(\theta - \sigma_t^2)dt + \delta\sigma_t dW'_t, \quad d\langle W, W' \rangle_t = \rho dt \\ \text{or equivalently} \quad dv_t &= \kappa(\theta - v_t)dt + \delta\sqrt{v_t}dW'_t \quad \text{with} \quad v_t = \sigma_t^2\end{aligned}$$

Here $\delta > 0$ is the volatility of variance $\theta > 0$ is the long-term mean of variance and $\kappa > 0$ is the rate of mean reversion. This model is a member of affine family of processes, which means that the characteristic function of log-price is known explicitly. If $\frac{2\kappa\theta}{\delta^2} > 1$, the process never touches zero, otherwise reflection at zero is possible.

The equation for the variance can be rewritten in the following form, which simplifies its analysis:

$$v_t = \theta + e^{-\kappa t}(v_0 - \theta) + \delta \int_0^t e^{-\kappa(t-s)} \sqrt{v_s} dW'_s.$$

In particular, the expectation of the variance is given by the first three terms, and it is clear that it converges to θ with speed κ :

$$\mathbb{E}[v_t] = e^{-\kappa t}(v_0 - \theta) + \theta.$$

Similarly, the variance of v_t may be computed by applying Itô isometry:

$$\text{Var}[v_t] = \frac{\delta^2\theta}{2\kappa} + \frac{\delta^2(v_0 - \theta)}{\kappa}e^{-\kappa t} + \frac{\delta^2(\theta - 2v_0)}{2\kappa}e^{-2\kappa t}.$$

If t is small,

$$\text{Var}[v_t] \approx v_0\delta^2 t,$$

and when $t \rightarrow \infty$, the variance of v_t converges to the limit $\frac{\delta^2\theta}{2\kappa}$.

Characteristic function of log stock price Making a change of variable: $S_t = S_0 e^{rt+X_t}$ allows to get rid of the interest rate and work with the log price:

$$dX_t = -\frac{1}{2}\sigma_t^2 dt + \sigma_t dW_t$$

By Itô's lemma, the characteristic function of X_T , defined by $f(t, v, x) = \mathbb{E}[e^{iuX_T} | X_t = x, v_t = v]$, satisfies:

$$\frac{v}{2} \frac{\partial^2 f}{\partial x^2} + \rho \delta v \frac{\partial^2 f}{\partial x \partial v} + \frac{\delta^2 v}{2} \frac{\partial^2 f}{\partial v^2} - \frac{v}{2} \frac{\partial f}{\partial x} + \kappa(\theta - v) \frac{\partial f}{\partial v} + \frac{\partial f}{\partial t} = 0,$$

with terminal condition $f(T, v, x) = e^{iux}$.

Thanks to the affine structure of the coefficients and the exponential terminal condition, the equation can be solved explicitly:

$$\mathbb{E}[e^{iuX_T} | X_t = x, v_t = v] = \exp\{iux + \theta C(T-t) + vD(T-t)\},$$

$$\text{où } D(t) = -\frac{u^2 + iu}{\gamma \coth \frac{\gamma t}{2} + \kappa - i\rho \delta u},$$

$$\text{et } C(t) = \frac{\kappa t(\kappa - i\rho \delta u - \gamma)}{\delta^2} - \frac{2\kappa}{\delta^2} \ln \left(\frac{1}{2}(1 + e^{-\gamma t}) + \frac{\kappa - i\rho \delta u}{2\gamma}(1 - e^{-\gamma t}) \right),$$

avec $\gamma = \sqrt{\delta^2(u^2 + iu) + (\kappa - i\rho \delta u)^2}$.

Knowledge of the explicit form of the characteristic function allows to compute the prices of European options quickly by Fourier transform. In terms of the log-price, the call option price writes:

$$C(K) = e^{-rT} \mathbb{E}[(S_0 e^{rT+X_T} - K)^+] = S_0 \mathbb{E} \left[\left(e^{X_T} - \frac{K e^{-rT}}{S_0} \right)^+ \right]$$

One can then deduce:

$$C(K) = (S_0 - K e^{-rT})^+ + \frac{S_0}{2\pi} \int_{-\infty}^{+\infty} e^{-iv \log \frac{K e^{-rT}}{S_0}} \frac{\Phi_T(v-i) - 1}{iv(1+iv)} dv.$$

or, to reduce the truncation error,

$$C(K) = C^{BS}(S_0, K, T, \Sigma) + \frac{S_0}{2\pi} \int_{-\infty}^{+\infty} e^{-iv \log \frac{K e^{-rT}}{S_0}} \frac{\Phi_T(v-i) - \Phi_T^\Sigma(v-i)}{iv(1+iv)} dv,$$

where $\Phi_T(u) = \mathbb{E}[e^{iuX_T}]$ and Φ_T^Σ is the characteristic function of log-price in the Black-Scholes model with volatility Σ .

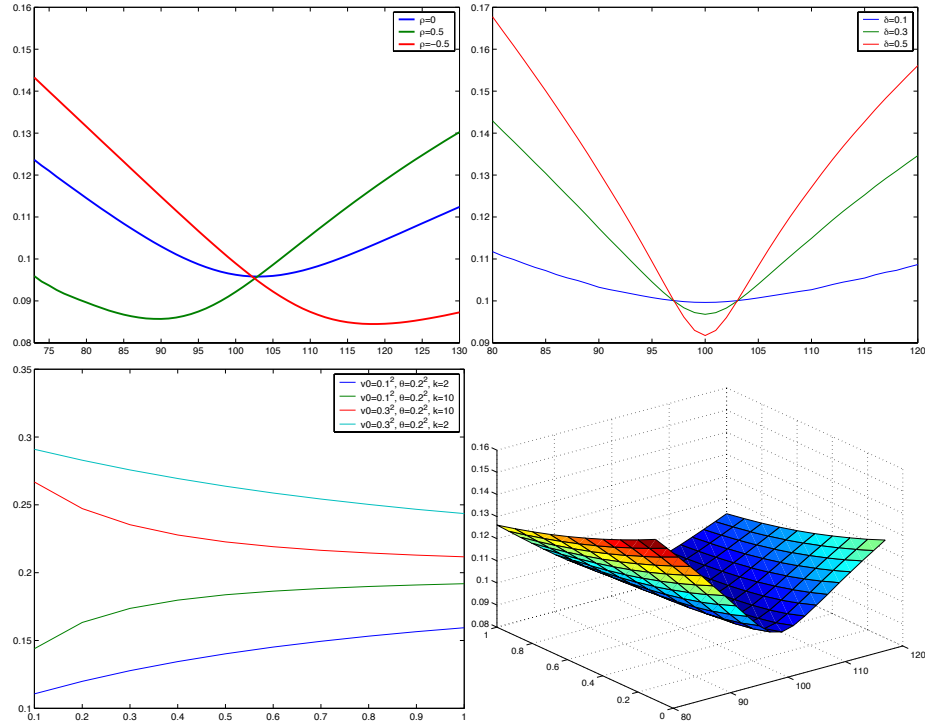


Figure 7.2: Top graph: Effect of correlation ρ (left) and the vol of vol δ on the smile. Bottom graph: term structure of ATM implied volatility (left) and the overall shape of the implied volatility smile (right).

Implied volatility smile in Heston's model The success of Heston's model is due to the fact that each parameter has a well-defined effect on the volatility smile (see Figure 7.2), which allows for easy calibration

- The correlation ρ determines the slope (risk reversal).
- The vol of vol δ determines the convexity (butterfly).
- The parameters v_0 , θ and k determine the ATM term structure

In the Heston model with constant parameters, one can therefore more or less precisely calibrate the term structure of ATM volatility and the smile at a given date.

The behavior of the implied volatility smile has been analyzed theoretically by many authors, who established explicit formulas for the implied volatility smile in

various asymptotic regimes. For instance, in Heston's model, clearly $I(T, K) \rightarrow \sqrt{v_0}$ as $T \rightarrow 0$. In addition, according to [12], in terms of the log strike $k = \log \frac{K}{S_0}$,

$$\left. \frac{\partial I}{\partial k} \right|_{ATM, T \rightarrow 0} = \frac{\rho \delta}{4\sqrt{v_0}}$$

The short-term ATM skew is thus inversely proportional to volatility, a behavior which is not observed in the markets. The ATM convexity in small time satisfies

$$\left. \frac{\partial^2 I}{\partial k^2} \right|_{ATM, T \rightarrow 0} = \left(\frac{1}{12} - \frac{5}{24} \rho^2 \right) \frac{\delta^2}{v_0^{3/2}}.$$

Interpretation of the short-term ATM skew Differentiating the Black-Scholes formula with respect to the log forward strike $k = \log \frac{K e^{-rT}}{S_0}$, we get:

$$\begin{aligned} \partial_k I(T, k) &= \frac{\partial_k c(k, T) - \partial_k c^{BS}(k, T, I(T, k))}{\partial_I c^{BS}(k, T, I(T, k))} \\ &= \frac{\mathbb{P}^{BS}[X_T \geq k] - \mathbb{P}[X_T \geq k]}{e^{-k} n(d_1) \sqrt{T}}, \end{aligned}$$

where under $\mathbb{P}^{BS}$, S_T follows the Black-Scholes model with volatility $I(T, k)$. Next, substituting $k = 0$, we obtain the formula for at the money forward (ATMF) skew:

$$\left. \frac{\partial I}{\partial k} \right|_{ATMF} = \frac{\mathbb{P}^{BS}[X_T \geq 0] - \mathbb{P}[X_T \geq 0]}{n\left(\frac{I(T)\sqrt{T}}{2}\right) \sqrt{T}}.$$

As $T \rightarrow 0$, under above conditions,

$$\left. \frac{\partial I}{\partial k} \right|_{ATM, T \rightarrow 0} = \frac{\sqrt{2\pi}}{\sqrt{T}} \{ \mathbb{P}^{BS}[X_T \geq 0] - \mathbb{P}[X_T \geq 0] \} \{ 1 + O(I^2(T, 0)T) \}$$

The ATM skew thus appears as a measure of *asymmetry of the stock price distribution function around the money*, compared to the Black-Scholes model. Similarly, it can be shown that the ATM convexity measures the difference of the stock price *density* around the money compared to the Black-Scholes model.

The long-maturity behavior of implied volatility in Heston's model has been analyzed theoretically in [13]. Taking a fixed forward log strike $k = \log \frac{K e^{-rT}}{S_0}$, one can show that the implied volatility converges to a constant independent from k :

$$\lim_{T \rightarrow \infty} I(T, k) = \frac{2\sqrt{\kappa\theta(\eta + \rho\delta - 2\kappa)}}{\delta\sqrt{(1 - \rho^2)}} = \sqrt{\theta} \left(1 + \frac{\delta\rho}{4\kappa} + O(\delta^2) \right)$$

with $\eta = \sqrt{\delta^2 + 4\kappa^2} - 4\rho\delta\kappa$, for any value of k .

The skew $\frac{\partial I}{\partial k}$ flattens as $\frac{1}{T}$

$$\frac{\partial I(T, k)}{\partial k} \sim \rho \left(\frac{\eta + \rho\delta - 2\kappa}{\kappa\theta(1 - \rho^2)} \right)^{\frac{1}{2}} \frac{1}{T} = \frac{\delta\rho}{2\kappa\sqrt{\theta}} \frac{1}{T} + O\left(\frac{\delta^2}{T}\right), \quad T \rightarrow \infty.$$

Calibration of Heston's model to European options

- The short-term smile behavior is determined by v_0 , ρ and δ . These parameters may be pre-calibrated from ATM level, skew and convexity as $T \rightarrow 0$.
- The long-term behavior is determined by θ , κ , ρ and δ . θ and κ may be pre-calibrated on the long-term level and skew.
- Alternatively, one may calibrate θ and κ from the volatility term structure, for example the variance swap term structure.
- A precise calibration can then be performed with nonlinear least squares:

$$\Theta^* = \arg \inf_{\Theta} \sum_{i=1}^N w_i (C^\Theta(T_i, K_i) - C_t^M)^2,$$

where Θ is the vector of all parameters. Figure 7.3 illustrates the calibration of Heston's model to market data.

Heston model: summary

- ✓ Good static properties: intuitive calibration of level, skew, convexity and term structure of implied volatility, except the short-term wings of the smile.
- ✓ Fast pricing of Europeans and some exotics, fast calibration.
- ≈ Vega risk taken into account but dynamic properties are not fully consistent with the market: the skew falls when volatility goes up.
- ☹ Not a market model: hedging strategies are based on unobservable volatility $\Rightarrow$ *presence of model risk*.
- ☹ Not suitable for VIX options: the distribution of future volatility is not consistent with that of the market (see next chapter).

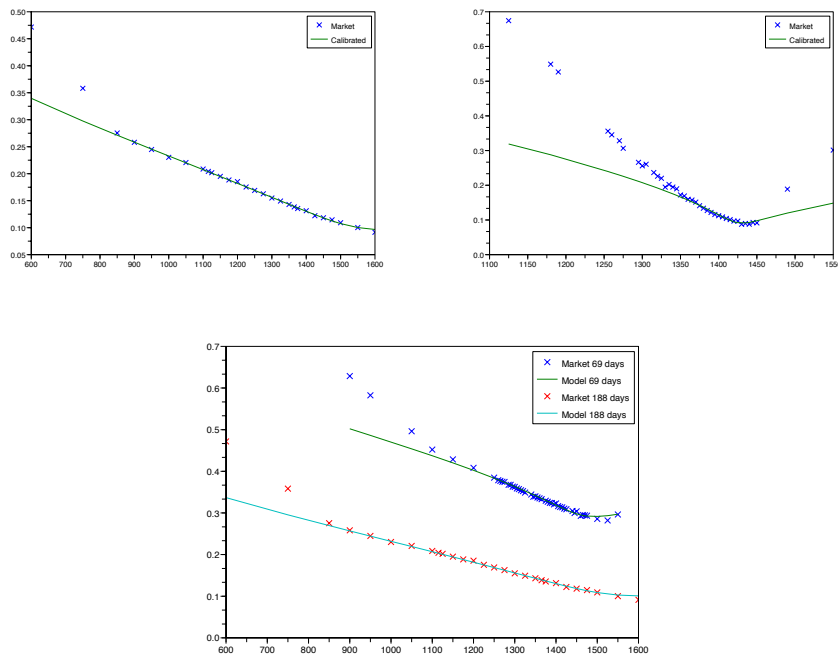


Figure 7.3: Calibration of Heston's model to market data. Top left: Maturity 188 days: the model reproduces correctly the long-maturity skew. Top right: Maturity 8 days: the wings of the short-term smile are not correctly calibrated Bottom: Simultaneous calibration to 2 maturities: the term structure is reproduced correctly

Chapter 8

Forward variance and variance curve models

Bird's eye view and key concepts of this chapter

- *Model-free replication of variance swaps*
- *Definition of forward variance*
- *VIX index, VIX futures and options, VIX implied volatility smile*
- *General log-normal variance curve models*
- *Fractional Brownian motion and rough volatility models*

8.1 Variance swaps

A variance swap is a contract to exchange the realized variance of a stock over a period against a fixed price:

$$\frac{A}{N} \sum_{i=0}^{N-1} \log^2 \frac{S_{i+1}}{S_i} - \hat{\sigma}_{V,S,T}^2$$

In a continuous model with stochastic volatility σ_t , one can use the approximation

$$\frac{1}{T} \int_0^T \sigma_t^2 dt - \hat{\sigma}_{V,S,T}^2.$$

By Itô formula,

$$\int_0^T \sigma_t^2 dt = 2 \int_0^T \frac{dS_t}{S_t} - 2 \log \frac{S_T}{S_0}$$

The realized variance can therefore be replicated by the log contract plus a dynamic portfolio invested in stock.

Let us consider first the replication of the log contract. Let f be a C^2 function. It is easy to see that for F, x positive,

$$f(x) = f(F) + f'(F)(x - F) + \int_0^F f''(K)(K - x)^+ dK + \int_F^\infty f''(K)(x - K)^+ dK.$$

To replicate a European pay-off $f(S_T)$, we let $F = S_0 e^{rT}$ and create a portfolio with

- The amount $f(S_0 e^{rT})$ in cash;
- The quantity of forward contracts given by $f'(S_0 e^{rT})$;
- A portfolio of put options with strikes between 0 and $S_0 e^{rT}$, with density $f''(K)$ per strike;
- A portfolio of call options with strikes between $S_0 e^{rT}$ and ∞ , with density $f''(K)$ per strike.

For the log-contract, we take $f(x) = \log(x/S_0)$,

$$f(S_0 e^{rT}) = rT, \quad f'(S_0 e^{rT}) = \frac{e^{-rT}}{S_0}, \quad f''(K) = -\frac{1}{K^2}.$$

On the other hand, the quantity $\int_0^T \frac{dS_t}{S_t}$ is the terminal value of a self-financing portfolio

$$V_t = e^{-r(T-t)} \int_0^t \frac{dS_u}{S_u} + e^{-r(T-t)} r(T-t),$$

which invests $\frac{e^{-r(T-t)}}{S_t}$ into stock at time t and has initial cost $V_0 = e^{-rT} rT$.

Model-free replication of the variance swap

In any stochastic volatility model, if all call/put options with maturity T can be traded, the variance swap can be replicated exactly at cost

$$P \left(\frac{1}{T} \int_0^T \sigma_t^2 dt \right) = \frac{2}{T} \int_0^F \frac{P(T, K)}{K^2} dK + \frac{2}{T} \int_F^\infty \frac{C(T, K)}{K^2} dK.$$

Model-free bounds for a variance swap If the number of traded options is finite it is no longer possible to replicate the variance swap exactly, and the upper and lower bounds defined in equations (6.2)-(6.3) satisfy $P_{min} < P_{max}$. Davis, Obloj and Raval [6] compute P_{min} and P_{max} via convex optimization. For the lower bound, this boils down to finding a lower bound of $-\log x$ in the form of a piecewise linear function. On the other hand, since $-\log x$ cannot be upper bounded by a piecewise linear function, for the standard variance swap $P_{max} = +\infty$. The pay-off of the variance swap cannot be upper bounded by a finite number of European options hence this product is very dangerous without variance cap.

Consider now a “corridor” swap with pay-off

$$\frac{1}{T} \int_0^T \mathbf{1}_{S \in [K_{min}, K_{max}]} \sigma_t^2 dt$$

When all put and call options with maturity T can be used,

$$P_{min} = P_{max} = \frac{2}{T} \int_{[0, F] \cap [K_{min}, K_{max}]} \frac{P(T, K)}{K^2} dK + \frac{2}{T} \int_{[F, \infty] \cap [K_{min}, K_{max}]} \frac{C(T, K)}{K^2} dK.$$

When only a finite number of options can be used, still $P_{min} < P_{max} < \infty$.

Forward variance A very important notion related to variance swaps is that of forward variance. The forward variance is defined as the price of the future realized variance, as determined by the variance swap formula, paid at maturity:

$$\hat{v}_t^T = (\hat{\sigma}_t^T)^2 = \mathbb{E}^{\mathbb{Q}} \left[\frac{1}{T-t} \int_t^T \sigma_s^2 ds \middle| \mathcal{F}_t \right].$$

One can define the *instantaneous forward variance curve*

$$\xi_t(T) = \frac{\partial}{\partial T} \{ (T-t) \hat{v}_t^T \} = \mathbb{E}^{\mathbb{Q}} [\sigma_T^2 | \mathcal{F}_t].$$

and the forward variance between two future dates:

$$\hat{v}_t^{T_1, T_2} = \mathbb{E}^{\mathbb{Q}} \left[\frac{1}{T_2 - T_1} \int_{T_1}^{T_2} \sigma_s^2 ds \middle| \mathcal{F}_t \right] = \frac{1}{T_2 - T_1} \int_{T_1}^{T_2} \xi_t(u) du.$$

The instantaneous forward variance is driftless under the risk-neutral measure.

8.2 The VIX index and VIX derivatives

The VIX index The VIX index is the volatility index of S&P 500 options, computed by CBOE. Figure 8.1, top left graph shows a trajectory of the VIX index in 2017–2018.

Since 2003, VIX is computed using the discretized variance swap formula.

$$VIX_T^2 = \frac{2e^{rT}}{T} \sum_i \frac{\Delta K_i}{K_i^2} Q_i(K_i) - \frac{1}{T} \left[\frac{F}{K_0} - 1 \right]^2,$$

for two shortest expiry dates T_1 and T_2 ($T_1 < 1$ month and $T_2 > 1$ month), where

- Q_i is the price of option (call or put) which is OTM for the given strike K_i ;
- F is the forward value computed from call-put parity;
- K_0 is the largest strike smaller than the forward;
- ΔK_i is the interval between the strikes.

The third term is a discretization error correction. To compute the VIX index, one interpolates between VIX_{T_1} and VIX_{T_2} to have a maturity of 30 days.

VIX derivatives The VIX itself cannot be traded but has given rise to an ecosystem of derivative products. VIX futures trade on CBOE and expire on Wednesday which is 30 days before the expiration dates of listed S&P options (see Figure 8.1, top right graph for an example of VIX futures term structure). The settlement value of VIX futures equals the VIX. Due to non-linearity of the VIX (square root), there is no model-free formula for VIX futures. VIX options are options on the VIX futures which have the same expiry dates as the futures.

Assuming that the underlying follows a continuous stochastic volatility model, and that sufficiently many options are available, the squared VIX may be represented as

$$VIX_T^2 \approx \mathbb{E} \left[\frac{1}{\tau} \int_T^{T+\tau} \sigma_s^2 ds \middle| \mathcal{F}_T \right] = \frac{2e^{r\tau}}{\tau} \int_0^F \frac{P_T(\tau, K)}{K^2} dK + \frac{2e^{r\tau}}{\tau} \int_F^\infty \frac{C_T(\tau, K)}{K^2} dK,$$

with $\tau = 1$ month.

The fair value of the VIX future contract is then

$$F_t^{VIX}(T) = \mathbb{E}[VIX_T | \mathcal{F}_t]$$

and price of the VIX call option is

$$P_t^{VIX}(T, K) = \mathbb{E} \left[e^{-r(T-t)} (VIX_T - K)^+ \right],$$

where the expectations are evaluated under the risk-neutral measure. The implied volatility of VIX options is defined assuming that the VIX future is lognormal, as the volatility to be used in the Black-Scholes formula to retrieve the given value of the VIX option. Figure 8.1, bottom graphs shows the typical shape of implied volatility smile in the VIX market - as one can see, it is very different from equity smiles.

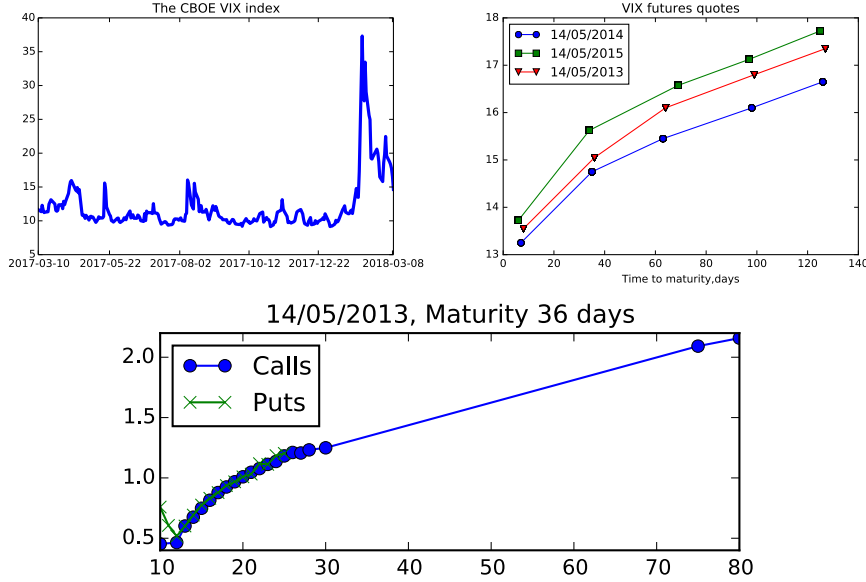


Figure 8.1: Top left: the CBOE VIX index. Top right: VIX Futures. Bottom: VIX options implied volatility smile

Volatility options in Heston model It is instructive to compute the VIX option implied volatility smile in Heston's model. The VIX index in Heston's model takes the form

$$VIX_T = \sqrt{a + bv_T}, \quad a = \theta \frac{\kappa\tau - 1 + e^{-\kappa\tau}}{\kappa\tau}, \quad b = \frac{1 - e^{-\kappa\tau}}{\kappa\tau}.$$

The VIX option price is therefore given by

$$P_t^{VIX}(T, K) = \mathbb{E}[e^{-r(T-t)}(\sqrt{a + bv_T} - K)^+ | \mathcal{F}_t]$$

The conditional density of v_T given v_t is known:

$$f(v_T | v_t, \kappa, \theta, \delta) = ce^{-u-v} \left(\frac{v}{u}\right)^{q/2} I_q(2\sqrt{uv})$$

where $q = \frac{2\kappa\theta}{\delta^2} - 1$, $c = \frac{2\kappa}{(1 - e^{-\kappa(T-t)})\delta^2}$, $u = cv_t e^{-\kappa(T-t)}$, $v = cv_T$

and I_q is the modified Bessel function of the 1st kind. Computing the option prices by numerical integration and then the implied volatility by inverting the Black-Scholes formula, one gets the curves shown in Figure 8.2. The shape of the smile is opposite to that observed in the market!

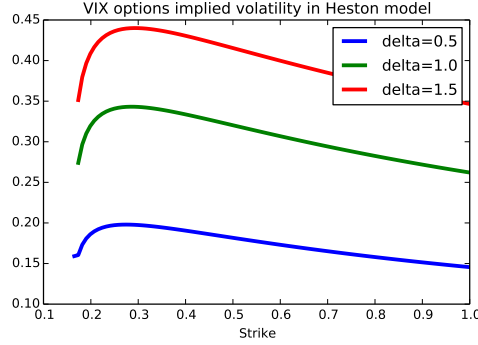


Figure 8.2: VIX options volatility smile in Heston's model for different values of vol of vol parameter.

8.3 Variance curve models

Liquid options being natural hedging instruments, ideally, one would like to have an arbitrage-free model for the dynamics of underlying and liquid option prices or implied volatilities. Such models have been proposed in the literature but are not used in practice due to the complexity of arbitrage constraints on the smile. By contrast, it is possible to specify an arbitrage-free dynamics for the curve of variance swap forward variances. Contrary to first generation stochastic volatility models, this gives a market model, that is, a model based solely on observable and tradable quantities

Let v_t^T be the variance of variance swap observed at t for expiry T . We define the forward variance $v_t^{T_1, T_2}$ by

$$v_t^{T_1, T_2} = \frac{(T_2 - t)v_t^{T_2} - (T_1 - t)v_t^{T_1}}{T_2 - T_1}$$

and the instantaneous forward variance by

$$\xi_t^T = \lim_{h \rightarrow 0} v_t^{T, T+h}.$$

The non-instantaneous forward variance can then be recovered from the instantaneous forward variance as follows:

$$v_t^{T_1, T_2} = \frac{1}{T_2 - T_1} \int_{T_1}^{T_2} \xi_t^T dT.$$

In particular, $\sqrt{\xi_t^t}$ is the instantaneous volatility of the underlying. By standard arguments, such a model is arbitrage-free if there exists a risk-neutral measure under which all forward variances ξ_t^T are martingales.

A log-normal model for the variance curve To build a usable model, one needs to specify the dynamics of the forward variances, and the simplest solution is to assume log-normal dynamics. Consider the model

$$d\xi_t^u = \xi_t^u g(u, t)^\top dW_t, \quad (8.1)$$

where W is a d -dimensional standard Brownian motion, and g is a kernel such that

$$\int_0^t \|g(t, s)\|^2 ds < \infty, \quad \forall t \geq 0.$$

We can introduce a dependence between stock price and volatility by taking

$$\frac{dS_t}{S_t} = \sigma_t(\rho^\top dW_t + \sqrt{1 - \rho^\top \rho} dB_t), \quad \sigma_t = \sqrt{\xi_t^t}.$$

where B is an independent Brownian motion and $\rho \in \mathbb{R}^d$ with $\rho^\top \rho \leq 1$. We have explicitly,

$$\xi_t^t = \xi_s^t e^{\int_s^t g(t, u)^\top dW_u - \frac{1}{2} \int_s^t \|g(t, u)\|^2 du} := \xi_s^t \mathcal{E}_{t, s}$$

Contrary to Heston's model, in this model there is no explicit formula for European option prices. For pricing options on variance only, since the variance is log-normal, one can use direct simulation without discretization. For pricing spot-dependent pay-offs, one needs to discretize the stock price process S_t via Euler scheme. Approximate formulas for ATM skew and convexity are available — see Bergomi's book. A recently popular class of *Rough volatility models* falls into the class of variance curve models.

8.4 Case study: pricing and hedging cliquets

The forward start (cliquet) option has pay-off at time $T + \tau$ given by

$$\left(\frac{S_{T+\tau}}{S_T} - K \right)^+.$$

In the Black-Scholes model with volatility $\sigma(t)$, its price is

$$P_t^{BS}(S_t, K, T, \tau) = e^{-r(T-t)} C^{BS}(1, K, \tau, \sigma_{T, T+\tau}), \quad t \leq T$$

$$\sigma_{T, T+\tau}^2 = \frac{1}{\tau} \int_T^{T+\tau} \sigma^2(s) ds.$$

In the Black-Scholes model (and any other model with no explicit dependence of volatility on S), before T , the cliquet option has zero delta, and its vega does not depend on S . The main risk of the cliquet is the risk of *future volatility*.

To hedge a cliquet, one therefore needs a portfolio of vanilla options with zero delta and vega independent of S . Such a portfolio can be constructed by taking the pay-off $f(S_{T+\tau}/S_T)$. The only such pay-off replicable by vanillas is the *forward log contract*

$$\log \frac{S_{T+\tau}}{S_T} = \log \frac{S_{T+\tau}}{S_0} - \log \frac{S_0}{S_T} = -\frac{1}{2} \int_T^{T+\tau} \sigma_s^2 ds + \int_T^{T+\tau} \frac{dS_t}{S_t}$$

The forward log contract can be represented as a forward variance swap plus a delta hedging portfolio between T and $T + \tau$.

Denoting by $I_T(\tau, k)$, the implied volatility at time T with time to maturity τ and moneyness K , the cliquet price is

$$P_t = e^{-r(T-t)} \mathbb{E}[C^{BS}(1, K, \tau, I_T(\tau, K)) | \mathcal{F}_t].$$

For ATM cliquets, when τ is small,

$$P_t \approx \frac{\sqrt{\tau}}{\sqrt{2\pi}} \mathbb{E} \left[I_T^{ATM}(\tau) | \mathcal{F}_t \right].$$

When τ is small, we may assume that ATM implied volatility is close to variance swap volatility, that is, $I_T^{ATM}(\tau) \approx \sqrt{\hat{v}_T^{T, T+\tau}}$, so that

$$P_t \approx \frac{\sqrt{\tau}}{\sqrt{2\pi}} \mathbb{E} \left[\sqrt{\hat{v}_T^{T, T+\tau}} | \mathcal{F}_t \right].$$

The pay-off of a cliquet is thus very similar to a VIX future.

Pricing and hedging ATM cliquets in a forward variance model To hedge the volatility risk, of an ATM cliquet / VIX future, the simplest formulation is the log-normal dynamics

$$\frac{d\hat{v}_t^{T, T+\tau}}{\hat{v}_t^{T, T+\tau}} = \gamma dW_t$$

The approximate price of the ATM cliquet in this model is given by

$$P(t, \hat{v}_t^{T, T+\tau}) \approx \frac{\sqrt{\tau}}{\sqrt{2\pi}} \mathbb{E} \left[\sqrt{\hat{v}_T^{T, T+\tau}} | \mathcal{F}_t \right] = \frac{\sqrt{\tau \hat{v}_t^{T, T+\tau}}}{\sqrt{2\pi}} e^{-\frac{\gamma^2 (T-t)}{8}},$$

and the dynamic hedge ratio is easily deduced from this formula. This model is very easy to calibrate and use but does not take into account the vol of vol risk, VIX option smile is flat.

Pricing and hedging ATM cliquets in the Heston model We use the same simplified pricing formula:

$$P(t, \hat{v}_t^{T, T+\tau}) \approx \frac{\sqrt{\tau}}{\sqrt{2\pi}} \mathbb{E} \left[\sqrt{\hat{v}_T^{T, T+\tau}} \middle| \mathcal{F}_t \right].$$

The forward variance in Heston's model is given by

$$\hat{v}_t^{T, T+\tau} = \theta + (v_t - \theta) e^{-k(T-t)} \frac{1 - e^{-\kappa\tau}}{\kappa\tau} \approx \theta + (v_t - \theta) e^{-k(T-t)}$$

when τ is small. The corresponding dynamics is the following one:

$$d\hat{v}_t^{T, T+\tau} \approx e^{-\kappa(T-t)} \delta \sqrt{\theta + (\hat{v}_t^{T, T+\tau} - \theta) e^{k(T-t)}} dW_t$$

This is a time-inhomogeneous model, that is very different from the log-normal one, and admits no explicit formula for cliquets. VIX option smile in this model is not flat but is different from the market smile. We conclude that the log-normal forward variance model is a more natural model for pricing and hedging cliquets than Heston's model.

8.5 Rough volatility

Recent studies [16, 3] have shown that fractional Brownian motion (see box below and Figure 8.3) with $H < \frac{1}{2}$ provides a very good fit to log-volatility time series across many markets and asset classes. Since fractional Brownian motion with $H < \frac{1}{2}$ has less regular paths than standard Brownian motion, these models have been termed *rough volatility models*. Rough volatility models perform well for *short-term volatility forecasting* and describe well the *short-term implied volatility smile* behavior (see Figure 8.4).

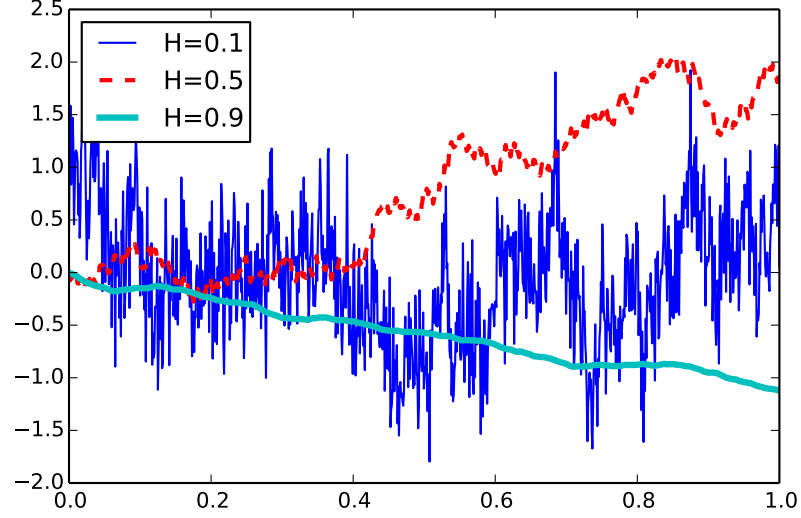


Figure 8.3: Sample trajectories of the fractional Brownian motion

Fractional Brownian motion

The fractional Brownian motion is a Gaussian process with covariance

$$\text{Cov}[W_t^H, W_s^H] = \frac{1}{2}(|t|^{2H} + |s|^{2H} - |t-s|^{2H})$$

where $H \in (0, 1)$ is the *Hurst parameter*

- $H = \frac{1}{2}$: standard Brownian motion
- $H > \frac{1}{2}$: positive autocorrelation, long-range dependence
- $H < \frac{1}{2}$: negative autocorrelation, very rough trajectories

The fractional Brownian motion may be related to the standard Brownian motion through the Mandelbrot - Van Ness representation: up to a multiplicative constant,

$$W_t^H = \int_{-\infty}^0 [(t-s)^{H-\frac{1}{2}} - (-s)^{H-\frac{1}{2}}] dW_s + \int_0^t (t-s)^{H-\frac{1}{2}} dW_s,$$

where W is a standard Brownian motion on the real line.

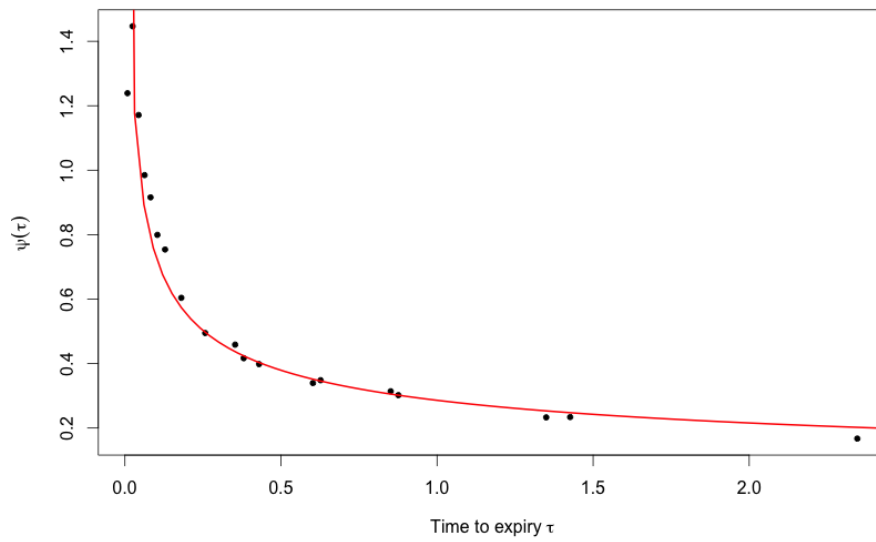


Figure 8.4: ATM skew of S&P options on Aug 14, 2013, and a power-law fit $\psi(\tau) = A\tau^{-0.407}$. This power law is compatible (at short maturities) with fractional Brownian motion volatility with $H = 0.093$. Source: [2].

Rough volatility as a variance curve model Let the instantaneous volatility be given by $\sigma_t = e^{\alpha W_t^H}$. The forward variance is

$$\xi_t^T = \mathbb{E}[\sigma_T^2 | \mathcal{F}_t] = \mathbb{E}[e^{2\alpha W_T^H} | \mathcal{F}_t] = e^{2\alpha \mathbb{E}[W_T^H | \mathcal{F}_t] + 2\alpha^2 \text{Var}[W_T^H | \mathcal{F}_t]}$$

Remark that, by the Mandelbrot - Van Ness representation,

$$\mathbb{E}[W_T^H | \mathcal{F}_t] = \int_{-\infty}^0 [(T-s)^{H-\frac{1}{2}} - (-s)^{H-\frac{1}{2}}] dW_s + \int_0^t (T-s)^{H-\frac{1}{2}} dW_s,$$

so that by Itô formula and the martingale property,

$$\frac{d\xi_t^T}{\xi_t^T} = 2\alpha(T-t)^{H-\frac{1}{2}} dW_t.$$

Volatility derivatives: a simple example Let us come back for the moment to the general log-normal variance curve model (8.1) and consider a call option on *instantaneous forward variance* $\xi_{T_0}^T$. Assuming zero interest rate, Its time- t price is

$$P_t = \mathbb{E}[(\xi_{T_0}^T - K)^+ | \mathcal{F}_t].$$

Since $(\xi_t^T)_{t \geq 0}$ is a log-normal martingale with dynamics

$$d\xi_t^T = \xi_t^T g(T, t)^\top dW_t,$$

the option may be prices using the Black's formula:

$$P_t = \mathbb{E}[(\xi_{T_0}^T - K)^+ | \mathcal{F}_t] = \xi_t^T N(d_t^1) - K N(d_t^2),$$

where N is the standard normal distribution function and

$$d_t^{1,2} = \frac{\log \frac{\xi_t^T}{K} \pm \frac{1}{2} v_{t,T_0}}{\sqrt{v_{t,T_0}}}, \quad v_{t,T_0} = \int_t^{T_0} \|g(T, s)\|^2 ds.$$

Applying the Itô formula, we get the hedging portfolio:

$$dP_t = N(d_t^1) d\xi_t.$$

VIX options in lognormal variance curve models Let us now turn to VIX options.

In terms of the forward variance, the VIX index writes:

$$VIX_T = \sqrt{\frac{1}{\tau} \int_T^{T+\tau} \xi_T^u du}.$$

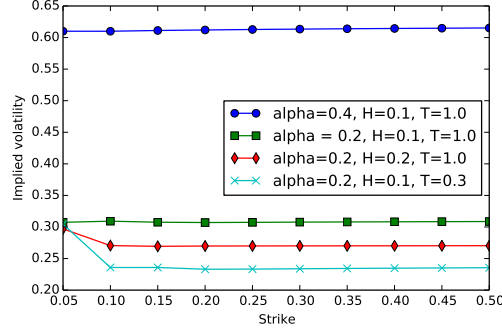


Figure 8.5: VIX Implied volatility smiles in the rough volatility model

The price of a VIX call option is therefore

$$\mathbb{E} \left[\left(\sqrt{\frac{1}{\tau} \int_T^{T+\tau} \xi_T^u du} - K \right)^+ \middle| \mathcal{F}_t \right]$$

There is no explicit formula for this price, however the variables ξ_T^u and ξ_T^v with u close to v are lognormal and strongly correlated. Since τ is small (1 month), the distribution of VIX_T is very close to lognormal, and therefore the smile of VIX options is very flat, as can be seen from Figure 8.5 in the case of a rough volatility model. We conclude that a log-normal model variance curve model leads to a very small correction with respect to constant volatility price, which is not consistent with strong VIX smiles observed in the market (see the bottom graph of Figure 8.1).

8.6 Case study: replication of VIX options

In this section we present the results of a back test, carried out in [14], and whose goal is to compare the hedging performance of the rough volatility model with that of two other more classical models. We consider three models for the VIX index: the Black-Scholes model, the CIR model, and the rough stochastic volatility model. In each model, we perform a back-test of dynamic hedging of a VIX option with a forward variance swap with the same maturity and 1 month duration. In all tests, the hedging portfolio is readjusted daily using the closing prices of the hedging instruments. The test is performed 1000 times, starting on each working day t between Jan 10, 2012 and Apr 29, 2016. On each date t ,

- The parameters are estimated on the 88-day period preceding day t ;
- The initial value of the hedging portfolio and the hedge ratio are initialized with model ATM VIX option price with maturity 1.5 months;
- For 29 working days following day t , the portfolio value / the hedge ratio are readjusted following the change in the value of the hedging asset;
- At the end of the 29-day period, P&L is recorded.

The back test uses synthetic forward variance curve data, computed from the historical prices of S&P index options.

Models: Black-Scholes Let VIX_t be the VIX index at time t , and let F_t^T be the T -forward variance swap at time t , which refers to the same period as the VIX index: $F_t^T = E_Q[VIX_T^2 | \mathcal{F}_t]$. Since we hedge short-maturity options, we simplify the model and let $VIX_t = Ce^{\gamma W_t}$. Then, forward variance swap has dynamics

$$dF_t^T = 2F_t^T \gamma dW_t.$$

We are hedging a VIX option with pay-off

$$(VIX_T - K)_+.$$

Neglecting the interest rate, the option price is given by

$$p(t, F_t^T) = \sqrt{F_t^T} e^{-\frac{\gamma^2}{2}(T-t)} N(d_1) - K N(d_2), \quad d_{12} = \frac{\log \frac{\sqrt{F_t^T} e^{-\frac{\gamma^2}{2}(T-t)}}{K} \pm \frac{\gamma^2(T-t)}{2}}{\gamma \sqrt{T-t}},$$

and the hedge ratio is $\frac{N(d_1)}{2\sqrt{F_t^T}} e^{-\frac{\gamma^2}{2}(T-t)}$.

Models: CIR Assume that the VIX index follows the driftless square root dynamics:

$$dVIX_t^2 = \gamma VIX_t dW_t.$$

The forward variance swap follows the dynamics

$$dF_t^T = \gamma \sqrt{F_t^T} dW_t$$

The price of the VIX option is given by

$$P(t, F_t^T) = \int_{K^2}^{\infty} (\sqrt{x} - K) p_{T-t}(F_t^T, x) dx,$$

where $p_T(v_0, x)$ is the density of the CIR process at time T with the starting value v_0 .

	Black-Scholes		CIR		RFSV	
	No hedge	Hedge	No hedge	Hedge	No hedge	Hedge
Mean	0.0144	0.00533	0.0136	0.00339	0.00991	0.000634
Std. dev.	0.0189	0.00355	0.0206	0.00650	0.0188	0.00414
RMSE	0.0238	0.00641	0.0247	0.00734	0.0212	0.00419
Red. factor	3.717		3.717		5.072	

Table 8.1: Empirical Performance of hedging strategies based on different models

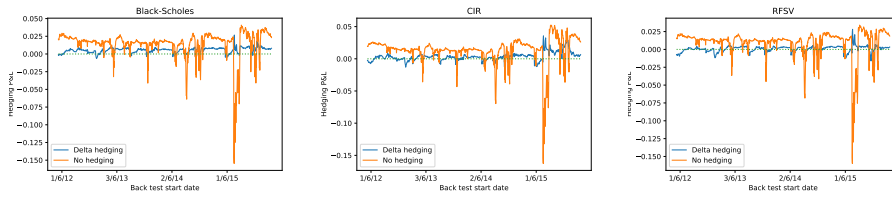


Figure 8.6: Back-testing PnL as function of time for the three models

Models: rough volatility Assuming that

$$\text{VIX}_t = C e^{\sigma W_t^H},$$

where W^H is the fractional Brownian motion with the Hurst parameter H , we get, after some computations:

$$P_t = \sqrt{F_t^T} e^{-\frac{C}{2}(T-t)^{2H}} N(d_t^1) - K N(d_t^2), \quad d_t^{1,2} = \frac{\log \frac{\sqrt{F_t^T} e^{-\frac{C}{2}(T-t)^{2H}}}{K} \pm \frac{C}{2}(T-t)^{2H}}{(T-t)^H \sqrt{C}},$$

and the option price dynamics takes the following form:

$$dP_t = \frac{N(d_t^1) e^{-\frac{C}{2}(T-t)^{2H}}}{2\sqrt{F_t^T}} dF_t^T \quad \text{where} \quad C = \frac{\sigma^2}{2H\Gamma^2(H + 1/2)}.$$

Contrary to the previous two models, this one formally has two parameters to be estimated: σ and H . We estimate H only once on the VIX index time series from April 17, 2001 to April 16, 2021, obtaining the value 0.377.

The results of the back-test are presented in Table 8.1 and Figure 8.6. It is clear that the rough volatility model has a much better performance than the other two models, with a 35% reduction of the root mean square error compared to the Black-Scholes model, and an even greater improvement compared to the CIR model.

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